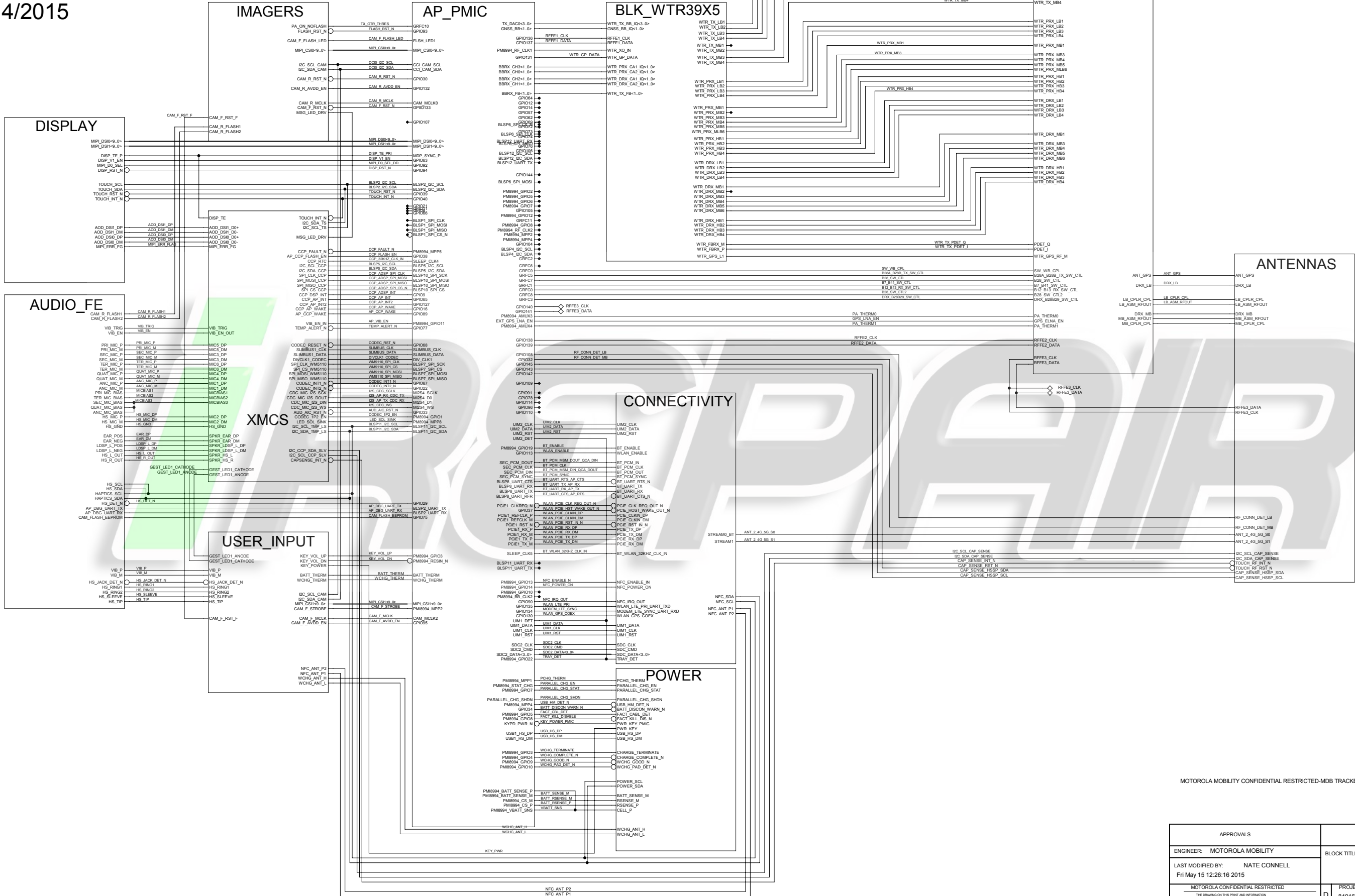


BOUNCE
SYSTEM 4.0
REV. A
05/14/2015



REVISIONS				
ZONE	LTR	DESCRIPTION	DATE	APPROVED

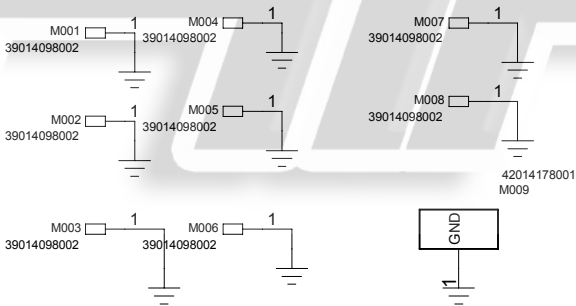
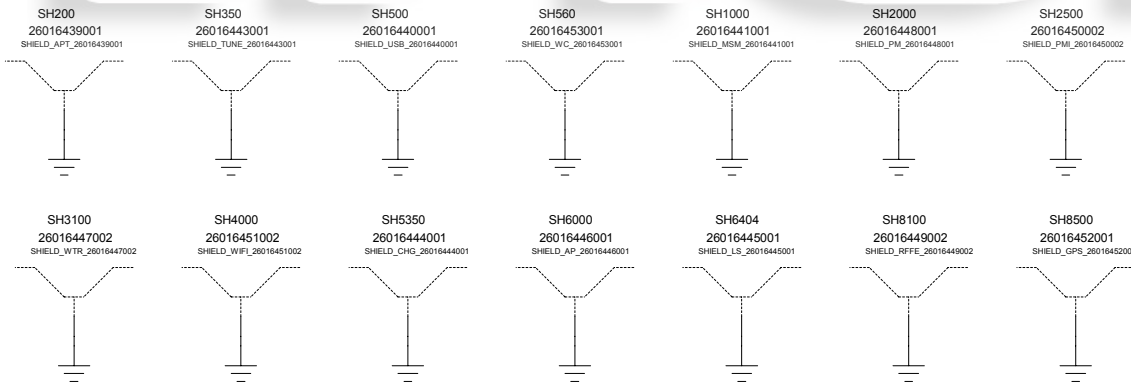
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MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
		84016327001_a_released.cpm	01
CNO FILE NAME:		PAGE 1 OF 76	

BOUNCE ME/SHIELDS

BOUNCE REFERENCE DESIGNATOR RANGES BY BLOCK

- 000 - 499: ANTENNA
- 500 - 599:: POWER
- 600 - 899: USER INPUT/PRODUCT FLEX
- 1000-2999: MSM8994+PM8994+PMI8994
- 3000-3199: WTR39X5

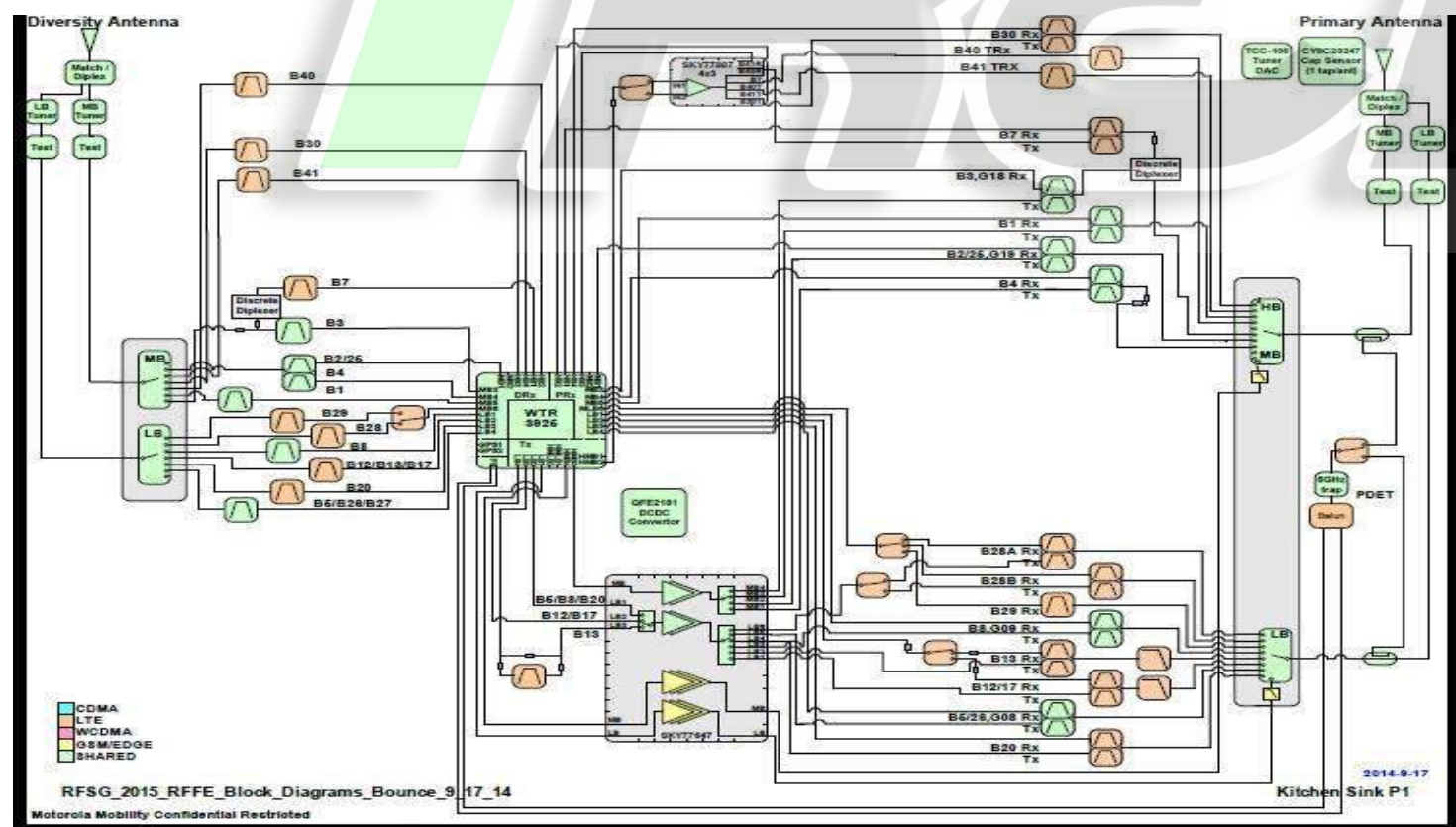
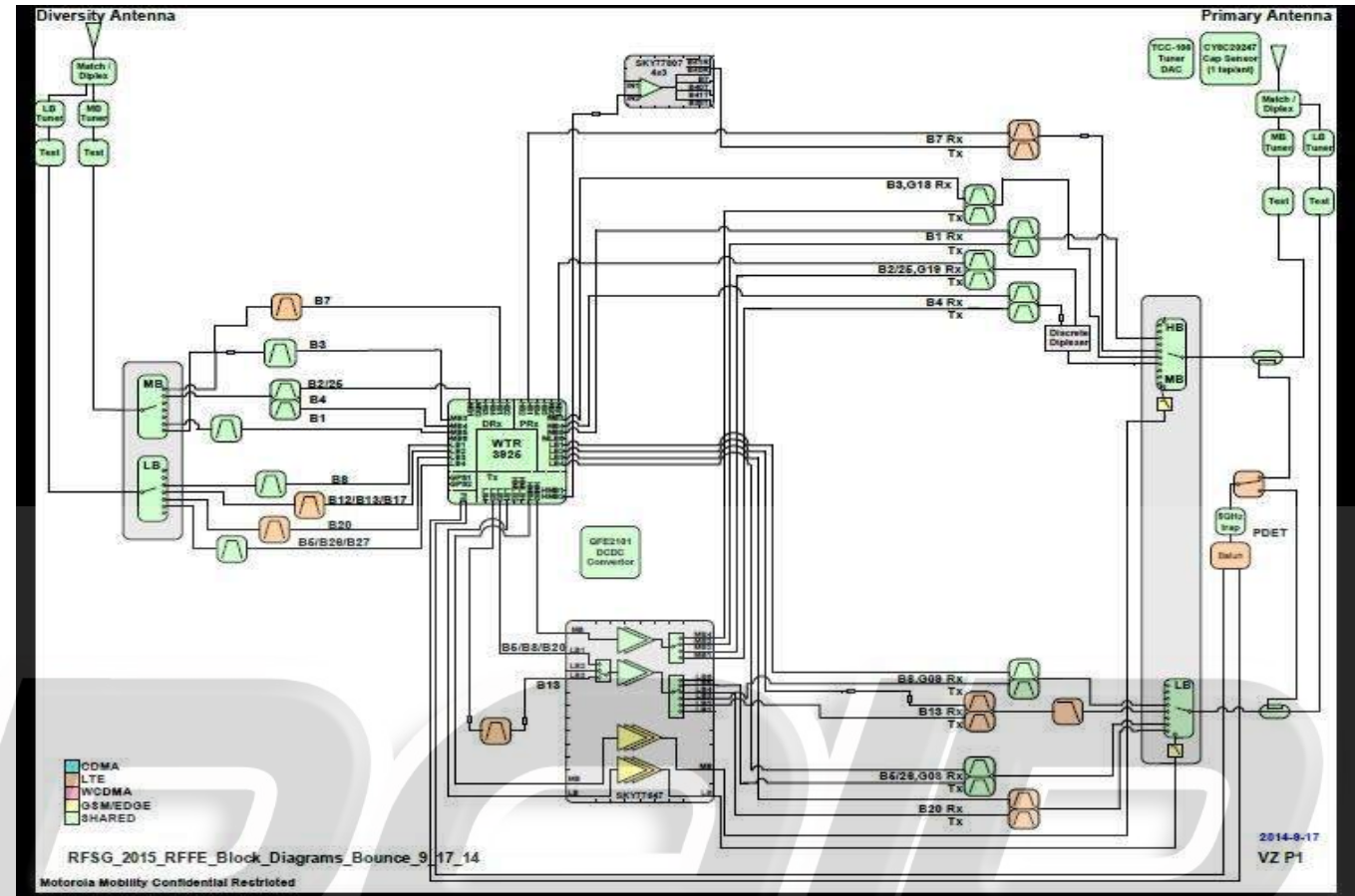
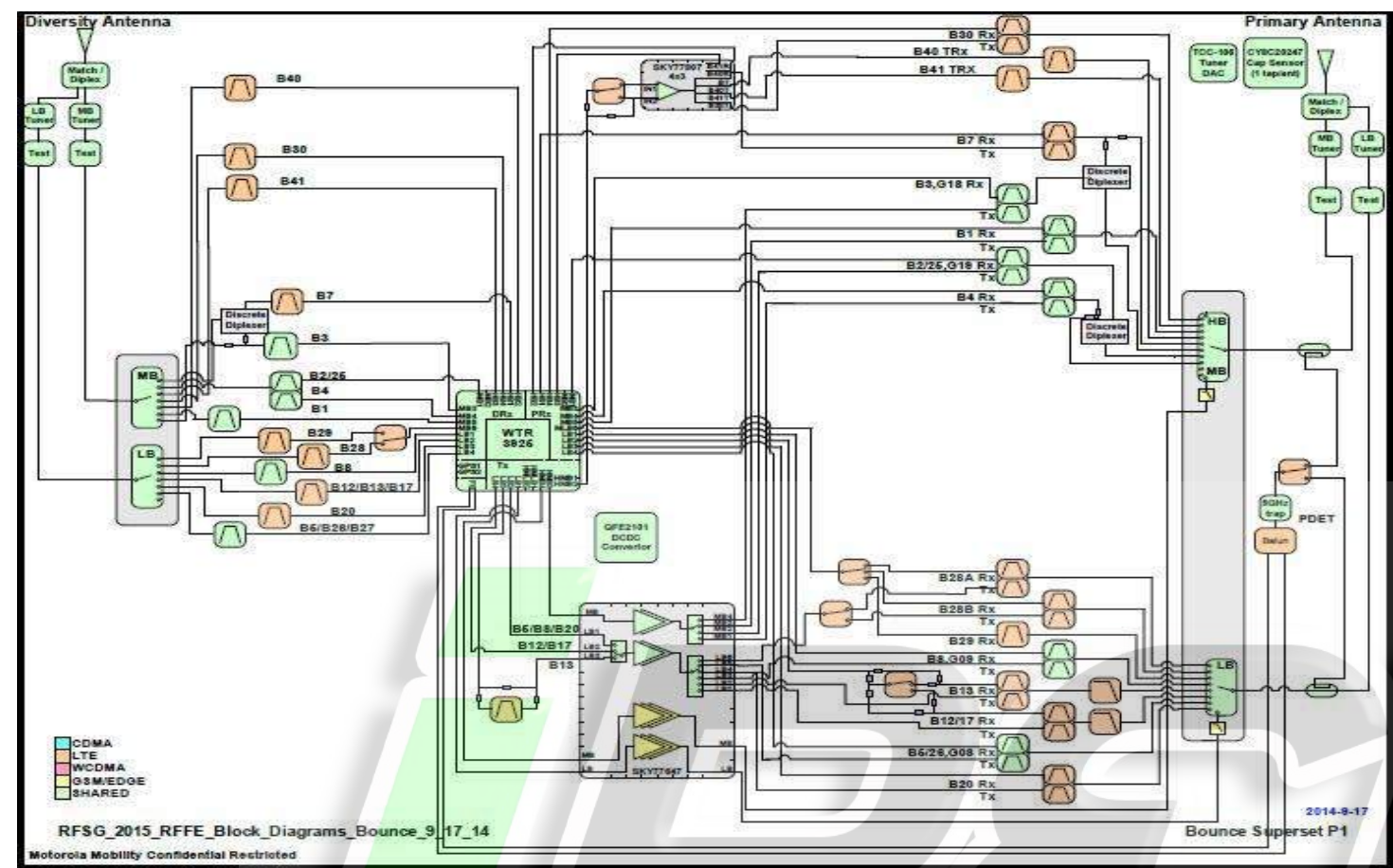
- 4000-4499: CONNECTIVITY
- 4500-4999: AUDIO FE
- 5000-5499: DISPLAY
- 5500-5999: CAMERA
- 6000-6999: XMCS
- 7000-8999: RFFE
- 9000-9999: PRODUCT-SPECIFIC (FILTERING, 100PF CAPS)



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Thu Jul 30 10:00:20 2015		84016327001_a_released	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		REV 01	
		PAGE 2 OF 76	

RFFE: BLOCK DIAGRAM

REFDES: 7000-8999



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE: block_rfe	
LAST MODIFIED BY: NATE CONNELL Thu Sep 18 09:29:12 2014		PROJECT NAME 84016327001_a_released.cpm	
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		PAGE 3 OF 76	

8

7

6

5

4

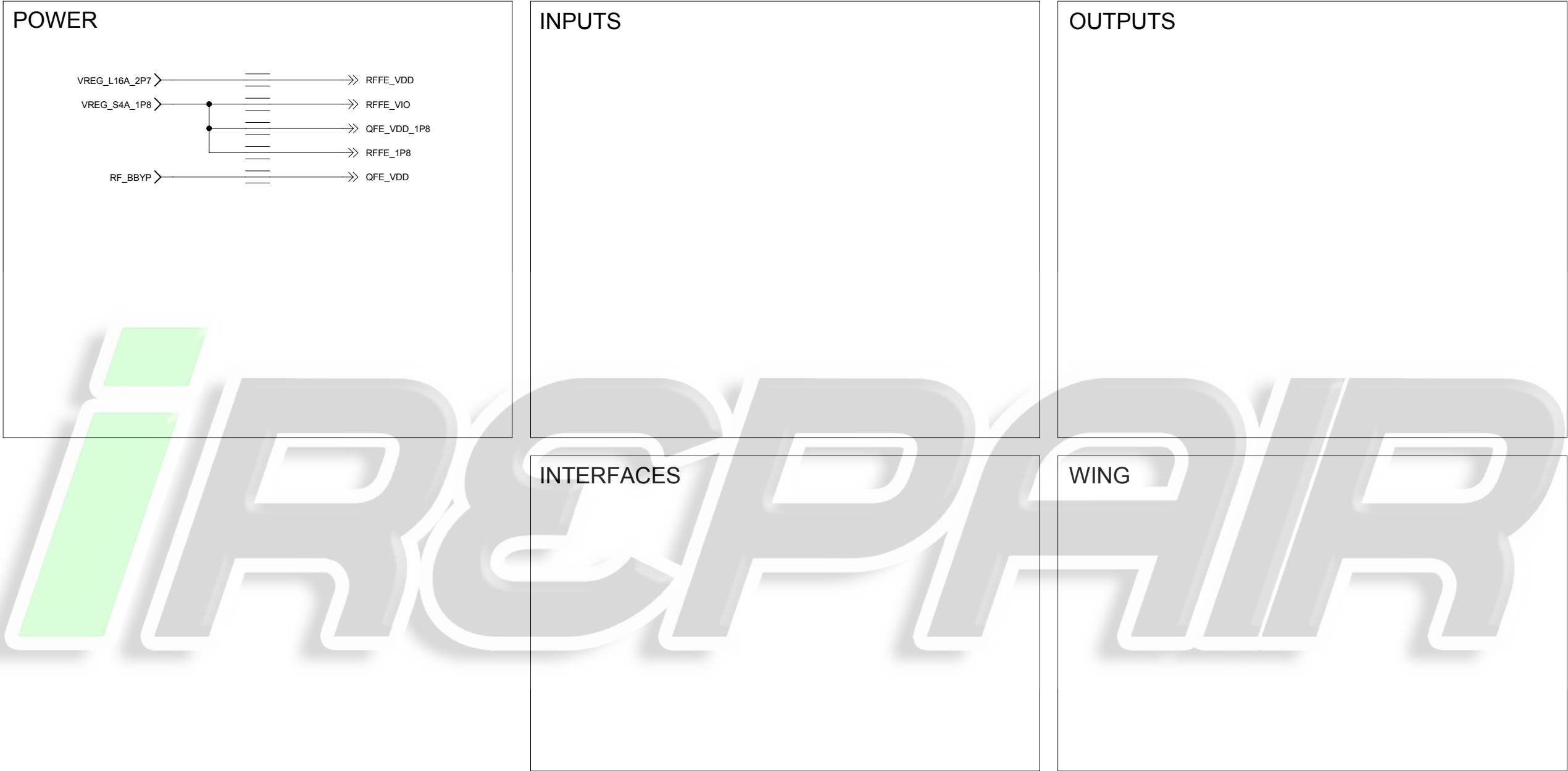
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2

1

RFFE: PORTS

REFDES: 7000-8999



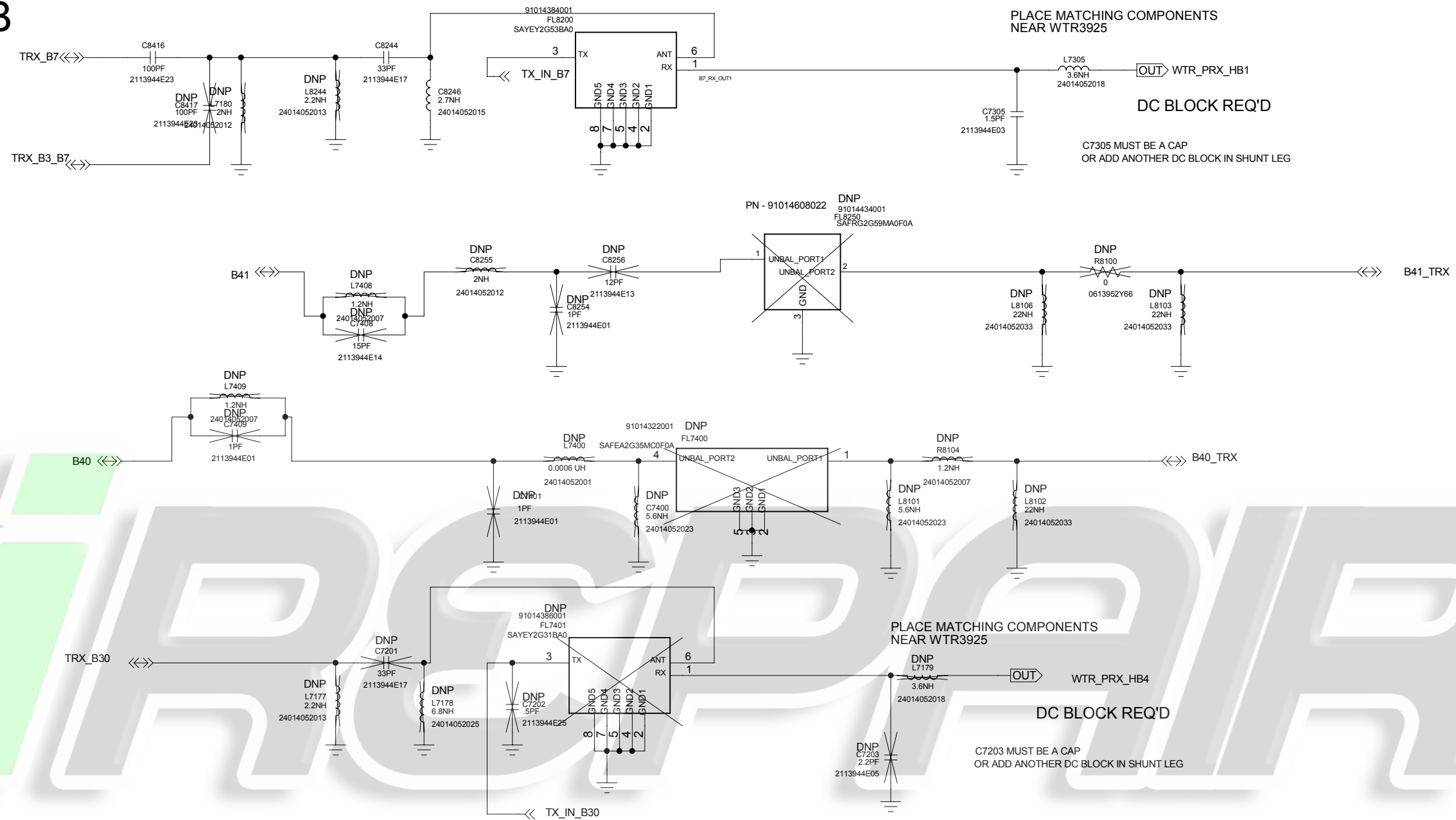
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ENGINEER: NATE CONNELL		BLOCK TITLE: block_rffe	
LAST MODIFIED BY: NATE CONNELL Fri May 15 09:11:01 2015			
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		CAD FILE NAME:	PAGE 4 OF 76

D



PAGE 5 OF 76

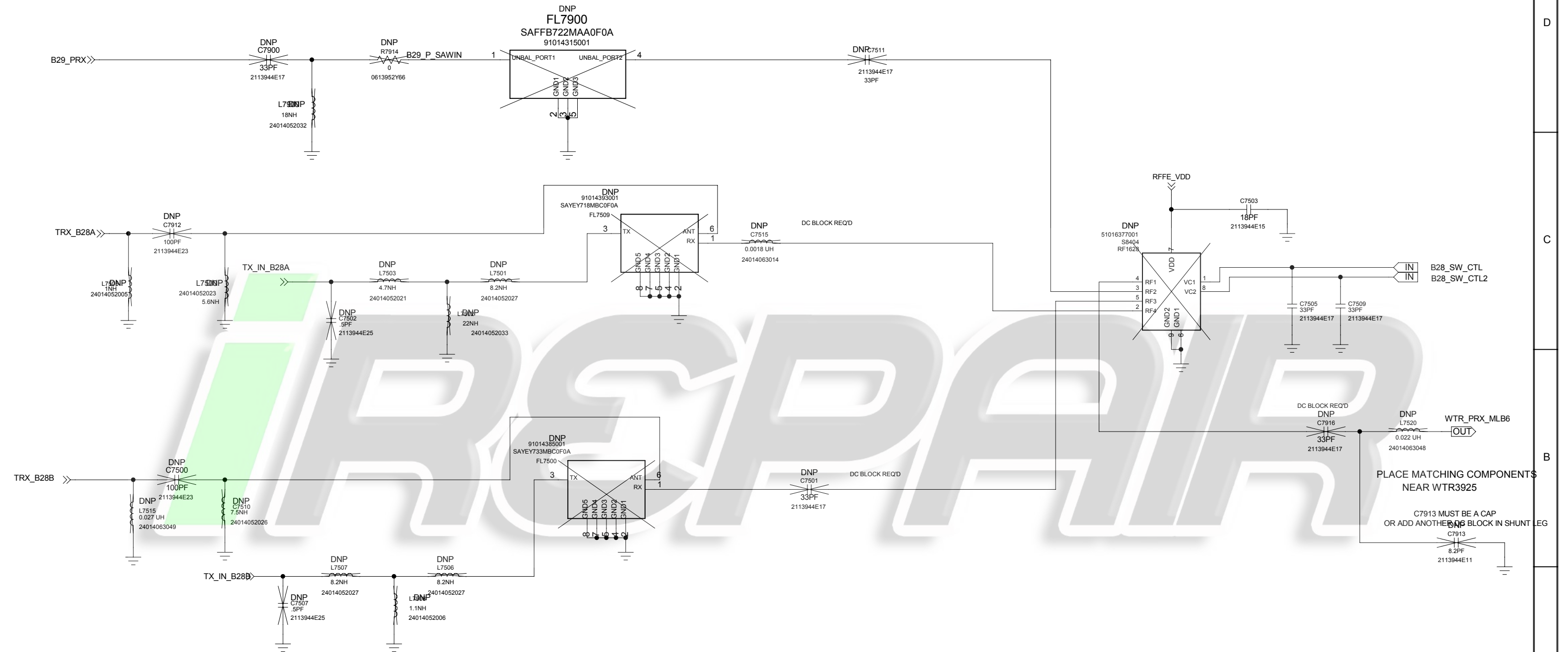
RFFE: TRX HB
REF 7100-8499



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: JATIN KULKARNI	BLOCK TITLE: block_rffe		
LAST MODIFIED BY: JATIN KULKARNI Thu Jul 30 10:00:28 2015			
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		CAD FILE NAME:	PAGE 6 OF 76

RFFE: PRX B29/B28A/B28B

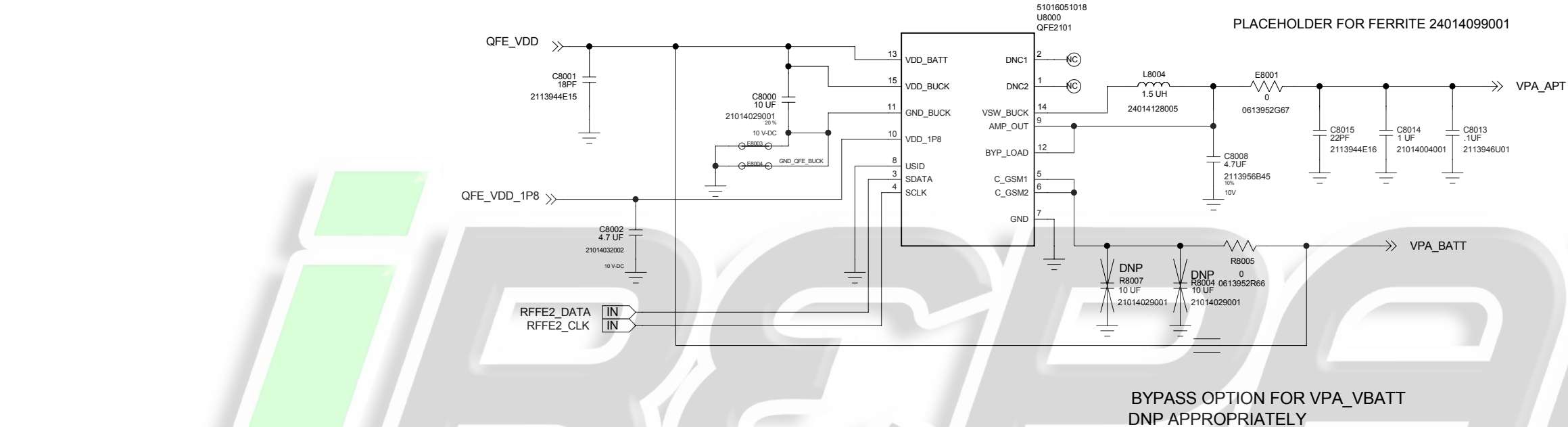
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APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: JATIN KULKARNI		BLOCK TITLE: block_rfe	
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		CAD FILE NAME: 80046327001_a_released.cpm	
		PAGE 7 OF 76	

RFFE: APT

REF 8000-8099



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: Thu Jul 30 10:00:31 2015		block_rfe	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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		CAD FILE NAME:	PAGE 8 OF 76

RFFE: TX MMBB PA

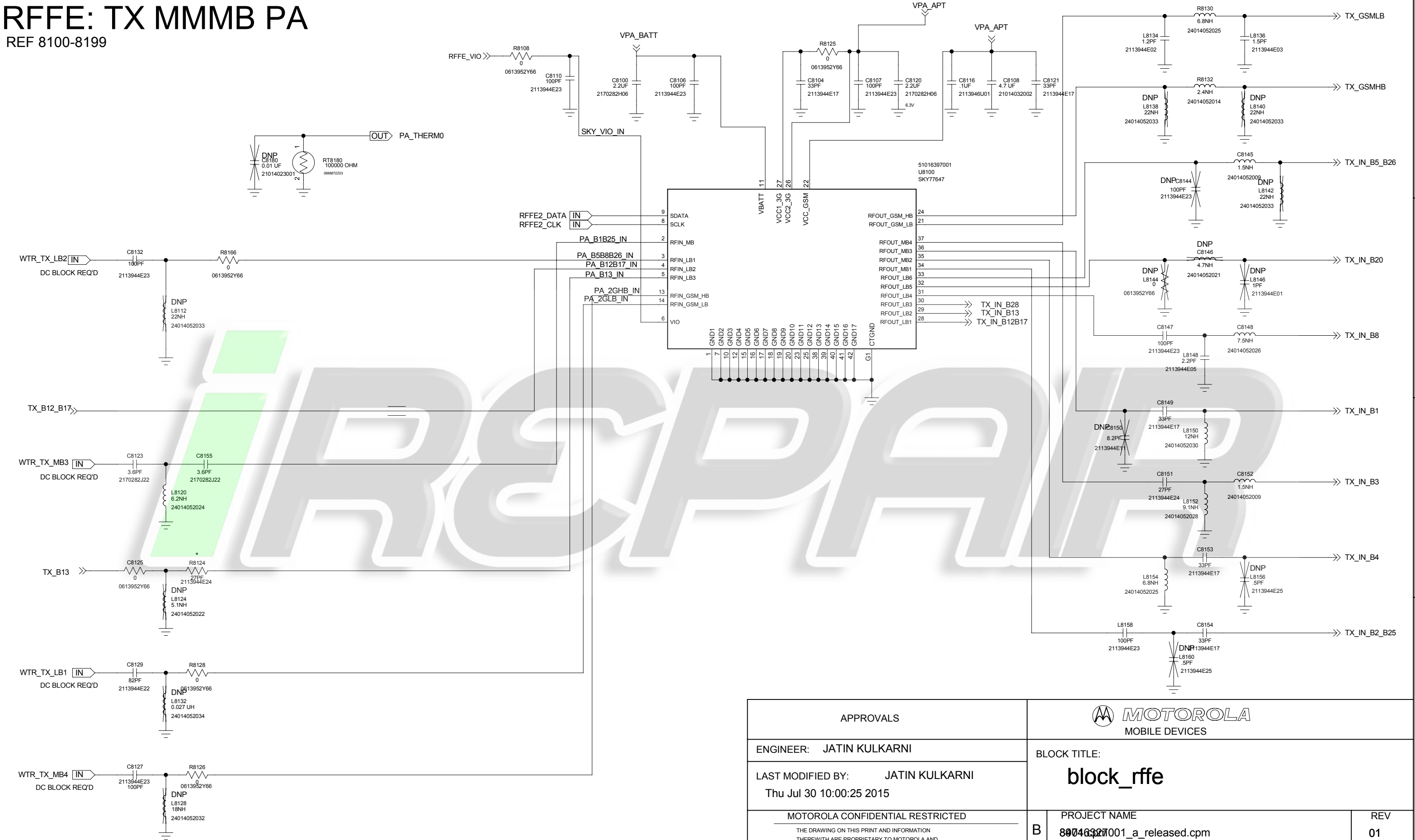
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C

B

A



D

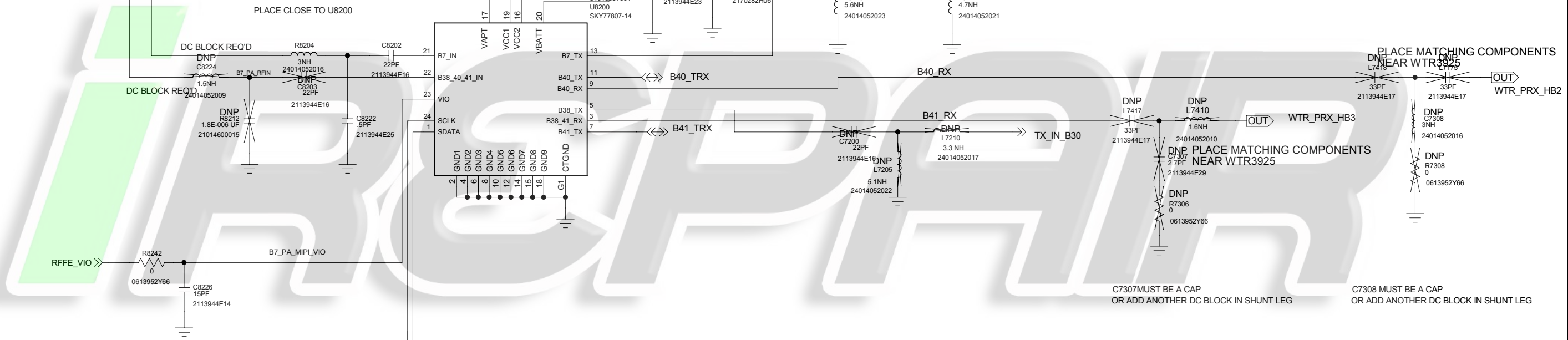
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B

A

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: JATIN KULKARNI		BLOCK TITLE: block_rfe	
LAST MODIFIED BY: JATIN KULKARNI Thu Jul 30 10:00:25 2015		PROJECT NAME 80046327_001_a_released.cpm	
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		PAGE 9 OF 76	

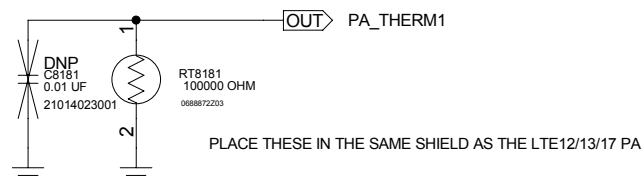
REF 7100-8299



DATE: 2/7/2013
- THE MATCHING COMPONENTS ARE TEMPORARY PLACE HOLDERS ONLY.
THE EXACT COMPONENTS WILL NEED TO BE DEFINED BEFORE RELEASED.

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: block_rffe		
LAST MODIFIED BY: Thu Jul 30 10:00:28 2015			
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		CAD FILE NAME:	PAGE 10 OF 76

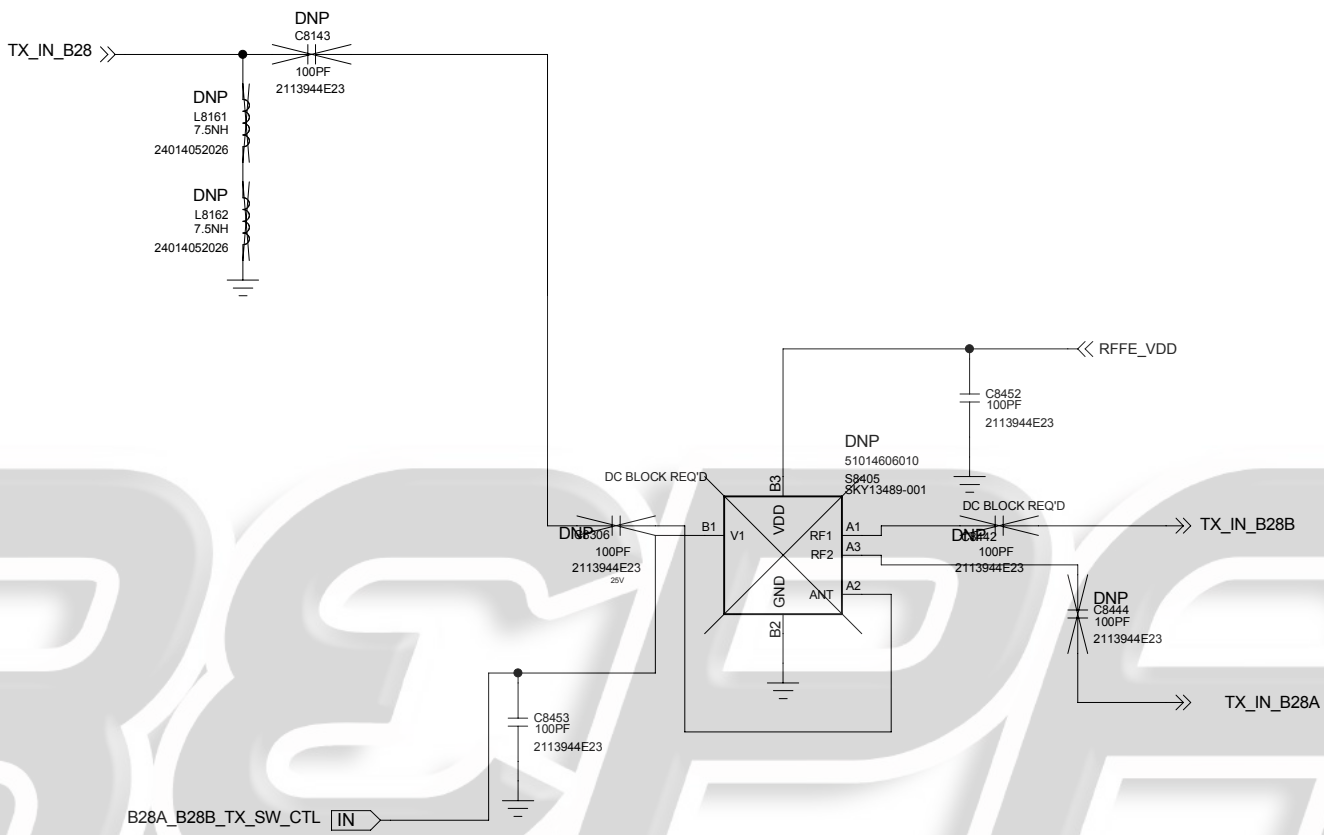
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APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: Thu Jul 30 10:00:25 2015			
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		CAD FILE NAME:	PAGE 11 OF 76

RFFE: TX B28A/B28B

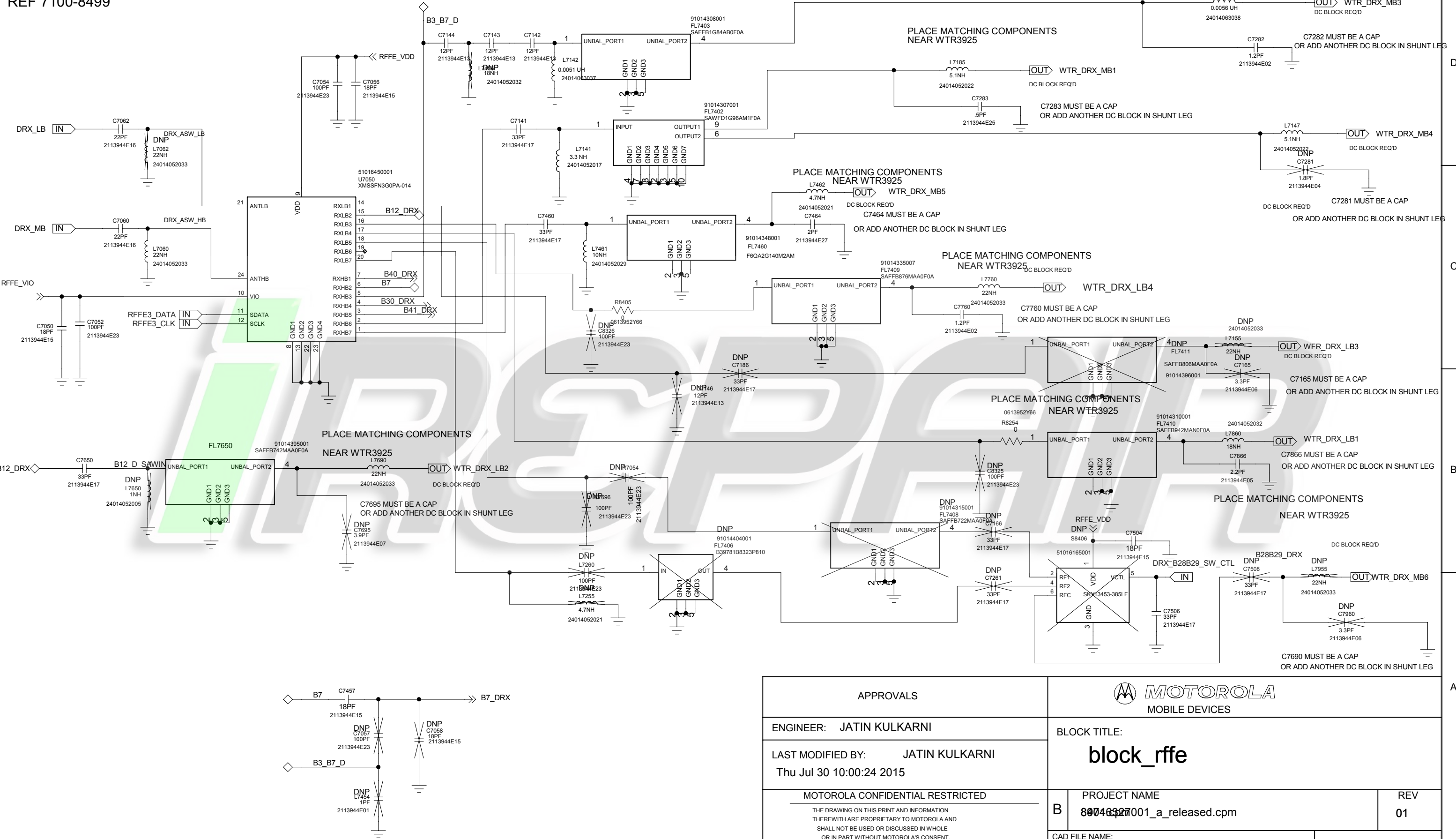
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ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		block_rfe	
Thu Jul 30 10:00:29 2015			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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© COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		REV 01	
		CAD FILE NAME:	
		PAGE 12 OF 76	

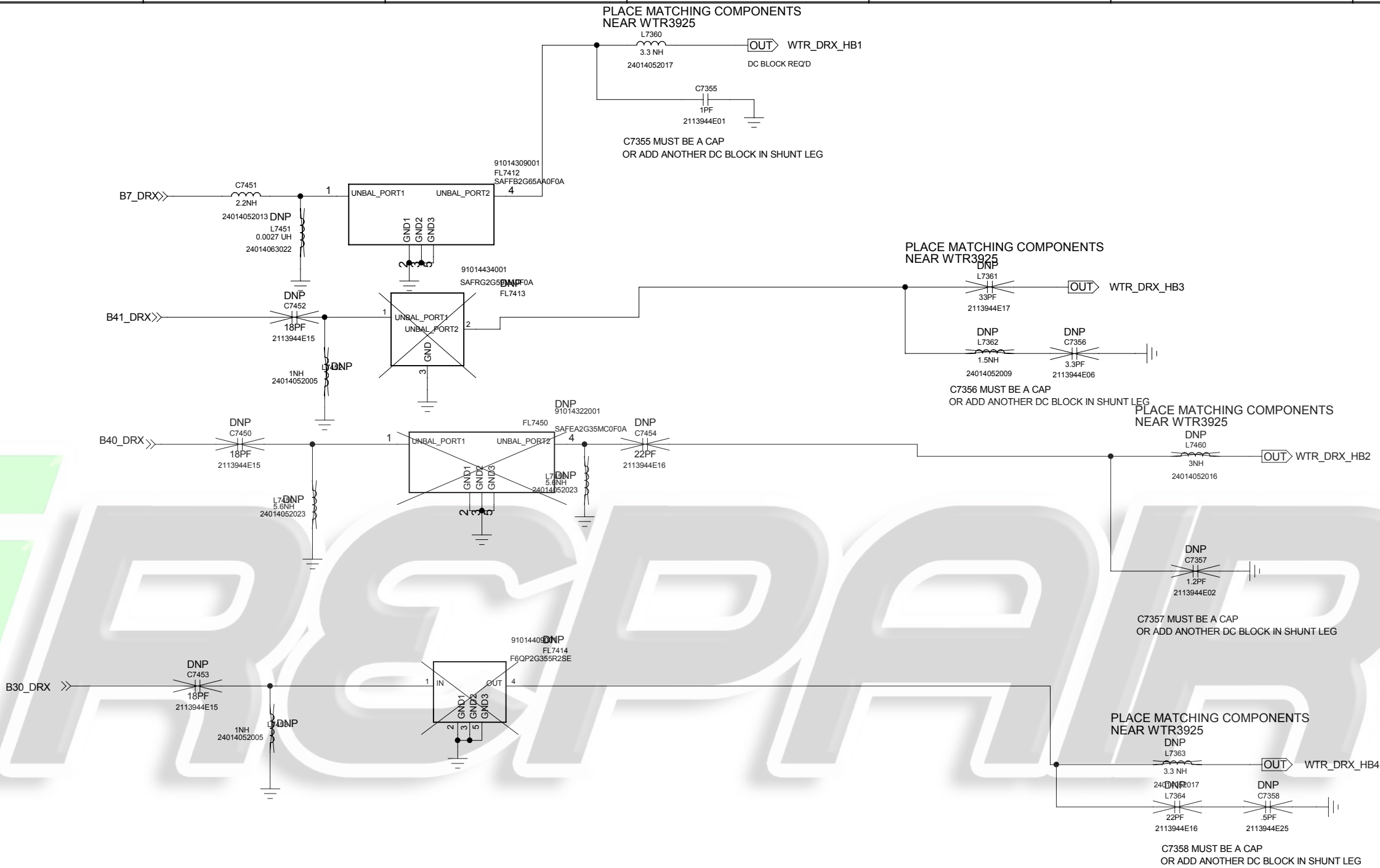
RFFE: DRX ANTENNA SWITCH

REF 7100-8499



RFFE: DRX HB

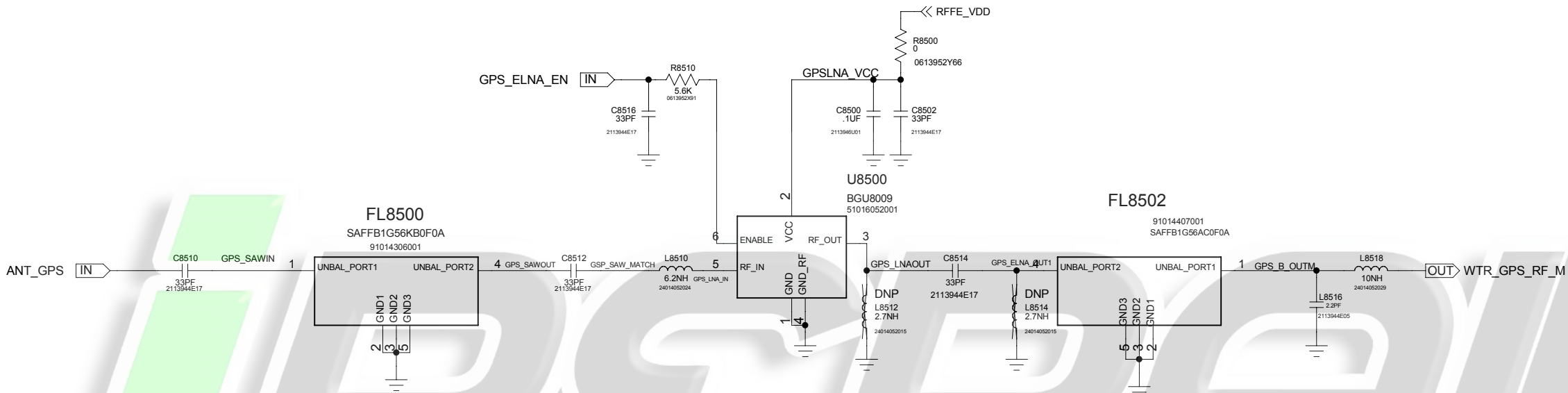
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ENGINEER: JATIN KULKARNI		BLOCK TITLE:	
LAST MODIFIED BY: JATIN KULKARNI Thu Jul 30 10:00:29 2015		block_rfe	
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		B 80046327001_a_released.cpm	
		REV 01	
		PAGE 14 OF 76	

RFFE: GPS

REF : 8500-8549

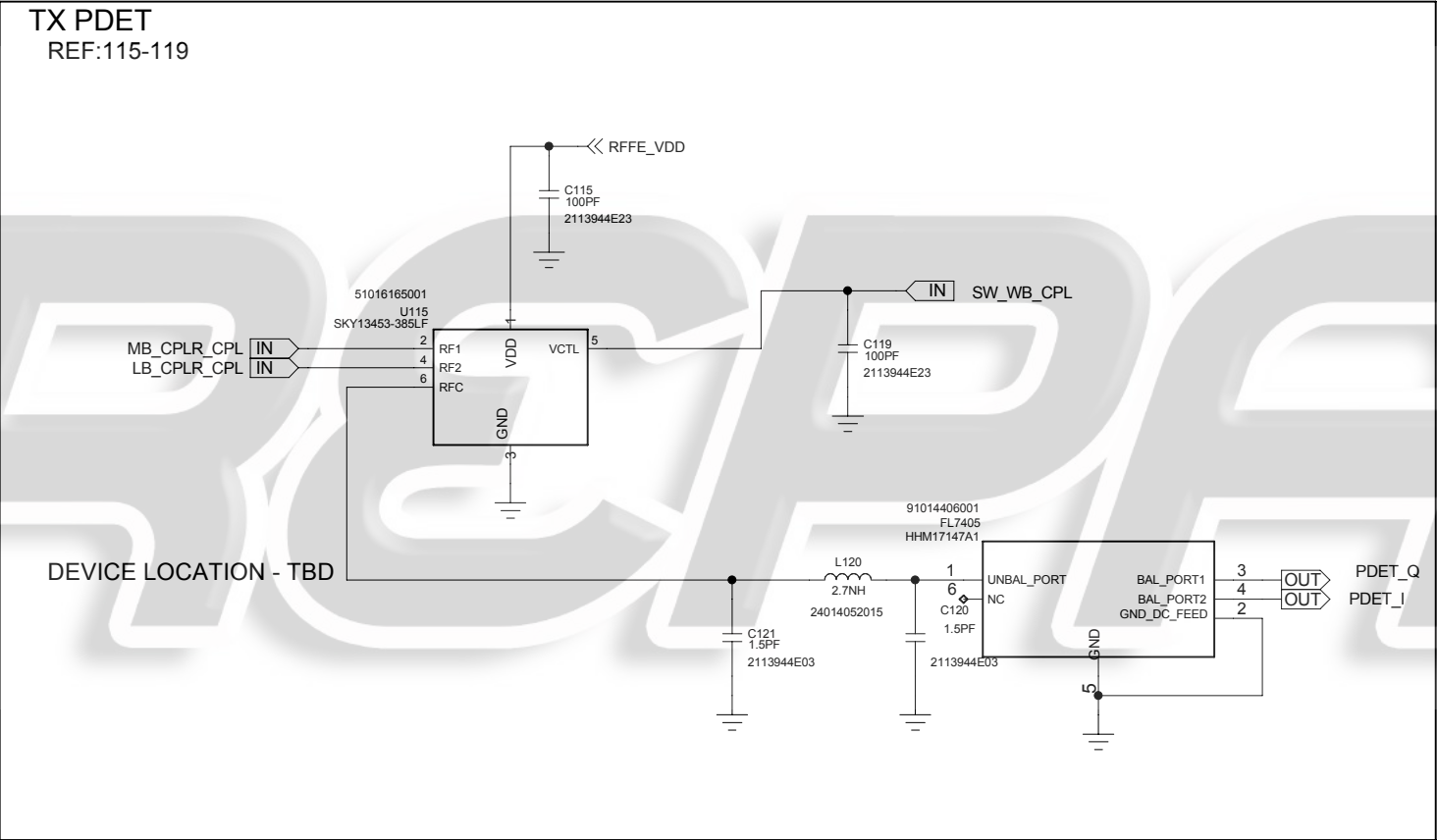


PLACE IN WTR1625 SHIELD

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: block_rfe	
LAST MODIFIED BY: Thu Jul 30 10:00:26 2015			
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RFFE: PDET

REF : 8500-8549

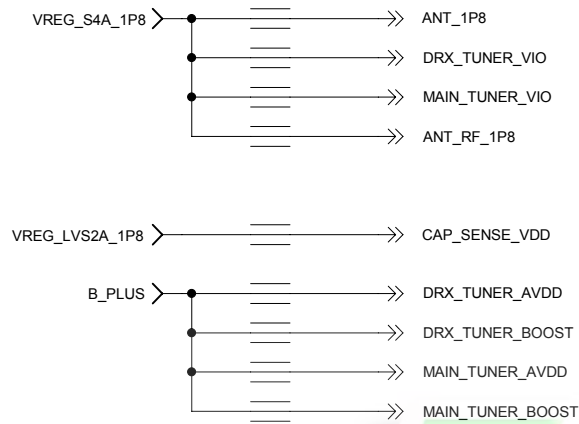


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LAST MODIFIED BY:		block_rfe	
Thu Jul 30 10:00:30 2015			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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		CAD FILE NAME:	PAGE 16 OF 76

ANT: PORTS

REFDES: 000 - 499

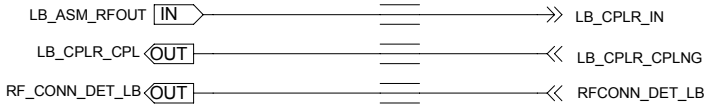
SUPPLIES



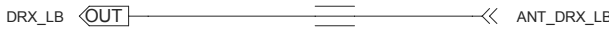
TX MID BAND ANTENNA



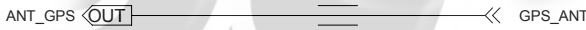
TX LOW BAND ANTENNA



DRX LOW BAND ANTENNA



GPS ANTENNA



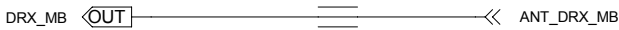
BT/WLAN ANTENNA



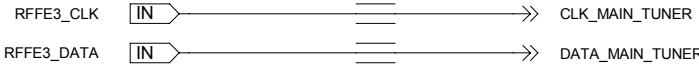
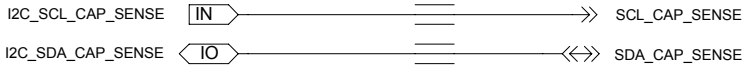
WLAN ANTENNA 2 (FOR 2X2 MIMO SUPPORT)



DRX MID BAND ANTENNA



INTERFACES



GPIO - INPUT



GPIO - OUTPUT



APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Fri May 15 09:10:52 2015

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

antennas

B

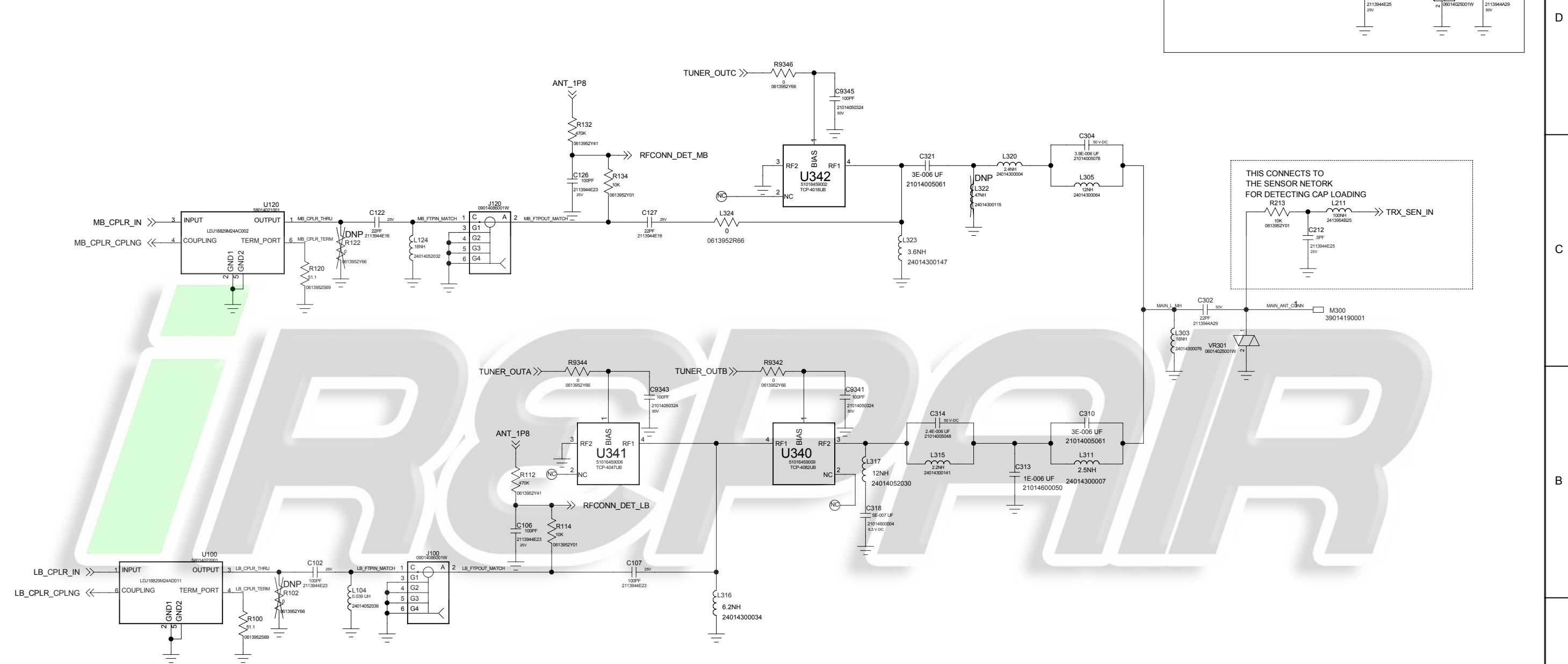
PROJECT NAME
84016327001_a_released.cpm

REV
01

CAD FILE NAME:

PAGE 17 OF 76

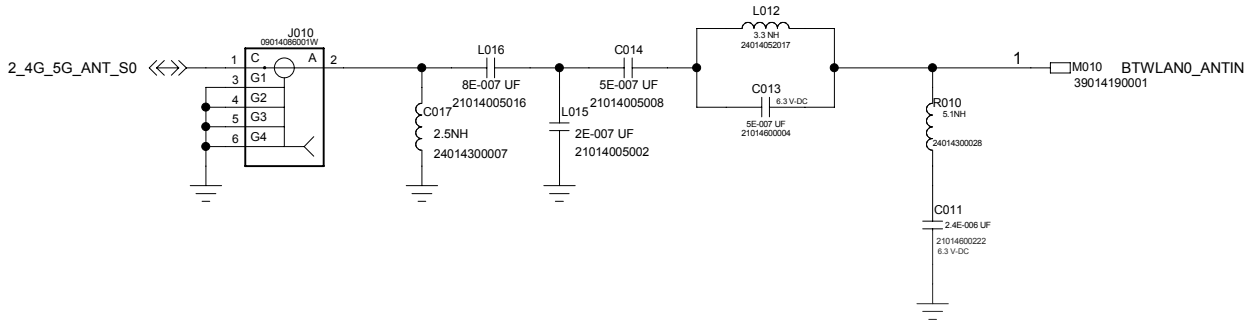
ANT: MAIN



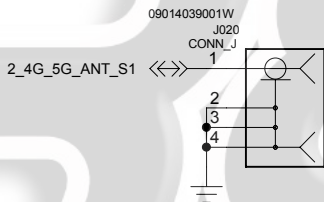
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ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Jul 30 10:00:33 2015		antennas	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		PAGE 18 OF 76	

ANT: WLAN

BT/WLAN CHAIN0 ANTENNA
REF:010-019



WLAN CHAIN1 ANTENNA
REF:020-029

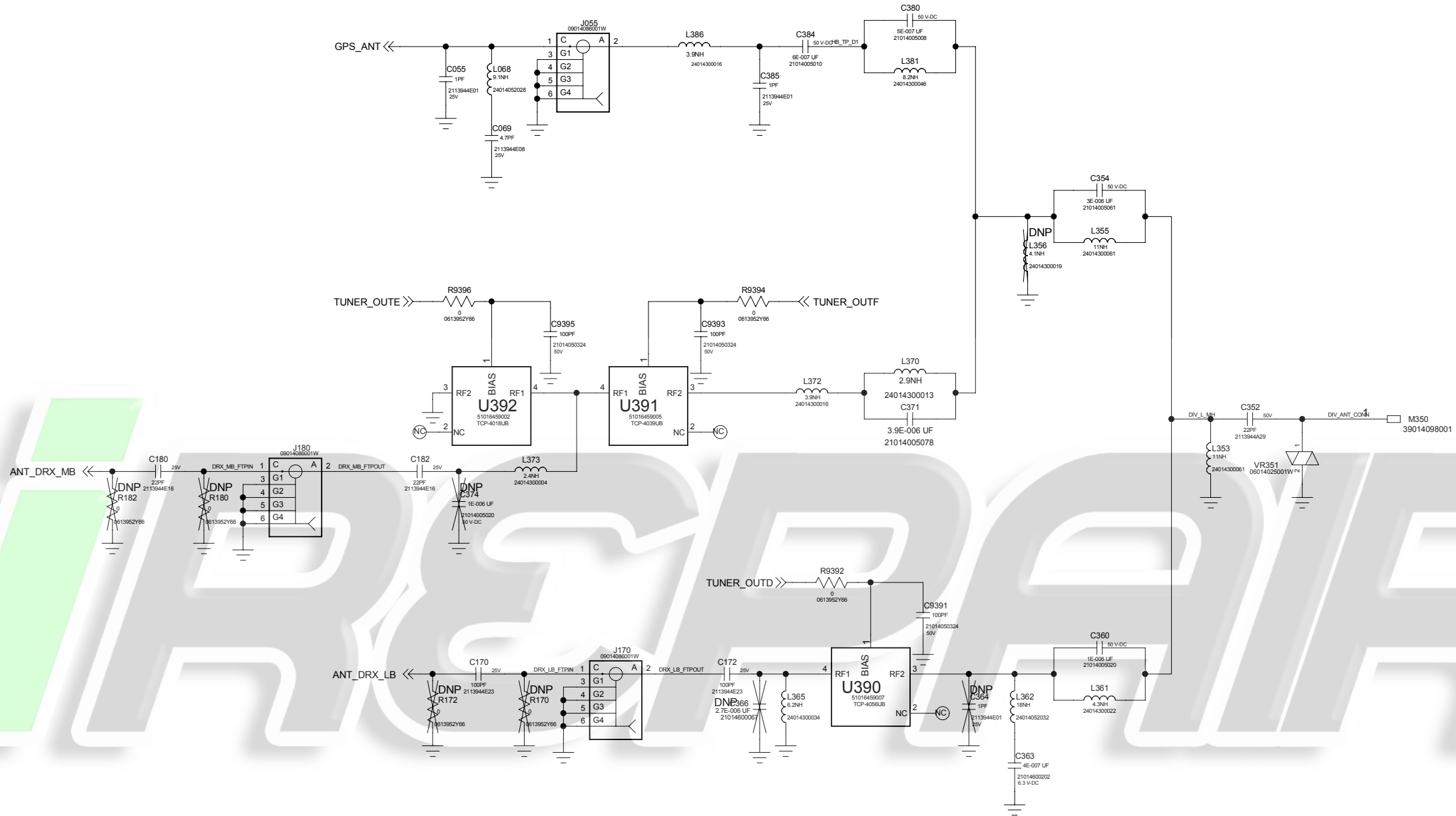


MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MOTOROLA Thu Jul 30 10:00:34 2015		antennas	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		CAD FILE NAME:	
		PAGE 19 OF 76	

ANT: DIV/GPS

REFDES: 300-399

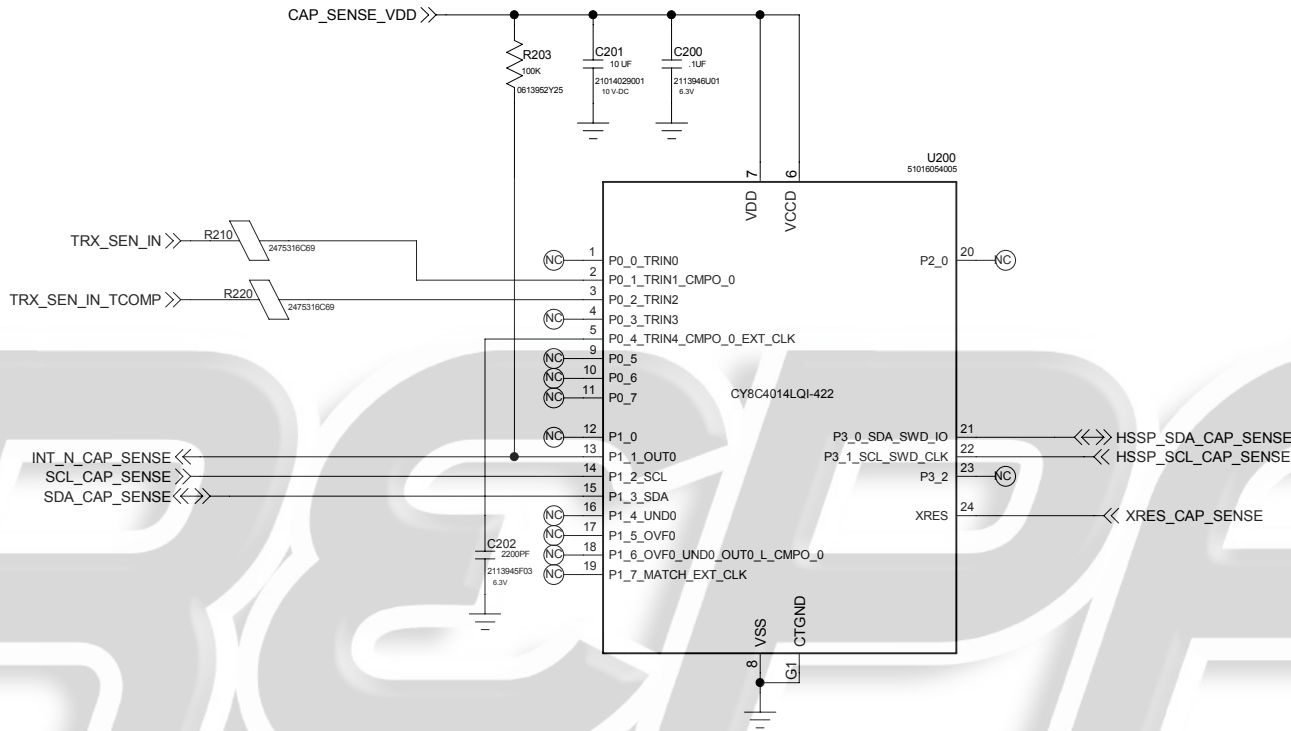


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APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Jul 30 10:00:33 2015		antennas	
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		84016327001_a_released.cpm	
		REV 01	
		PAGE 20 OF 76	

ANT: CAP SENSE

REFDES: 200-299



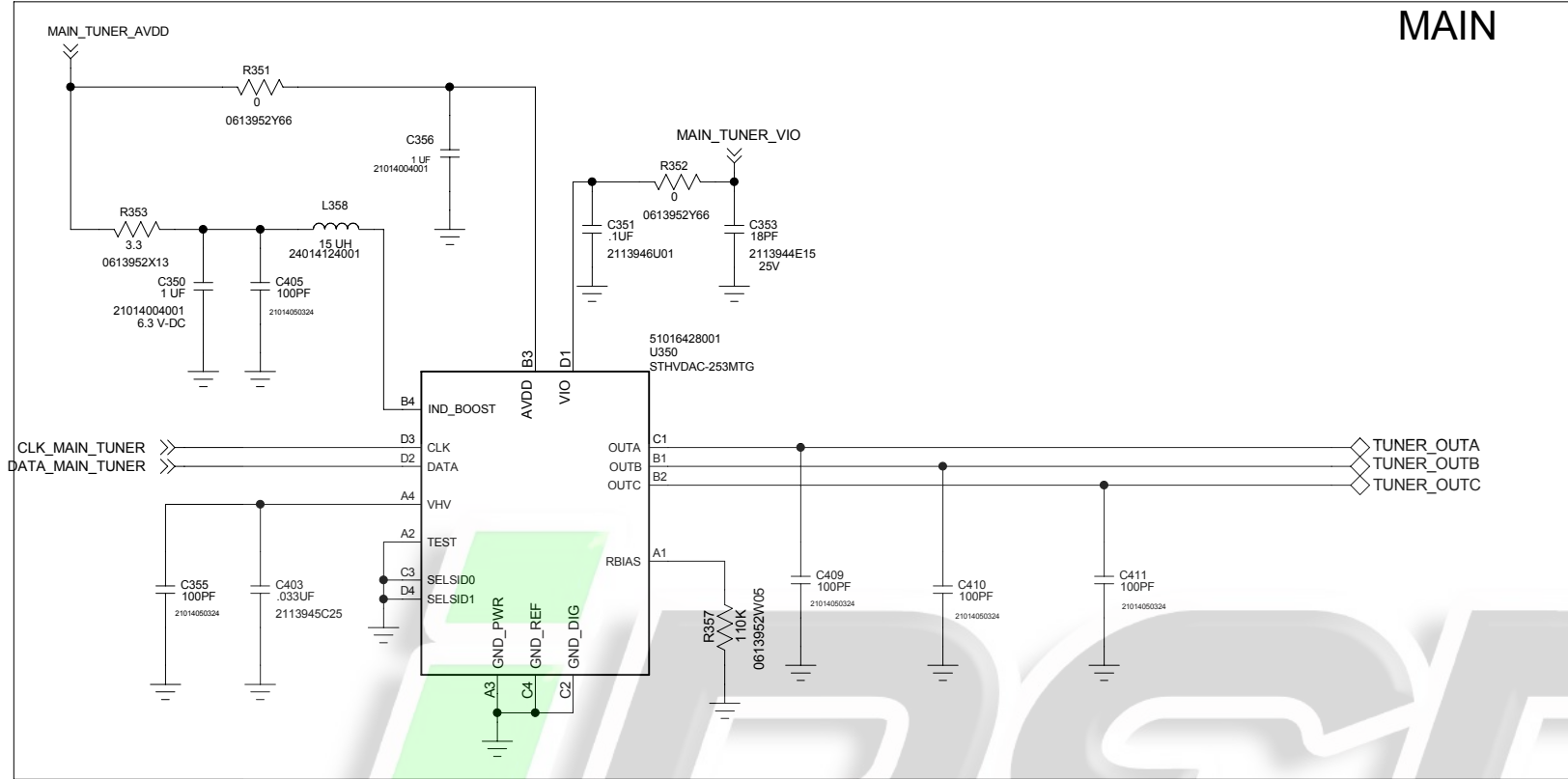
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Jul 30 10:00:34 2015		antennas	
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		B 84016327001_a_released.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 21 OF 76	

ANT: TUNERS

REFDES: 400-499

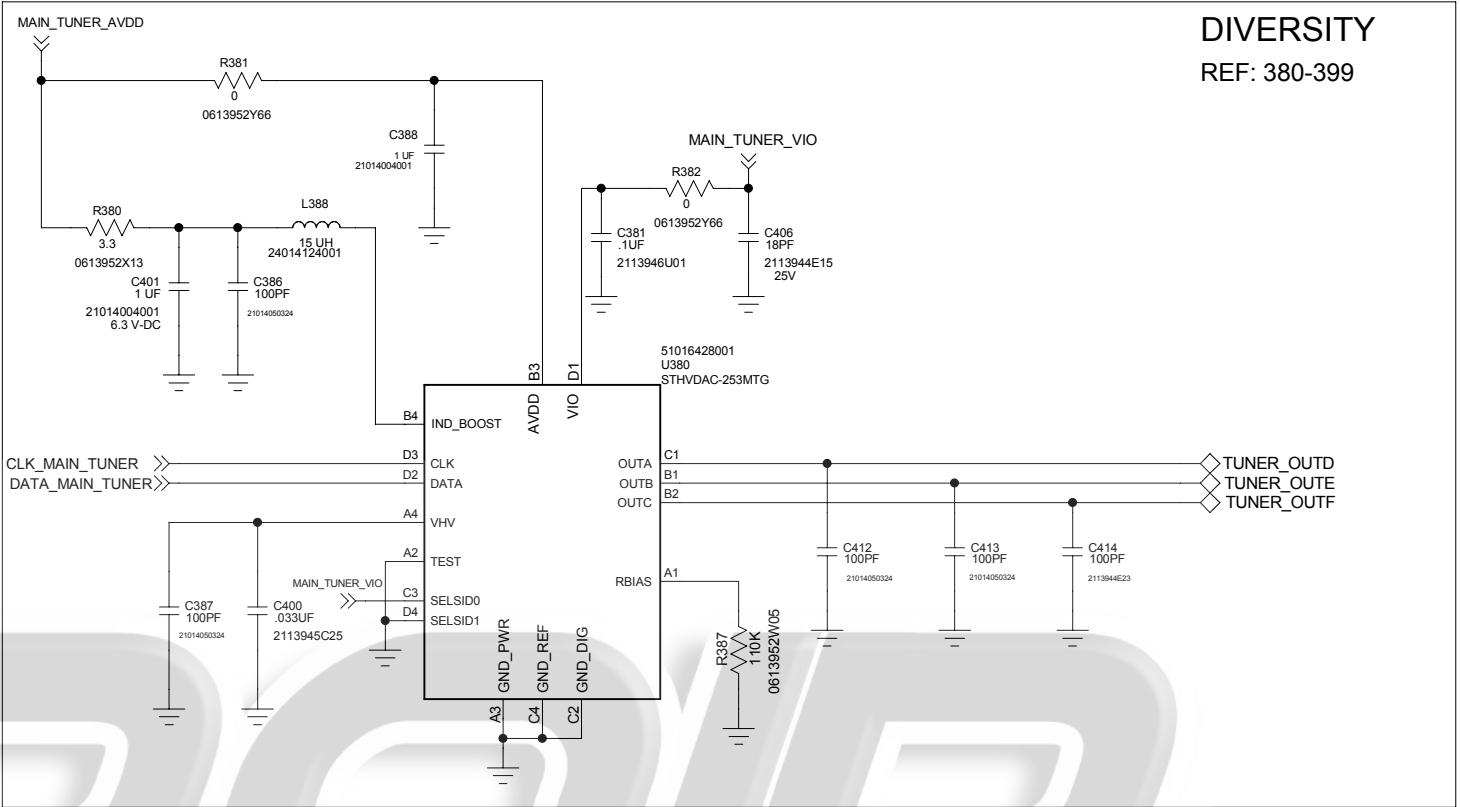
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MAIN



DIVERSITY

REF: 380-399

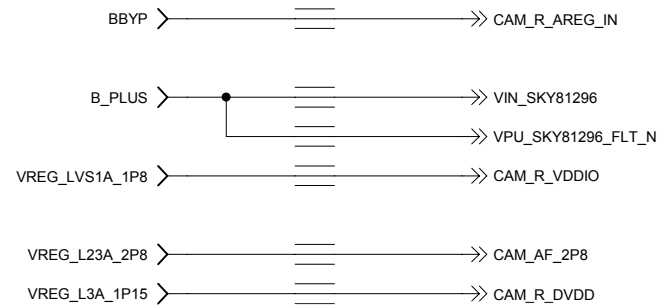


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Jul 30 10:00:34 2015		antennas	
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		PAGE 22 OF 76	

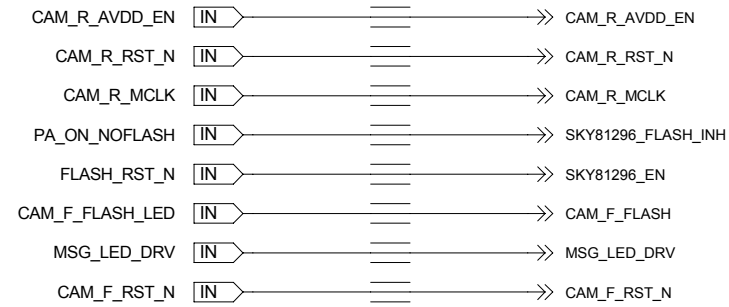
CAM: PORTS

REF 5500-5999

SUPPLIES



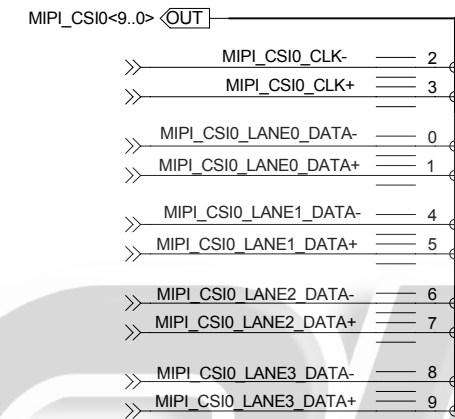
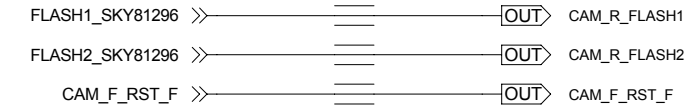
INPUTS



INTERFACES



OUTPUTS



APPROVALS

ENGINEER: NATE CONNELL

LAST MODIFIED BY: NATE CONNELL
Fri May 15 09:10:46 2015

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BLOCK TITLE:

imagers

B	PROJECT NAME 84016327001_a_released.cpm
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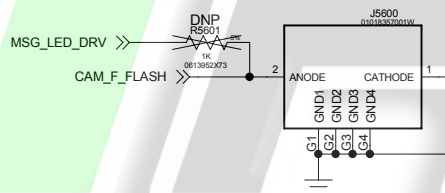
CAD FILE NAME:

REV
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CAM: FRONT
REF 5600-5699

FRONT CAMERA LIVES ON TOP FLEX
PLEASE SEE USR:TOP FLEX PAGE

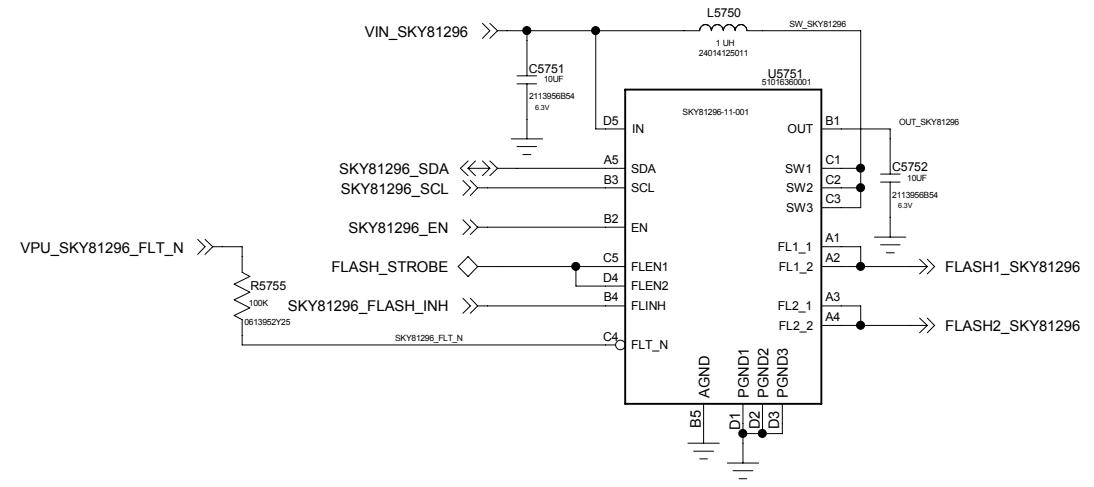


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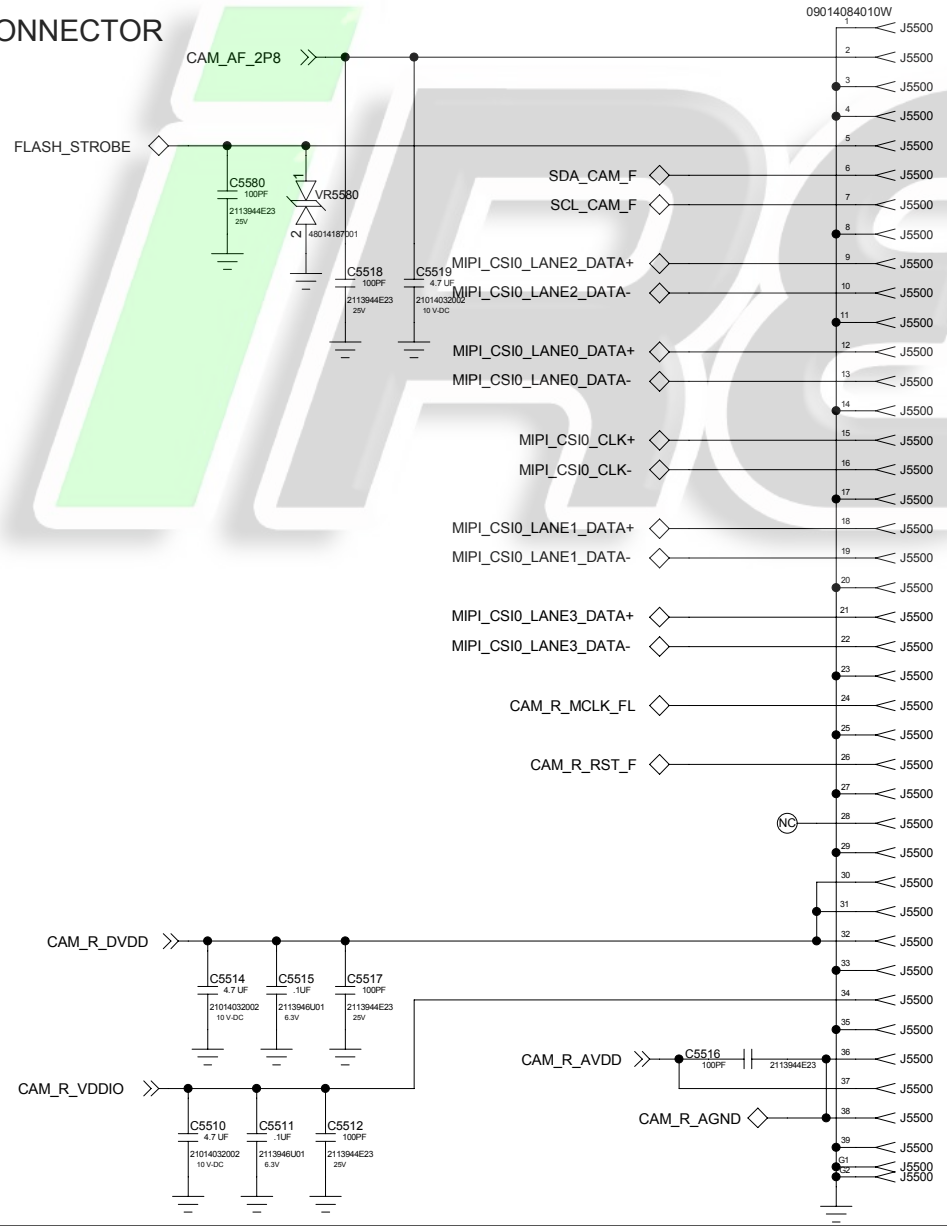
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	REV 01	
CAD FILE NAME:		PAGE 24 OF 76

CAM: REAR
REF:5500-5599

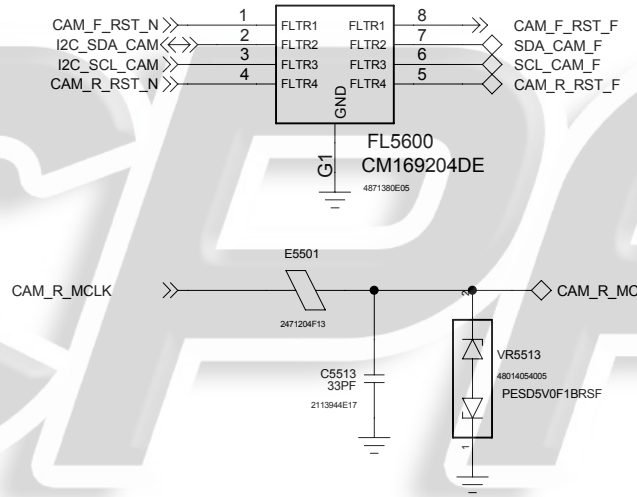
LED FLASH DRIVER
REF:5750-5799



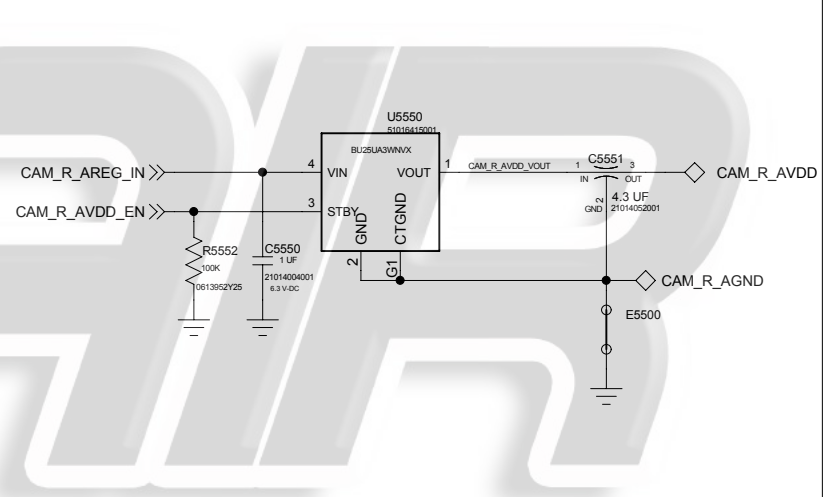
IMAGER FLEX CONNECTOR



EMI/ESD FILTER



ANALOG REGULATOR

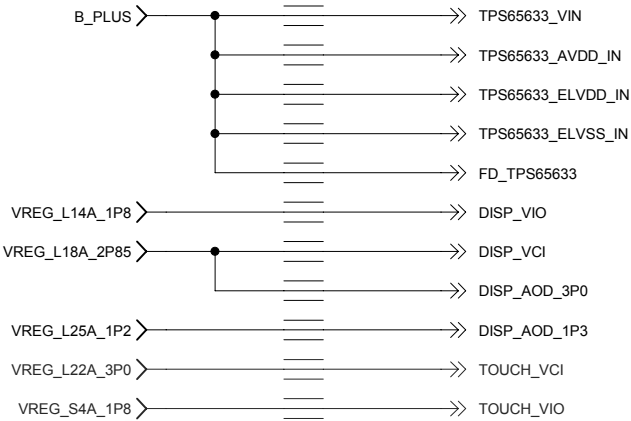


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Jul 30 10:00:36 2015		imagers	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		REV 01	
		PAGE 25 OF 76	

DISP: PORTS

REF 5000-5499

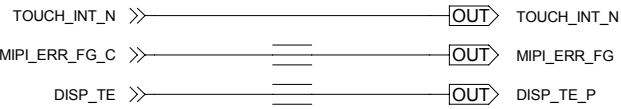
POWER



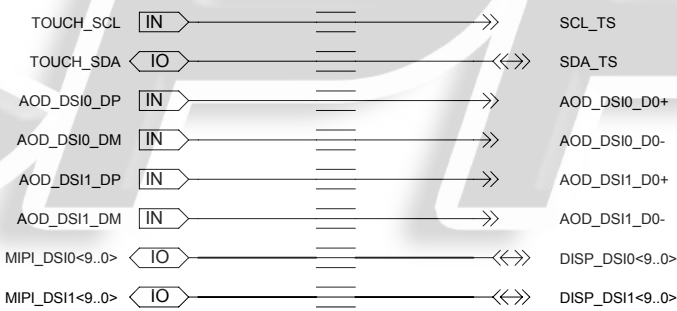
INPUTS



OUTPUTS



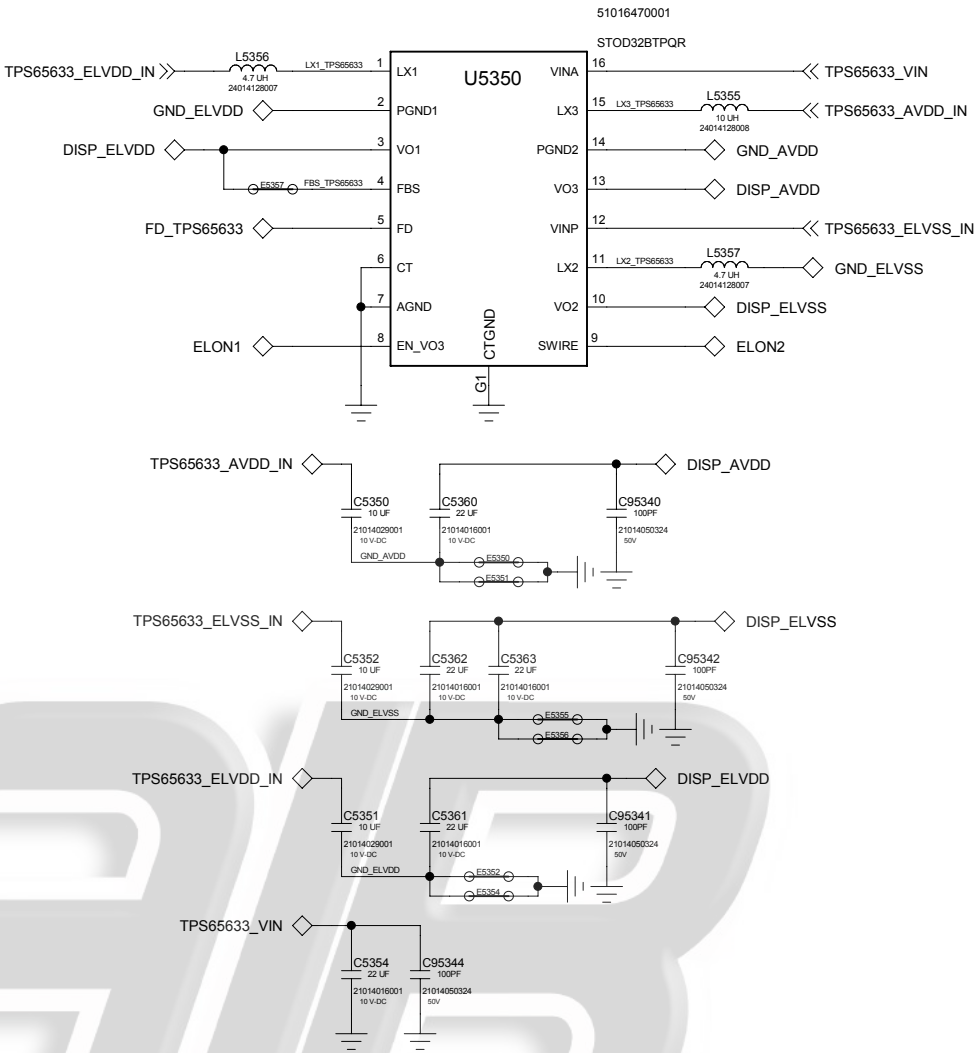
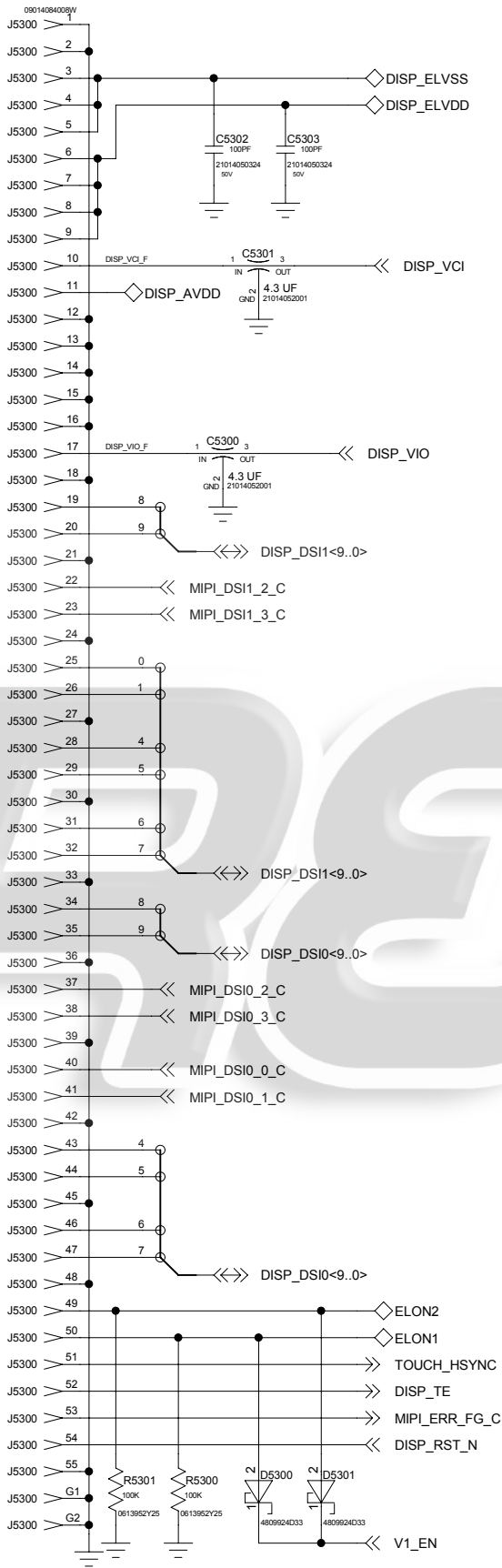
INTERFACES



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: NATE CONNELL		BLOCK TITLE: display		
LAST MODIFIED BY: NATE CONNELL Fri May 15 09:10:40 2015				
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		B 84016327001_a_released.cpm		01
		CAD FILE NAME:		PAGE 26 OF 76

DISP: CONNECTOR

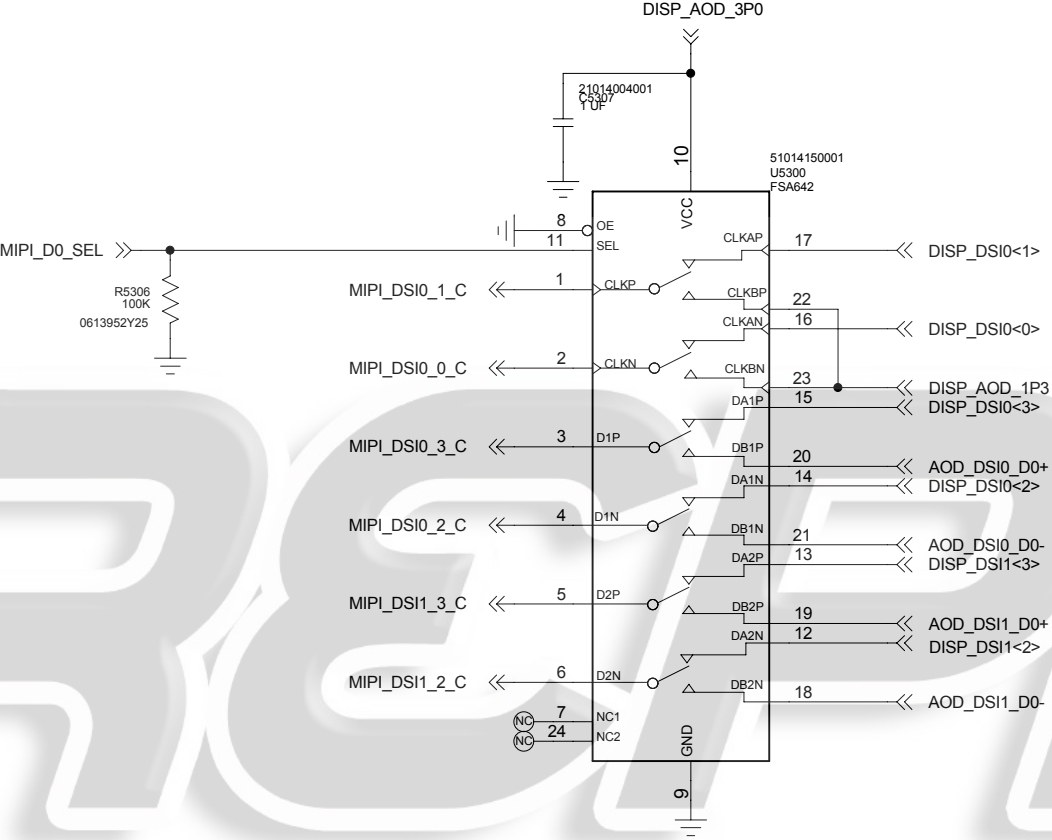
REF 5300-5349



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Thu Jul 30 10:00:38 2015		display	
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		01	
		PAGE 27 OF 76	

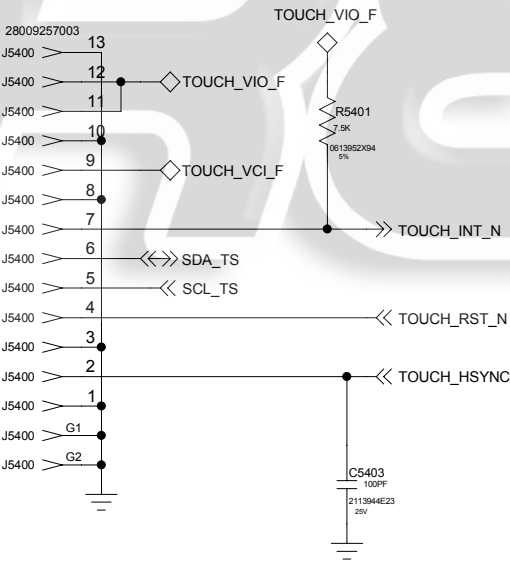
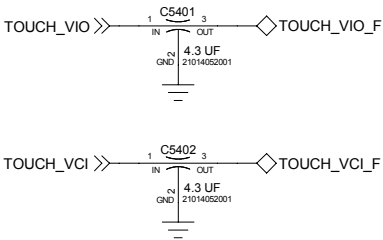
DISP: AOD



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MOTOROLA		BLOCK TITLE: display		
LAST MODIFIED BY: MOTOROLA Thu Jul 30 10:00:37 2015				
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		CAD FILE NAME:		PAGE 28 OF 76

TOUCH: CONNECTOR

REF 5400-5439

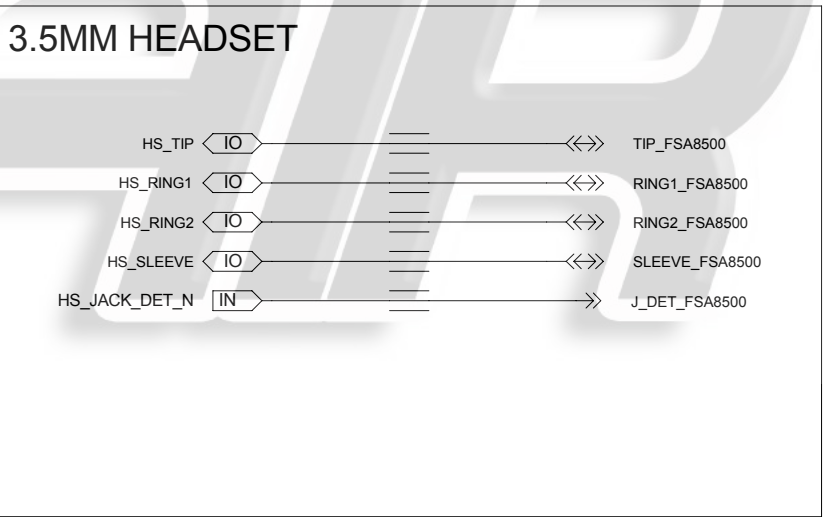
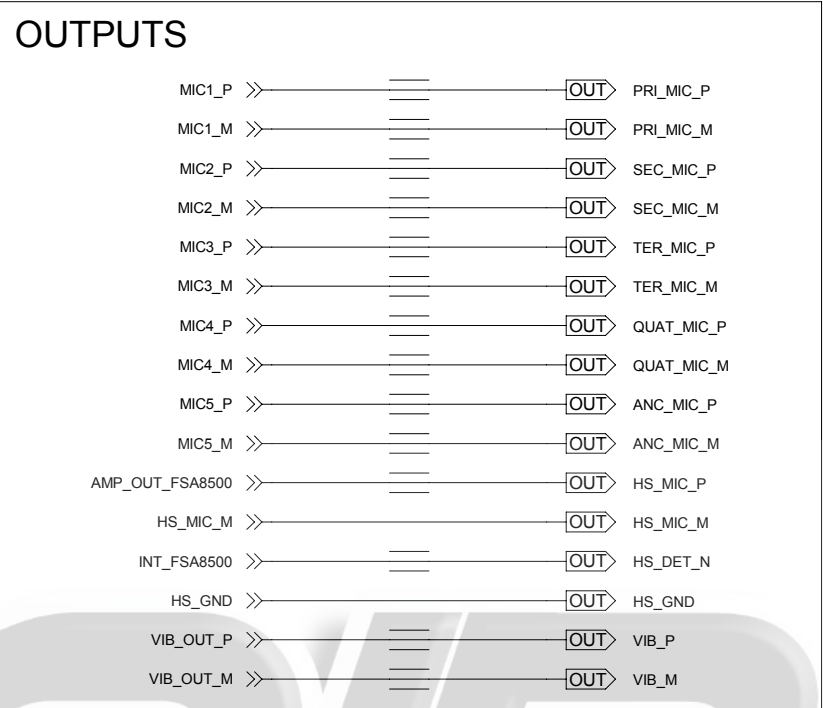
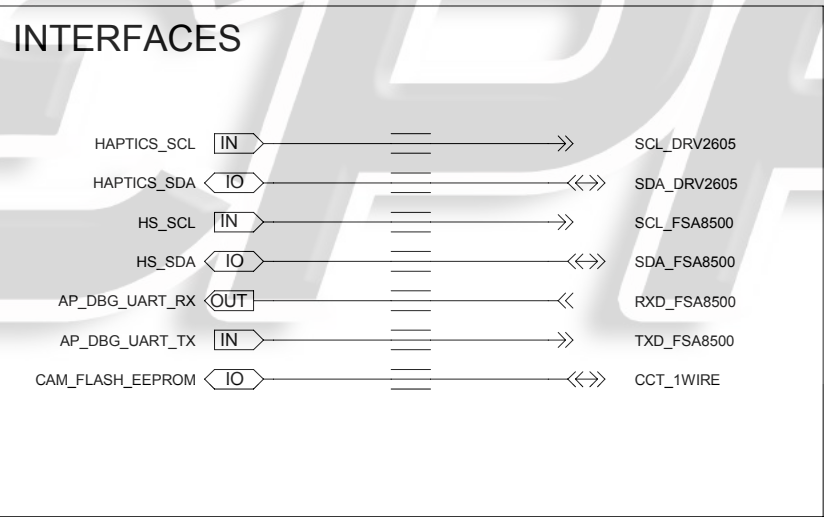
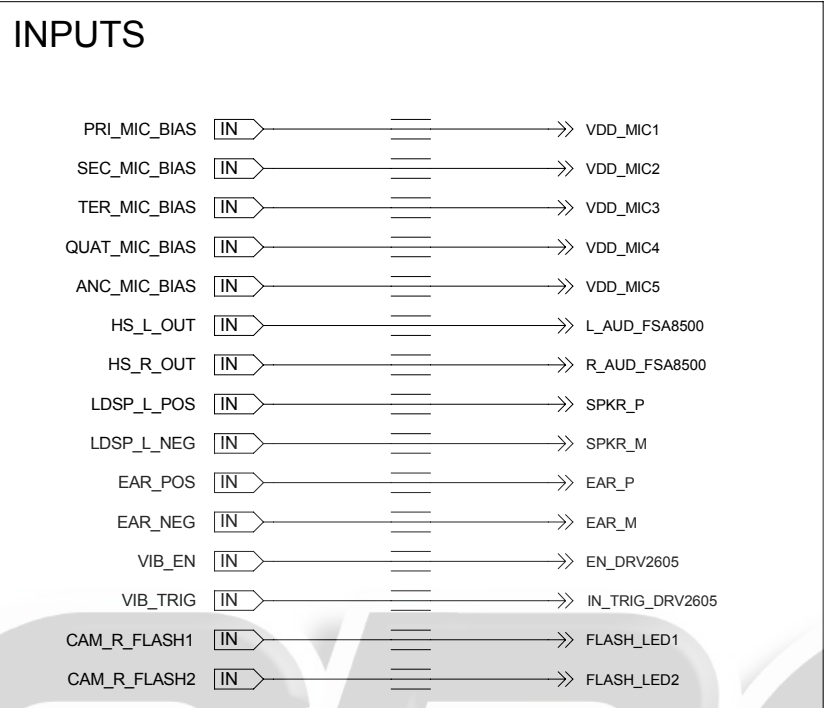
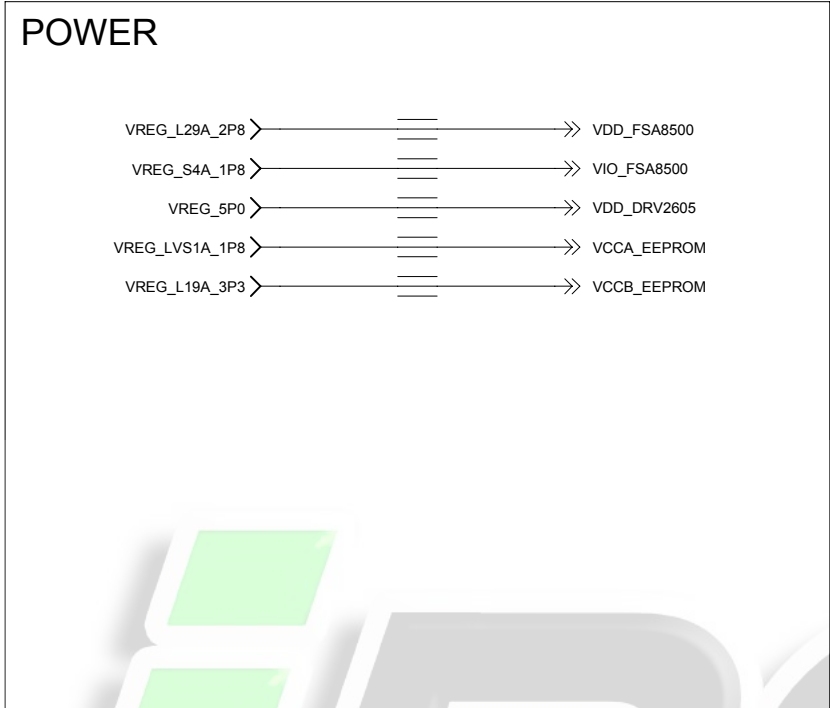


MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE: display	
LAST MODIFIED BY: NATE CONNELL Thu Jul 30 10:00:38 2015			
MOTOROLA CONFIDENTIAL RESTRICTED		B PROJECT NAME 84016327001_a_released.cpm REV 01	
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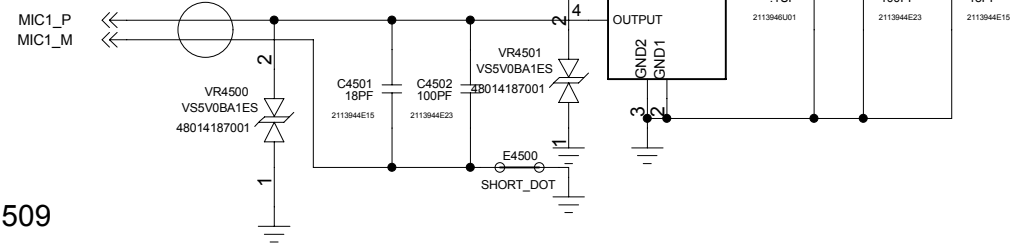
AFE: PORTS

REF 4500-4999



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: NATE CONNELL		BLOCK TITLE: audio_fe		
LAST MODIFIED BY: NATE CONNELL Fri May 15 09:10:31 2015				
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		B 84016327001_a_released.cpm		01
		CAD FILE NAME:		PAGE 30 OF 76

REF: 4500-4599



MIC3_P

VR4520
VSSV0BA1ES
48014187001

C4521
18PF
2113944E15

C4522
100PF
2113944E23

E4520
SHORT_DOT

OUTPUT

WM7121PE

C4524
100PF
2113944E23

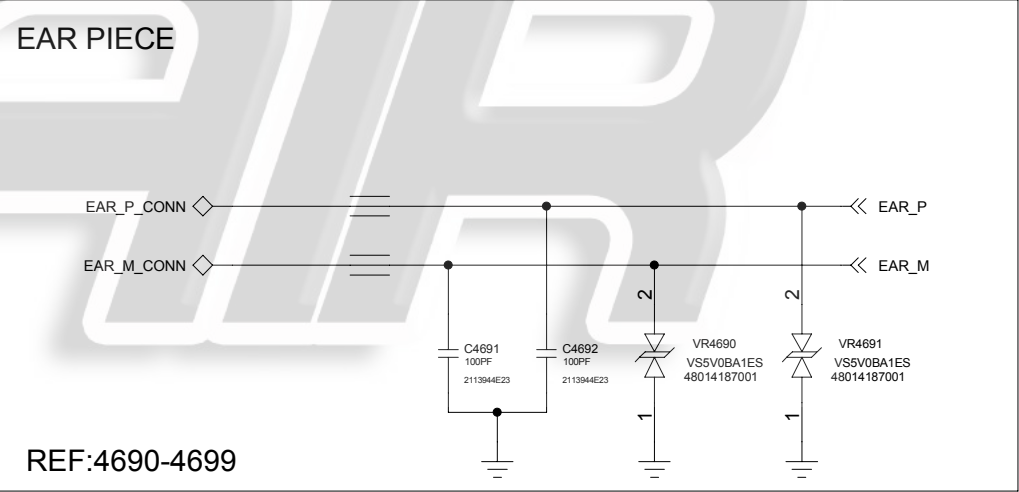
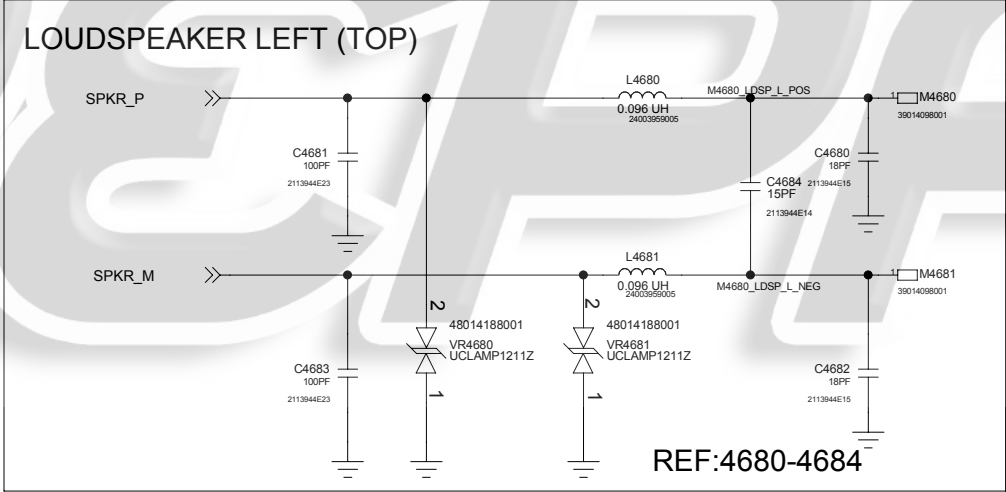
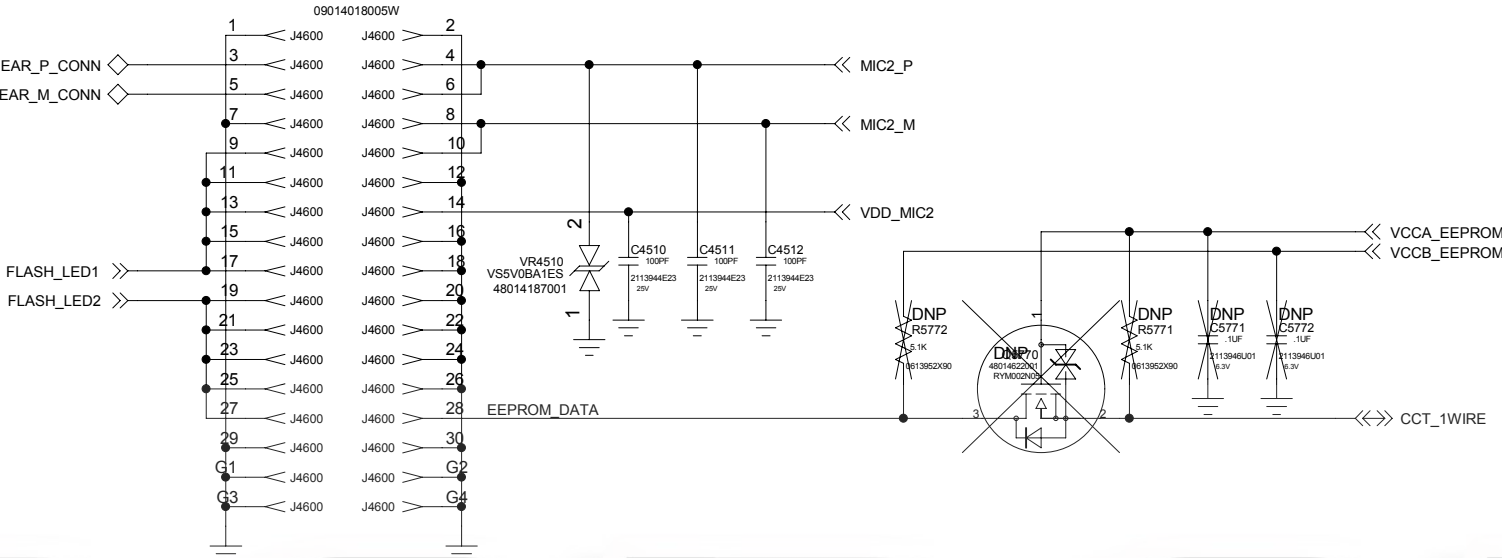
C4523
18PF
2113944E15

9

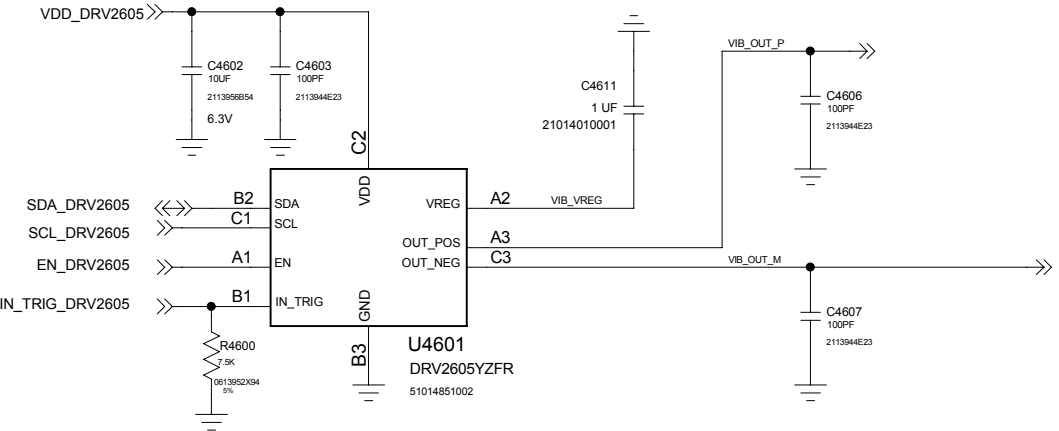
PAGE 31 OF 76

AFE: SPEAKERS

REF: 4600-4699



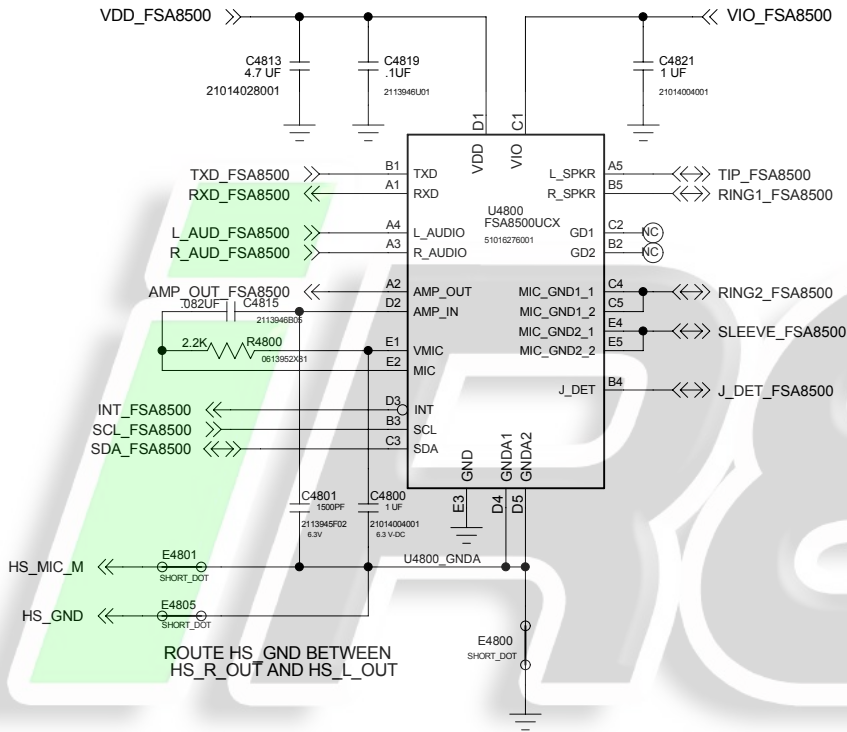
ROTARY VIBRATOR
REF DES 4600-4619



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Jul 30 10:00:41 2015		audio_fe	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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CAD FILE NAME:		PAGE 32 OF 76	

AFE: 3.5MM HEADSET

AUDIO JACK DETECTION
REF:4800-4849

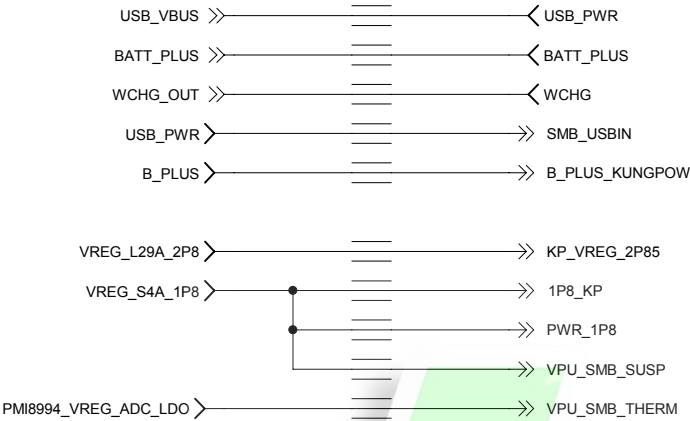


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MATT MOTTIER		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: NATE CONNELL Thu Jul 30 10:00:39 2015			
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		CAD FILE NAME:	PAGE 33 OF 76

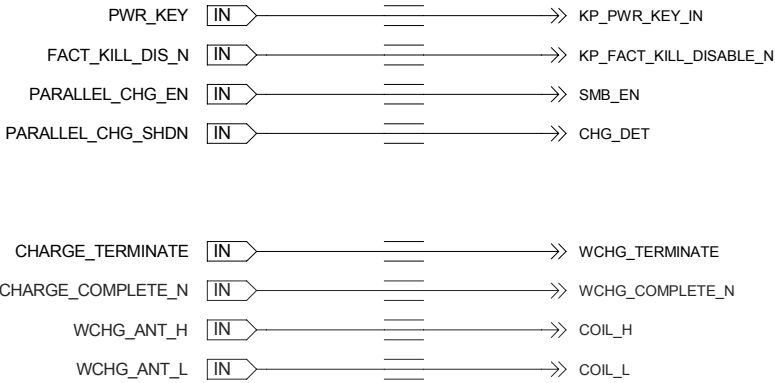
PWR: PORTS

REF 500-799

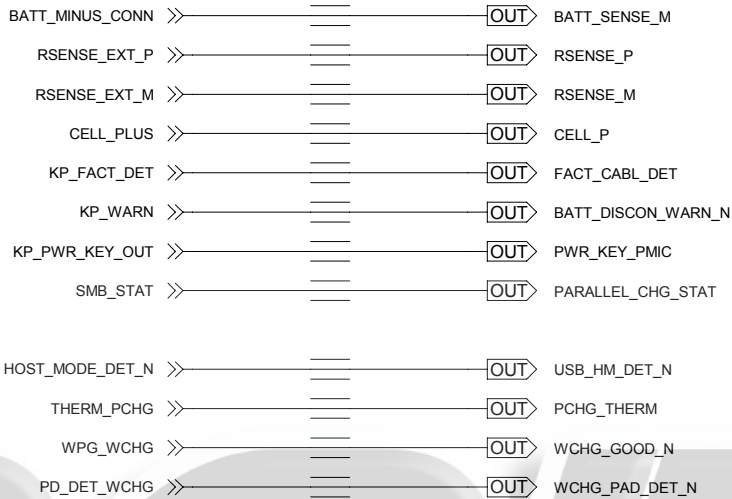
POWER



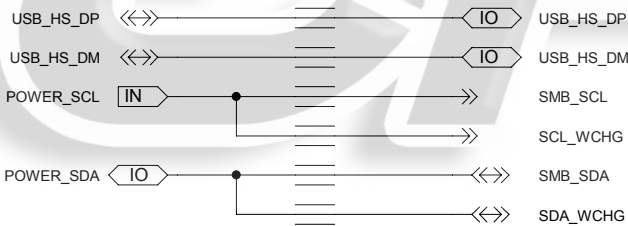
INPUTS



OUTPUTS



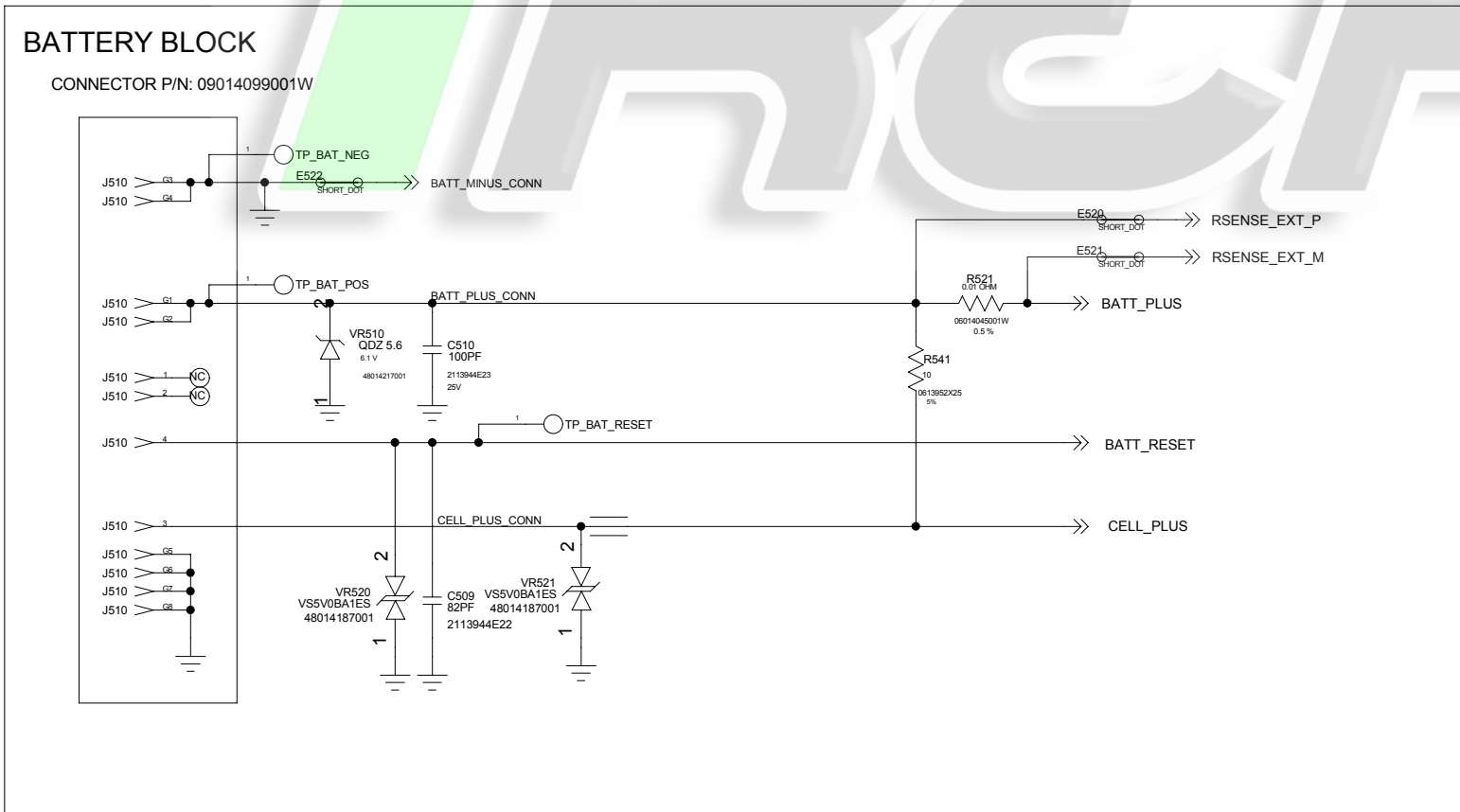
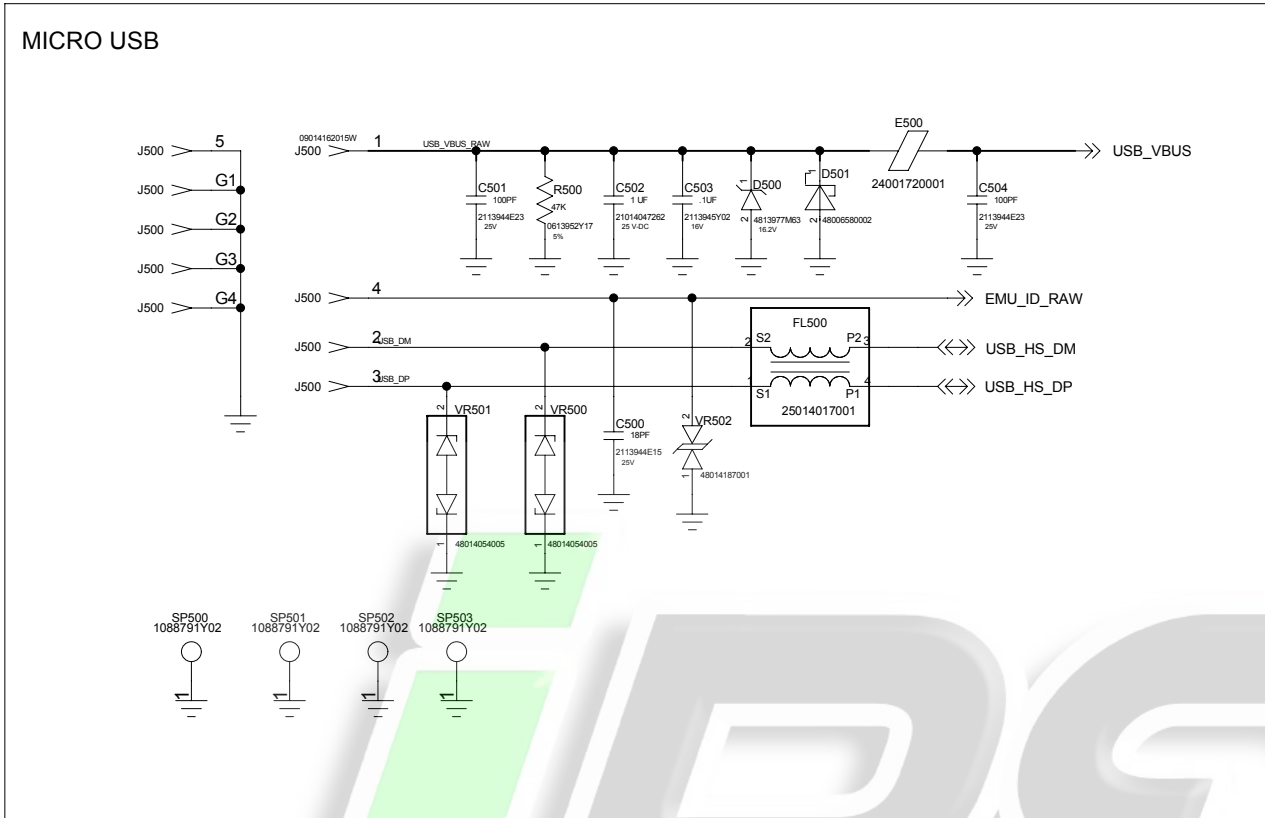
INTERFACES



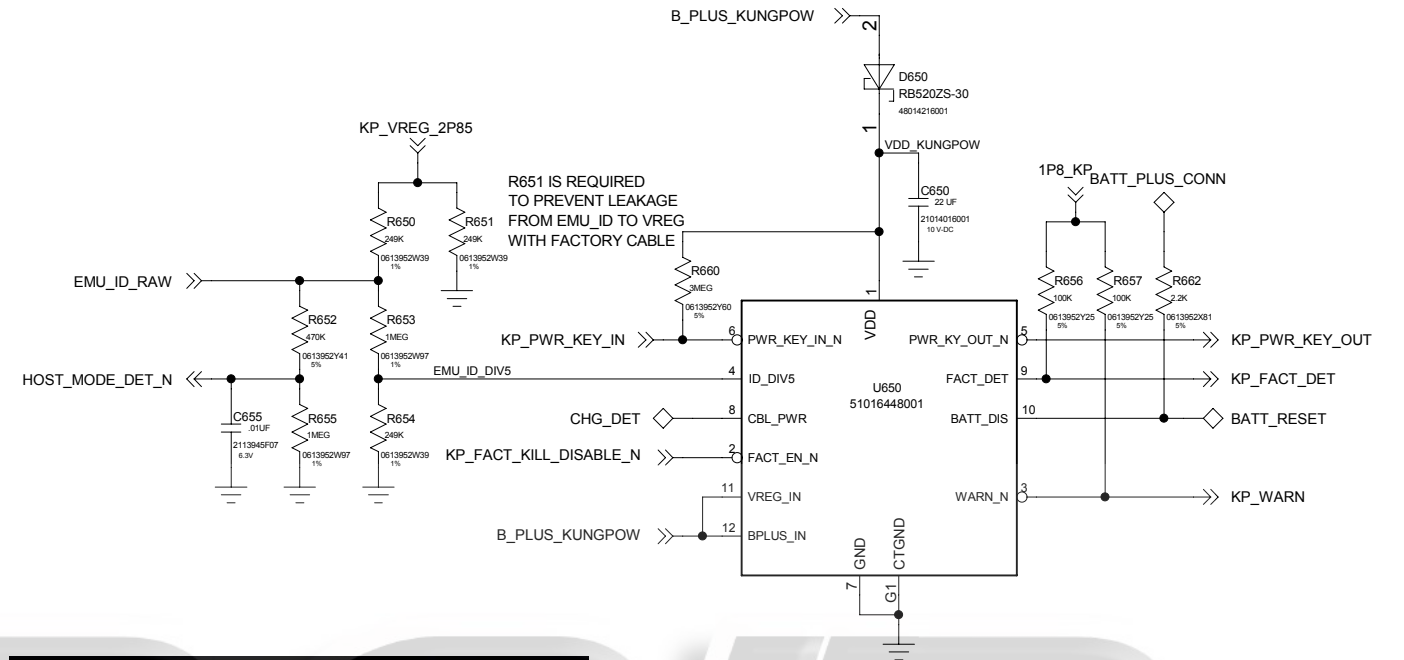
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ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Fri May 15 09:10:25 2015		power	
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		84016327001_a_released.cpm	
		REV 01	
		PAGE 34 OF 76	

PWR: USB2 / BATT

REF: 500-530



KUNG KOW BATTERY DISCONNECT

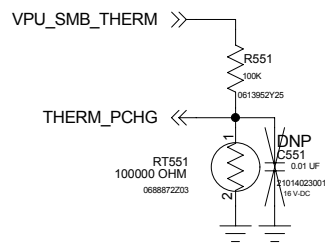
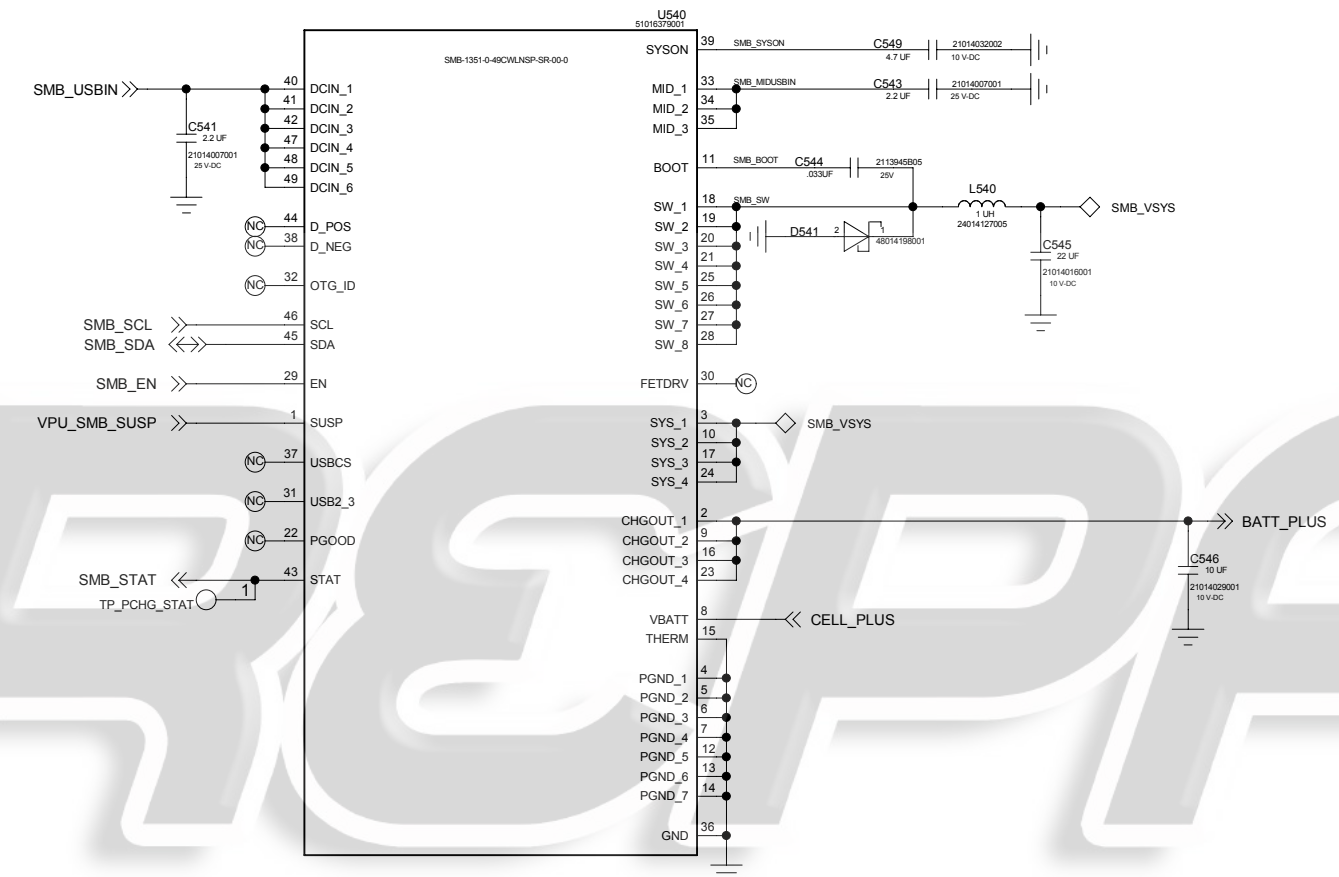


MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: RICH NG		BLOCK TITLE: power	
LAST MODIFIED BY: UDI ELRAN Thu Jul 30 10:00:42 2015			
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		CAD FILE NAME:	PAGE 35 OF 76

PWR: EXT CHG

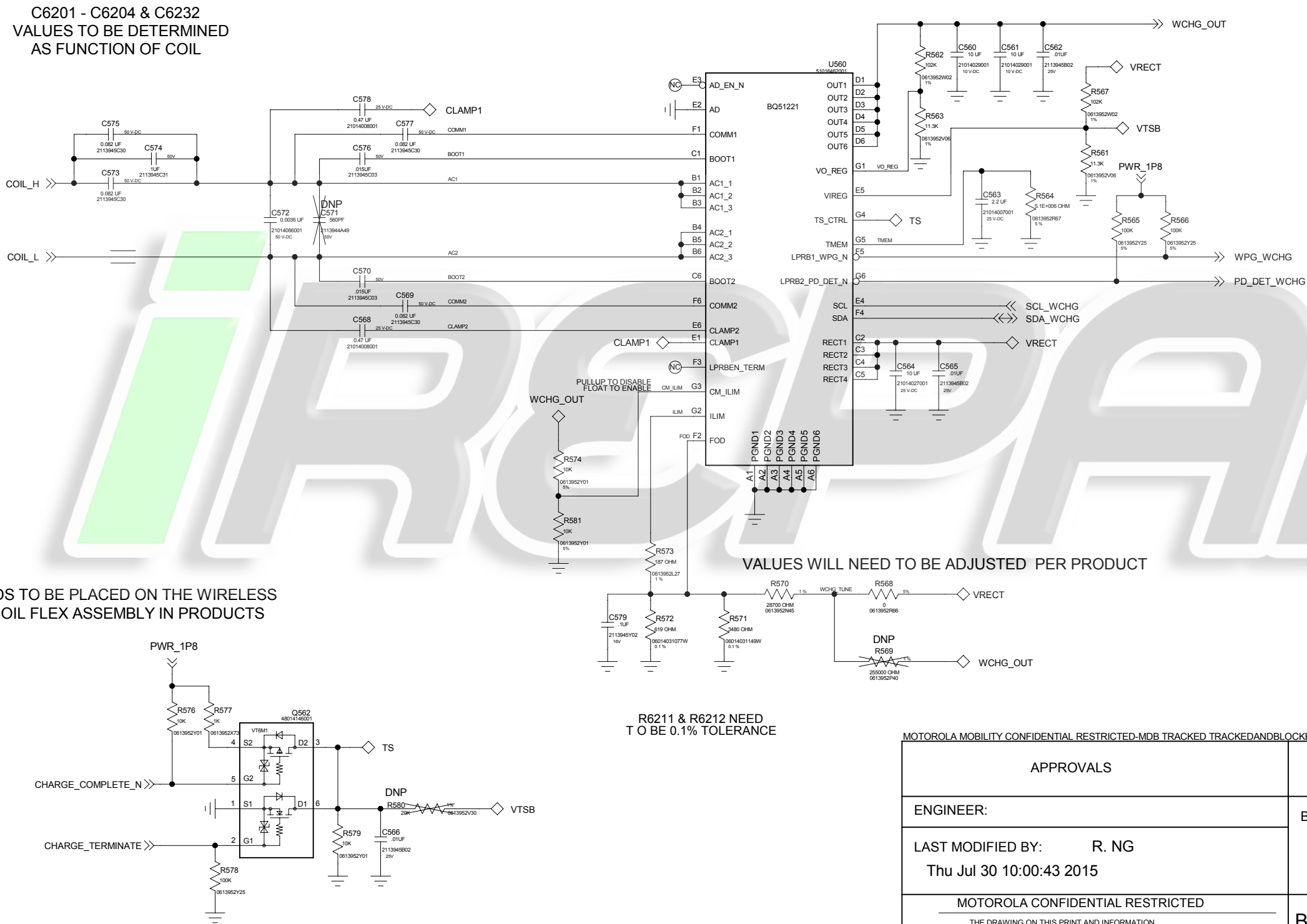
REF: 540-559



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: R. NG Thu Jul 30 10:00:43 2015		power	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		REV 01	
		PAGE 36 OF 76	

PWR: WCHG

REF: 560-599

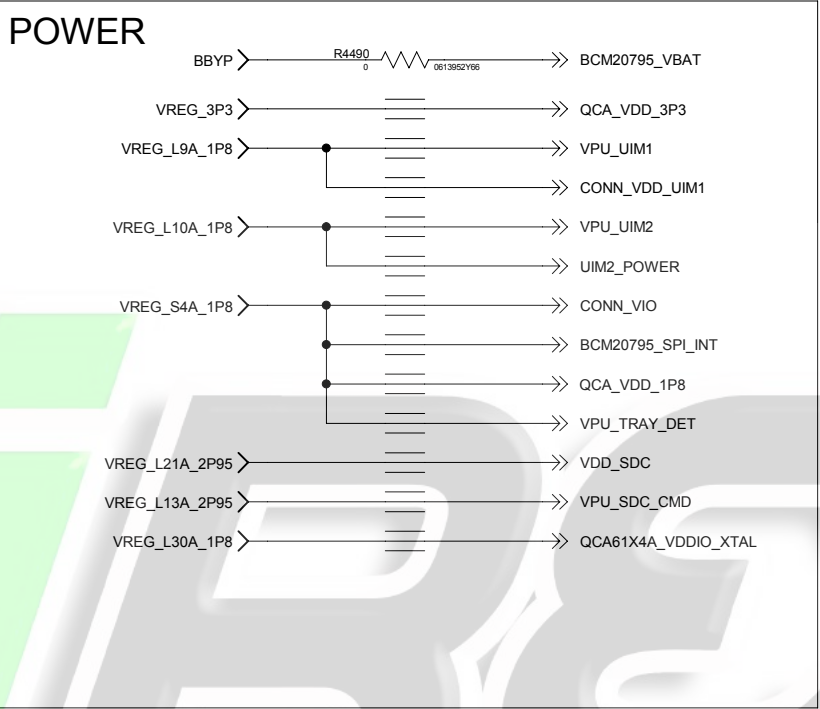


MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: R. NG Thu Jul 30 10:00:43 2015		power	
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		84016327001_a_released.cpm	
		REV 01	
		PAGE 37 OF 76	

CONN: POWER

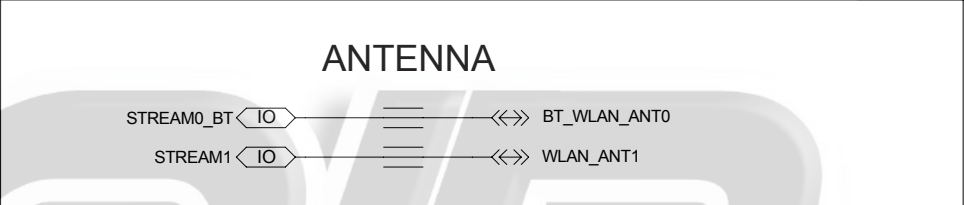
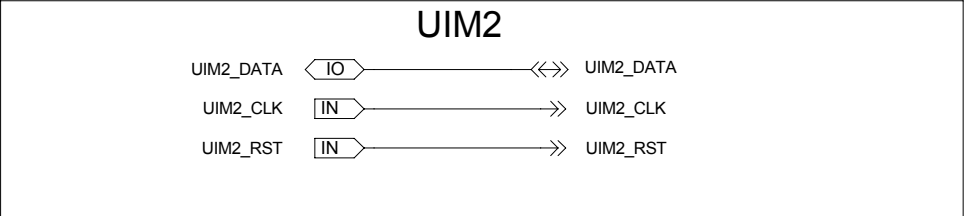
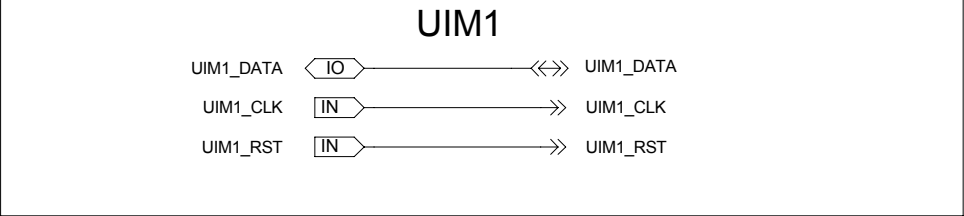
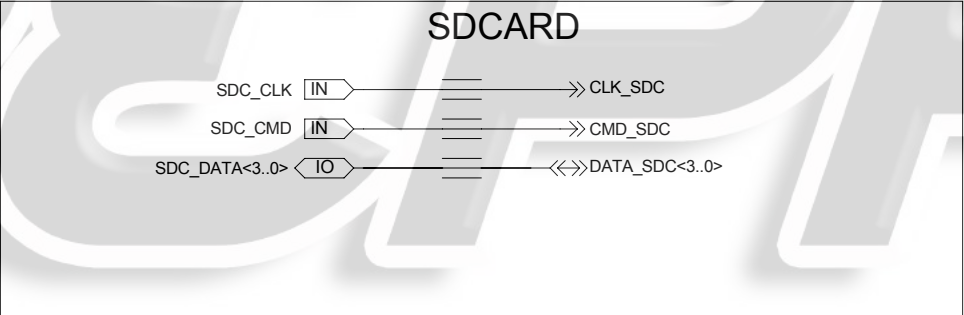
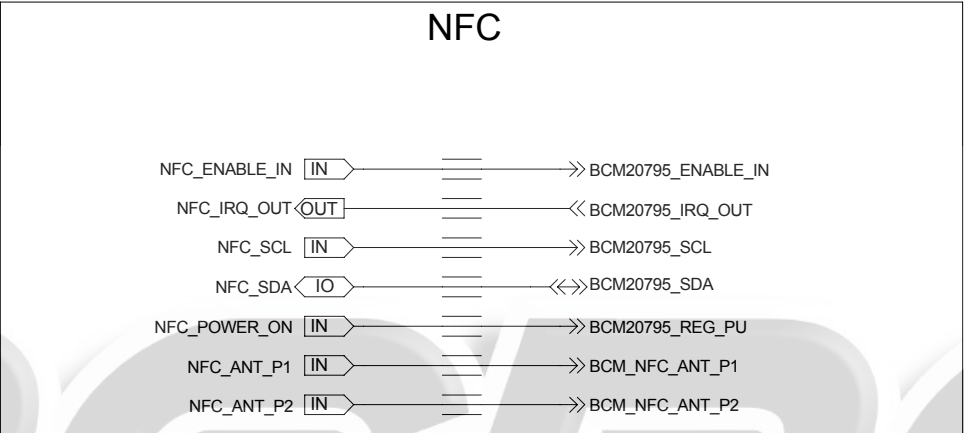
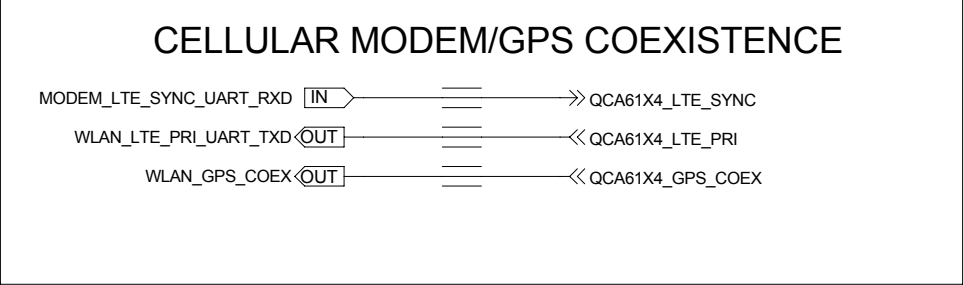
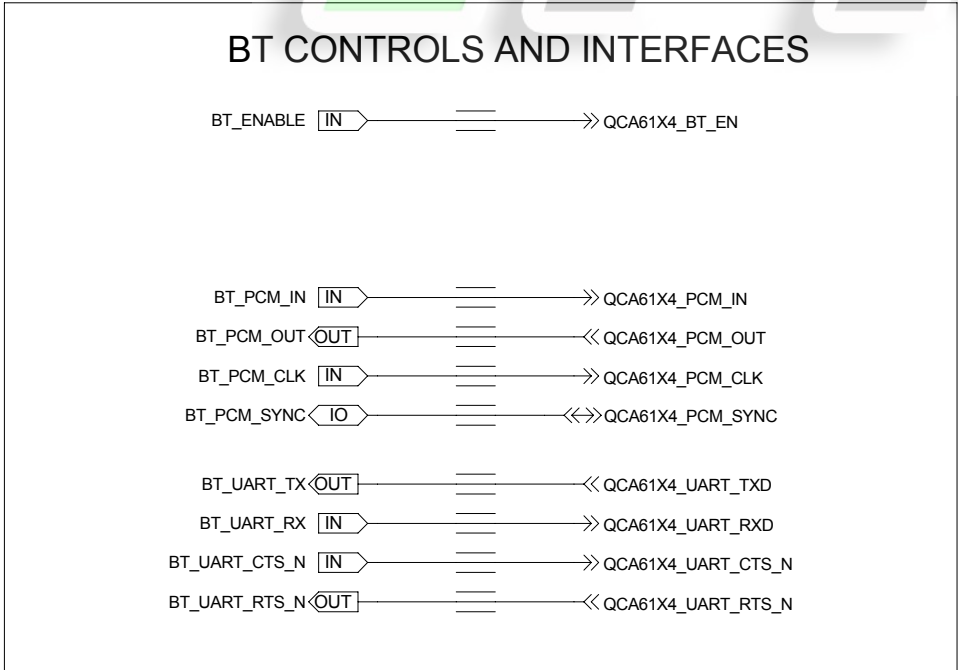
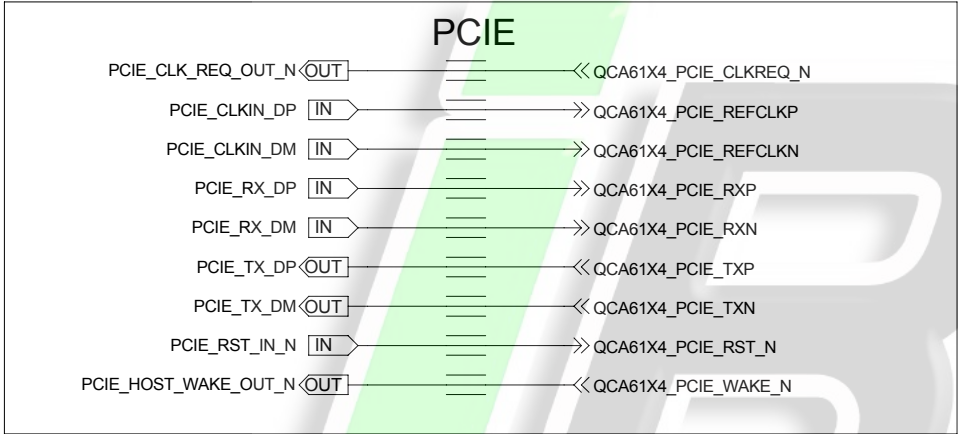
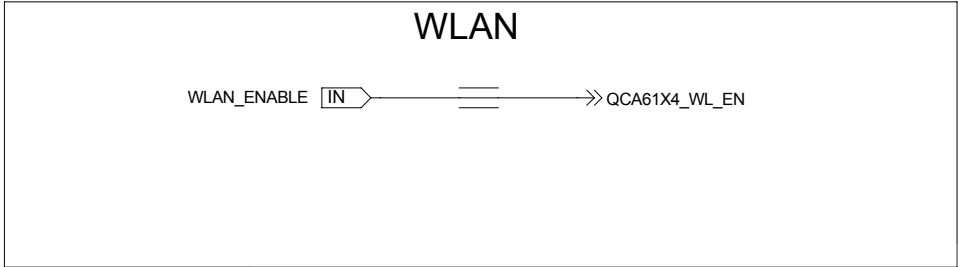
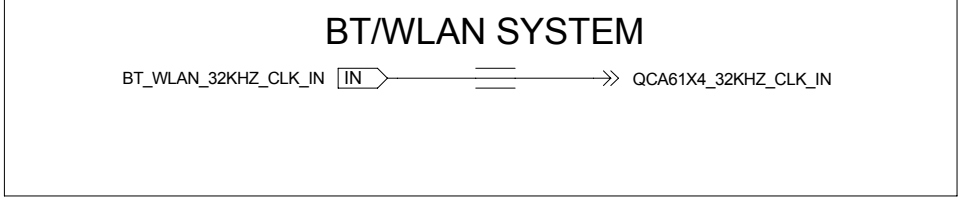
REF 4490-4499



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE: connectivity	
LAST MODIFIED BY: FROEHLING Thu Jul 30 10:00:48 2015			
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		CAD FILE NAME:	PAGE 38 OF 76

CONN: PORTS

REF 4000-4499



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: FROEHLING		BLOCK TITLE: connectivity		
LAST MODIFIED BY: FROEHLING Tue Mar 24 11:21:51 2015				
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		B 84016327001_a_released.cpm		01
		CAD FILE NAME:		PAGE 39 OF 76

□



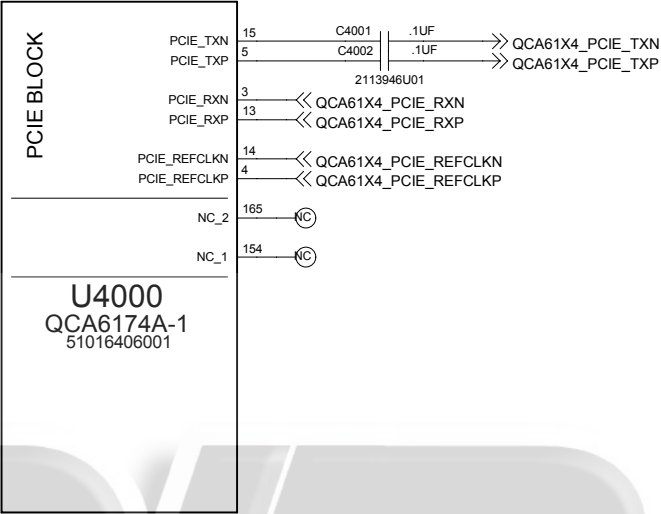
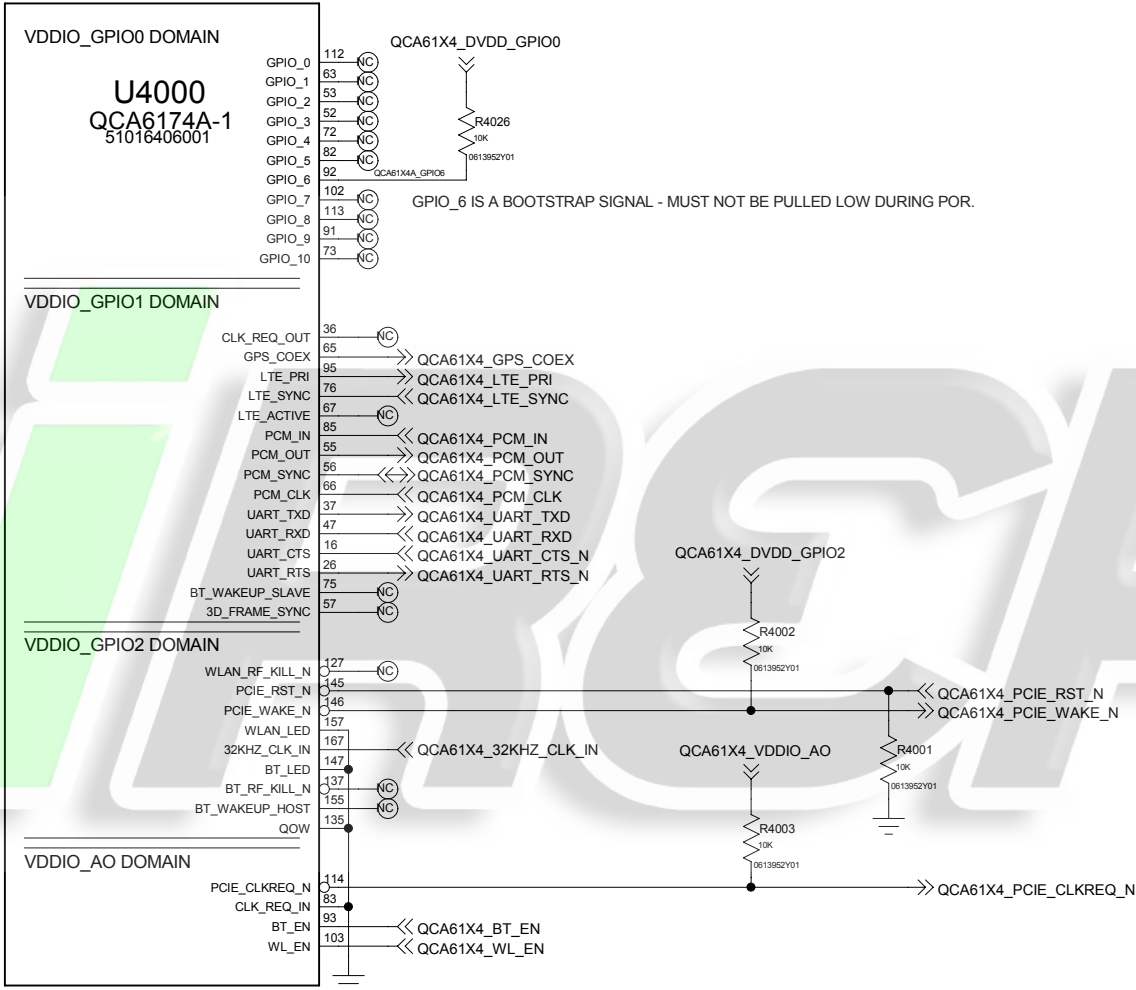
A



A

CONN: QCA61X4A HOST I/F

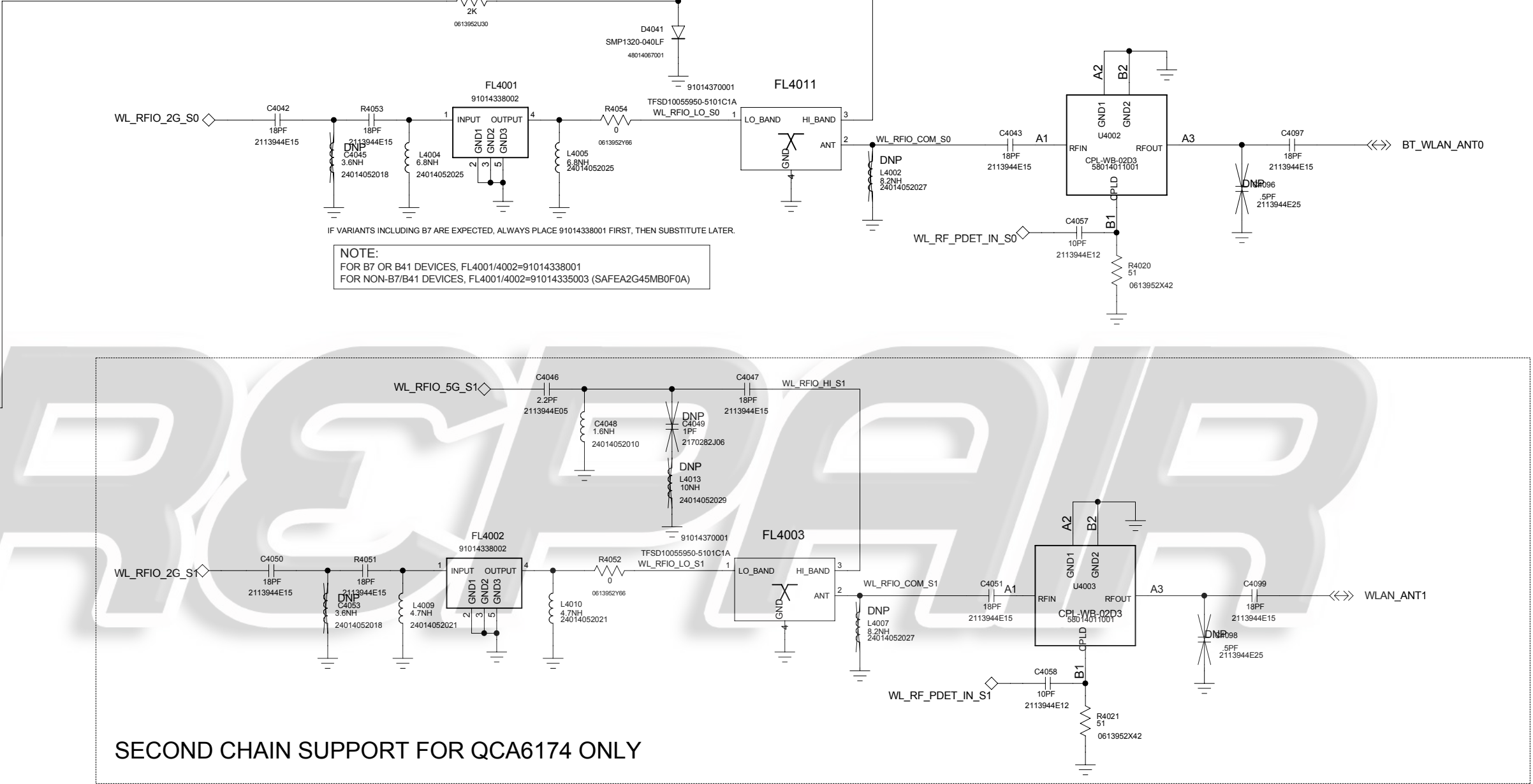
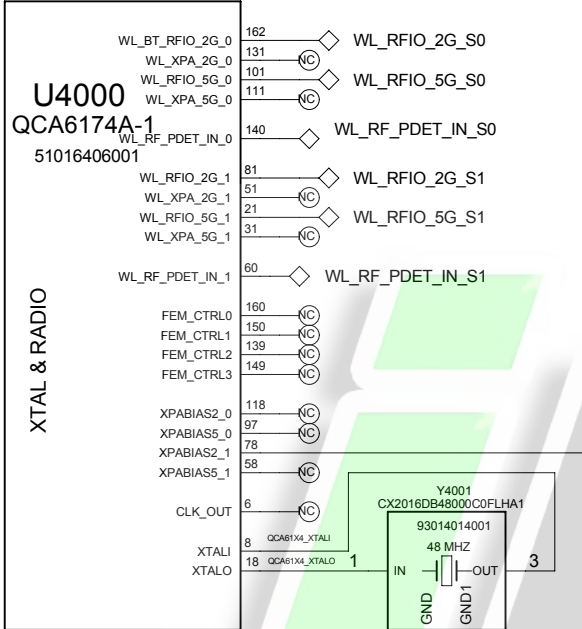
REF: 4000-4099



BASED ON 80-WL420-42 REV. D

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED		MOTOROLA MOBILE DEVICES	
APPROVALS		BLOCK TITLE:	
ENGINEER: FROEHLING		connectivity	
LAST MODIFIED BY: FROEHLING Thu Jul 30 10:00:45 2015		PROJECT NAME	
MOTOROLA CONFIDENTIAL RESTRICTED		REV	
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CONN: QCA61X4A RADIO/XTAL
REF 4000-4099



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Thu Jul 30 10:00:46 2015		connectivity	
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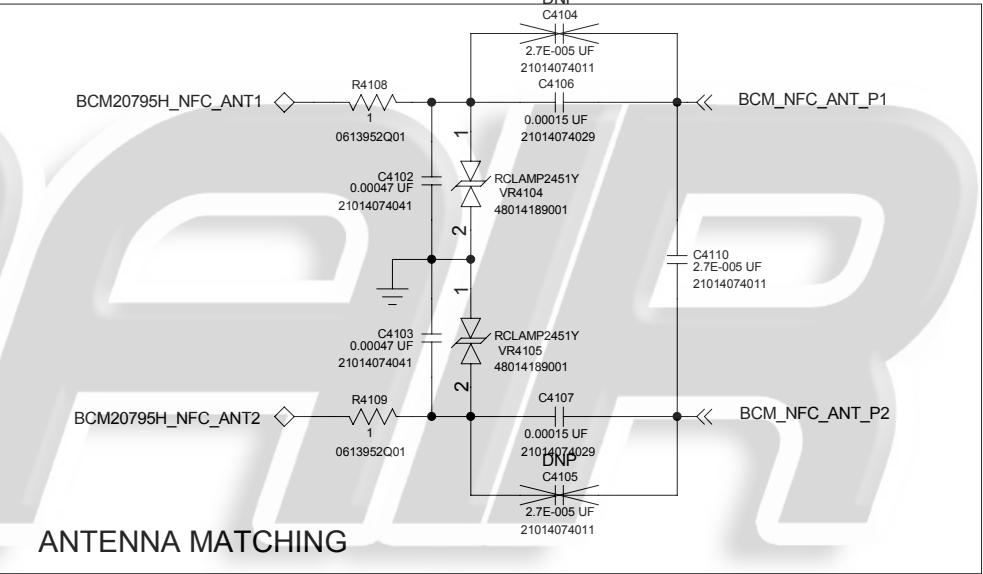
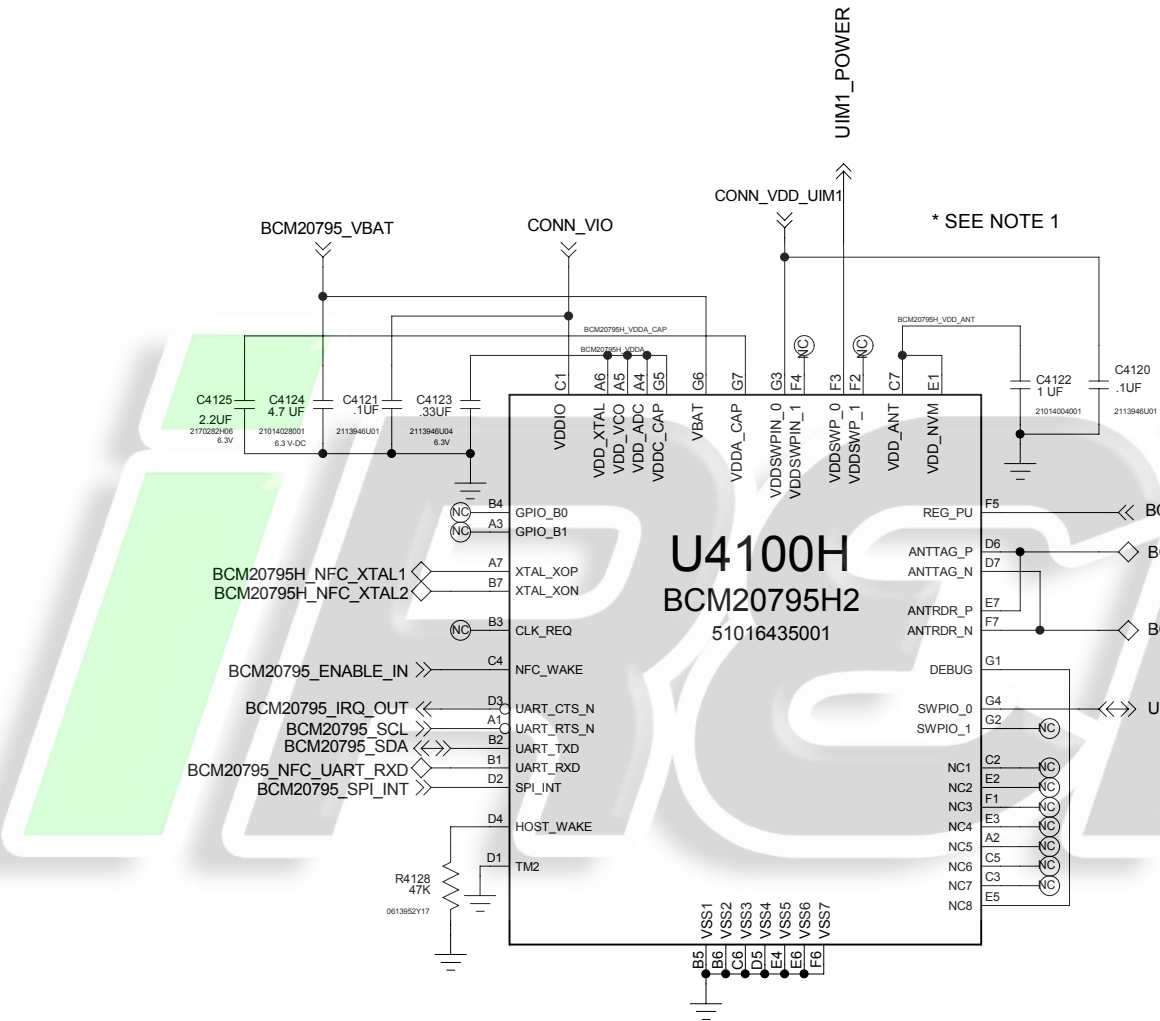
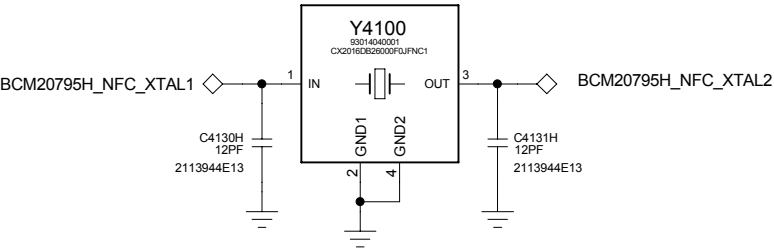
BASED ON 80-WL420-42 REV. D

CONN: BCM20795H NFC

REF: 4100-4199

I2C ADDR: 0X77 (BOOT DEFAULT WHEN PIN B1 IS PULLED TO GROUND)
OR
I2C ADDR: 0X76 (BOOT DEFAULT WHEN PIN B1 IS FLOATED/PULLED HIGH)

* SEE NOTE 3



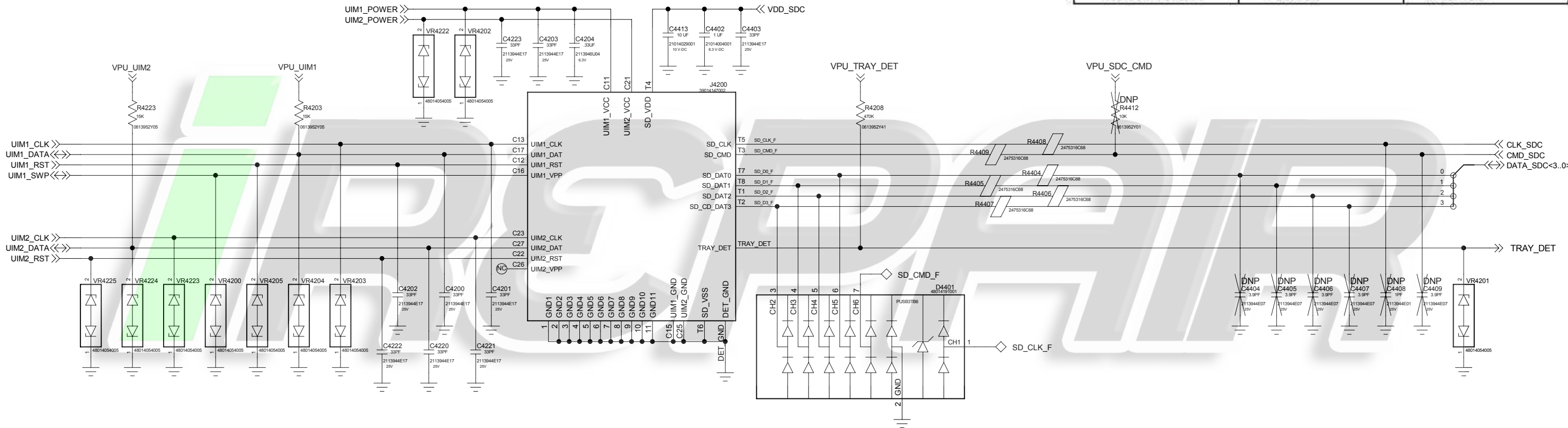
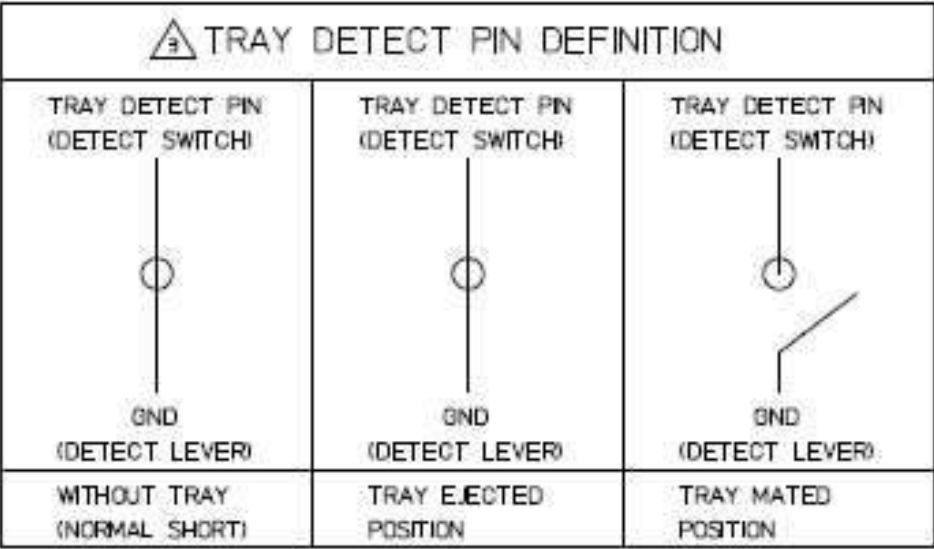
- NOTES:
- PER BROADCOM RECOMMENDATION, CAPACITANCE ON UIM1_POWER NET SHOULD BE 0.33UF. TOO LITTLE COMPROMISES REGULATOR STABILITY, TOO MUCH CAN CAUSE INADVERTENT POWER-ON RESETS AT VDD.
 - PLACEMENT FOR 20795G VS H:

REF	20795G (ESE)	20795H (EEPROM)
R4126H	DNI	PLACE
C4128H	PLACE	DNI
 - FOR SMALL NFC ANTENNAS, EXTERNAL CRYSTAL OSCILLATOR CIRCUIT MUST BE USED - SEE EE8994CORE-296 FOR DETAILS. IF USE OF EXTERNAL CLOCK IS DESIRED, CHECK WITH BROADCOM BEFORE COMMITTING.

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Thu Jul 30 10:00:47 2015		connectivity	
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		CAD FILE NAME:	
		PAGE 43 OF 76	

CONN: UIM1/UIM2/SDCARD

REF:4200-4219



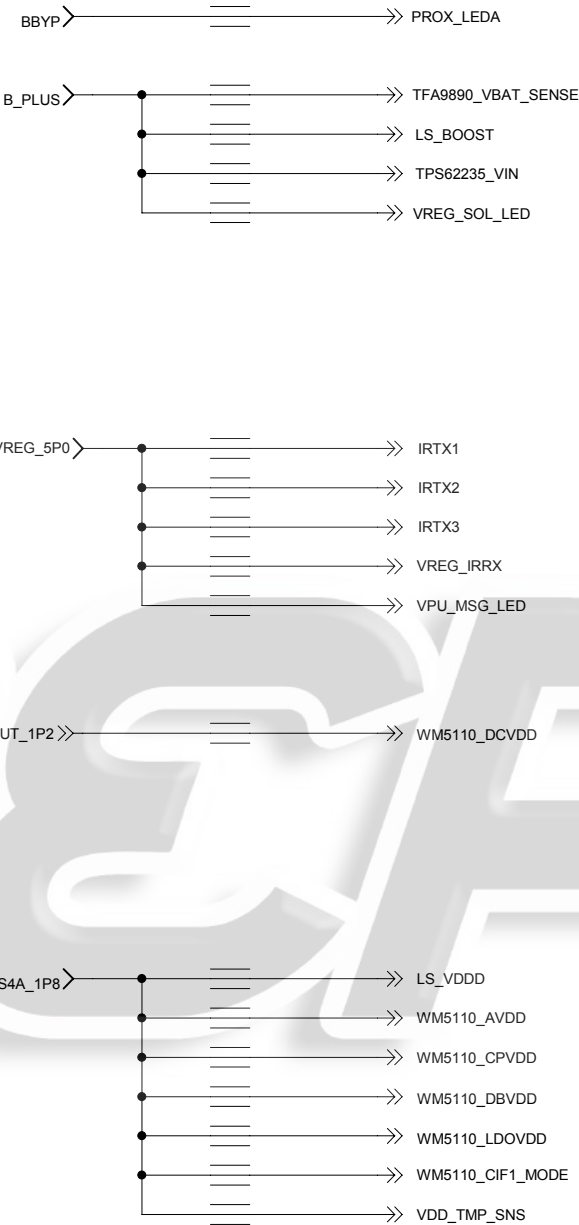
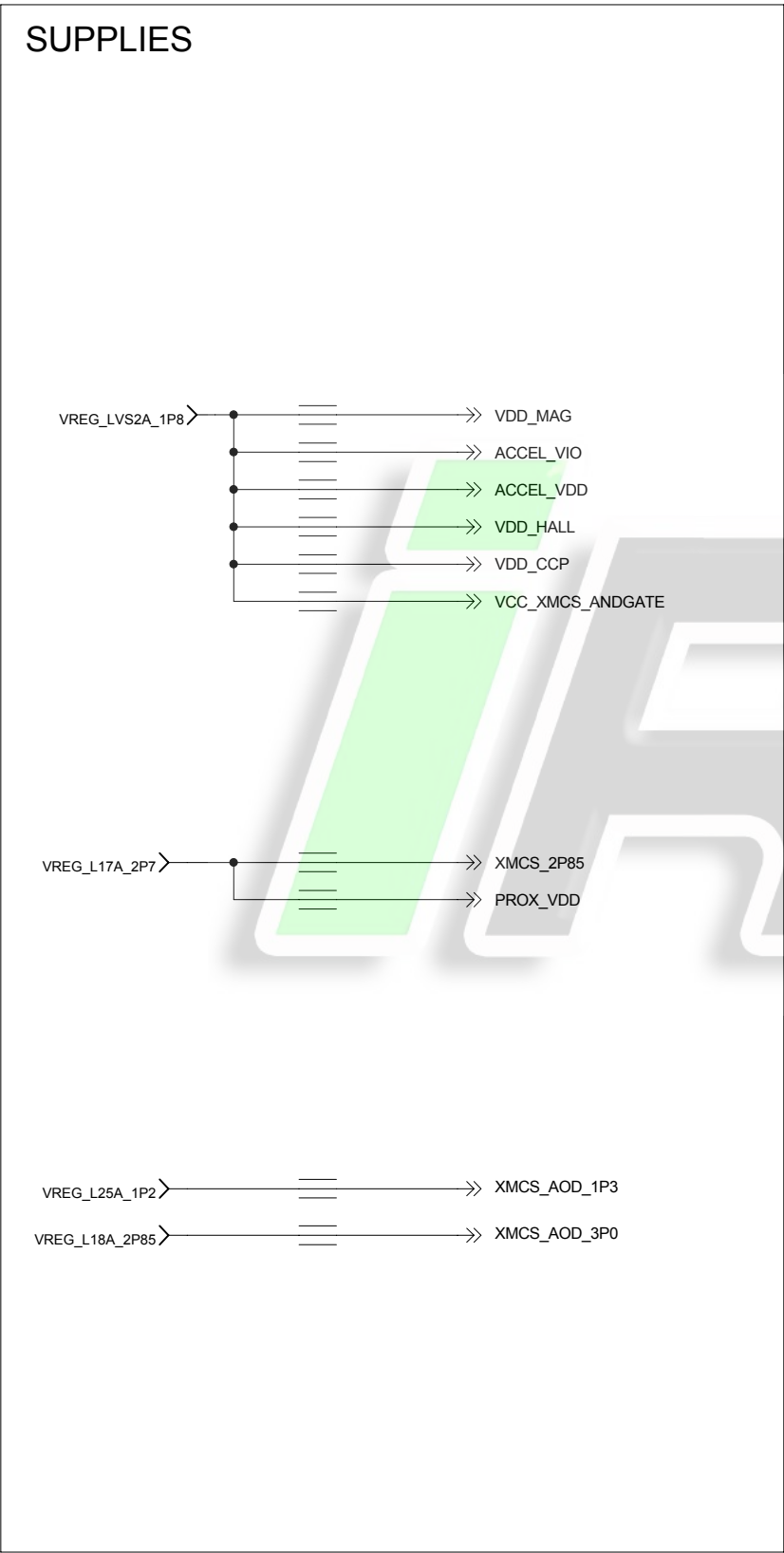
NOTE 1: USE 24.9K PULL-UP TO SUPPORT HOT SWAP WITH FREEZE-IO FEATURE.

NOTE 2: UIM_PWR BYPASS OF 0.33UF RECOMMENDED BY NFC IC SUPPLIER, PLACE NEAR CONNECTOR.

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Thu Jul 30 10:00:47 2015		connectivity	
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		CAD FILE NAME:	PAGE 44 OF 76

XMCS: POWER

REF 6000-6999

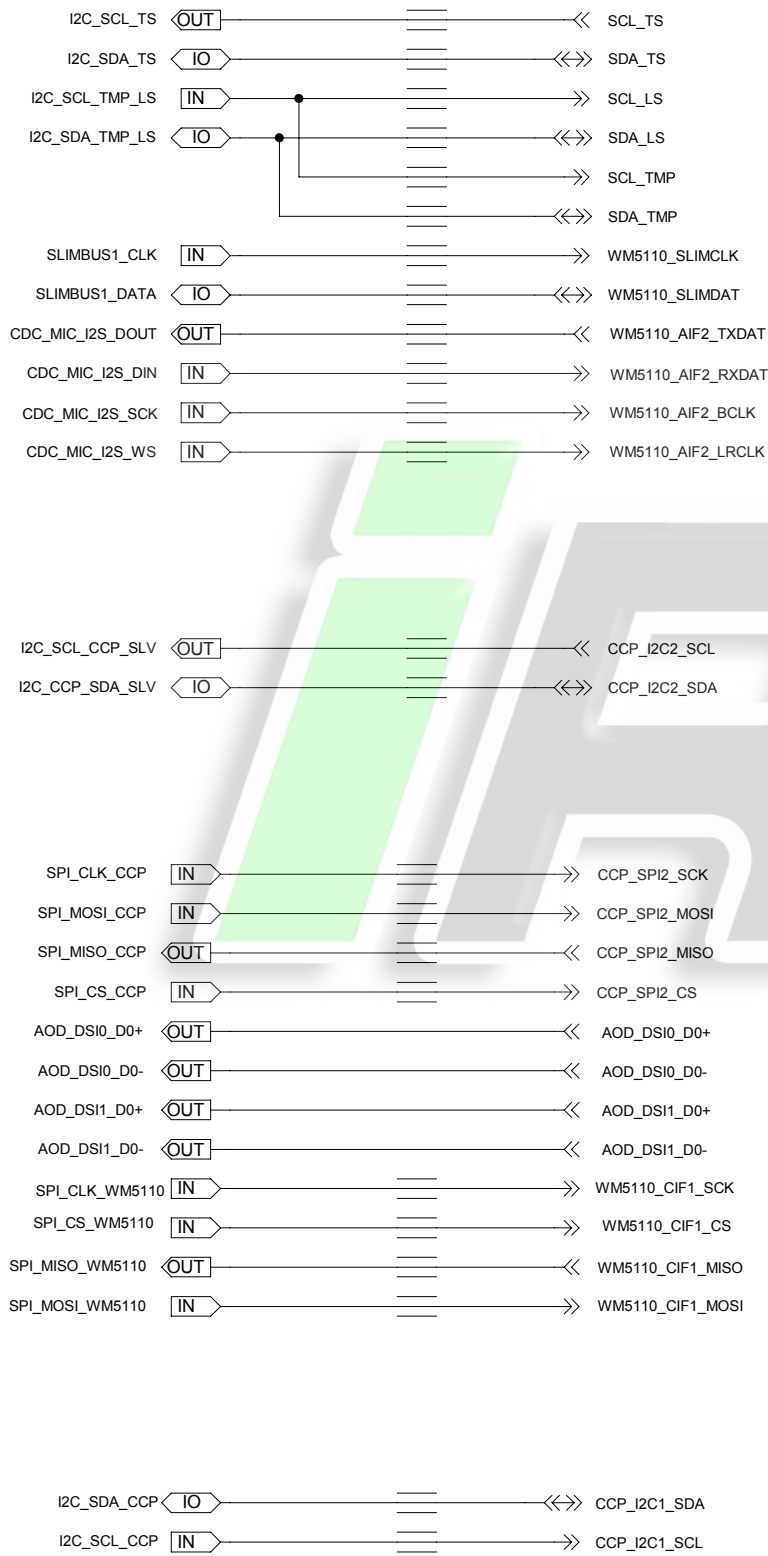


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ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Fri May 15 09:10:07 2015		xmcs	
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		84016327001_a_released.cpm	
		REV 01	
		PAGE 45 OF 76	

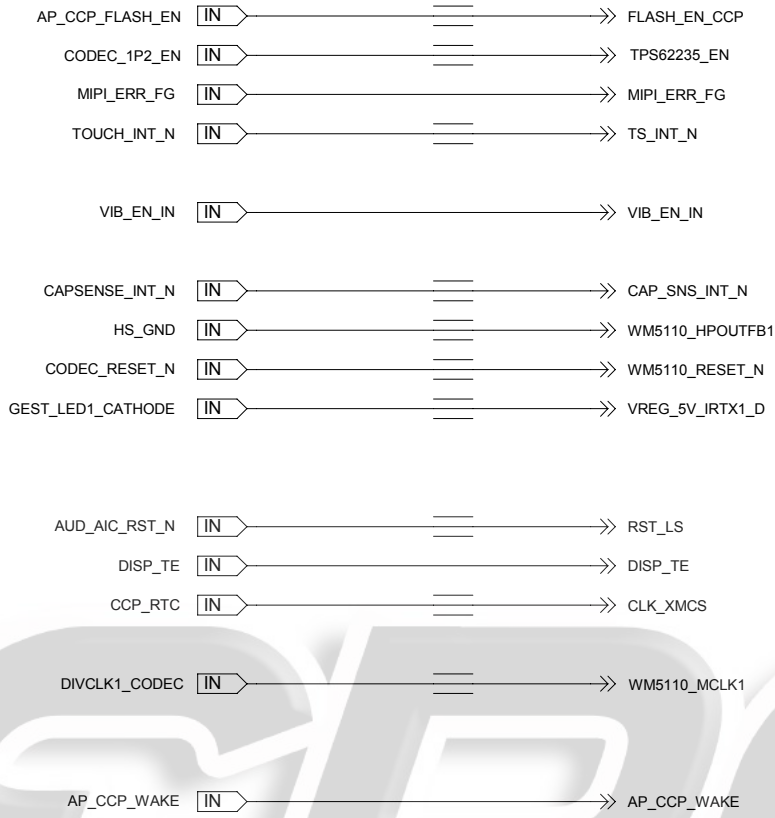
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REF 6000-6999

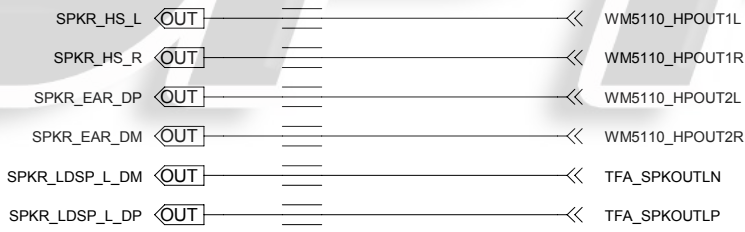
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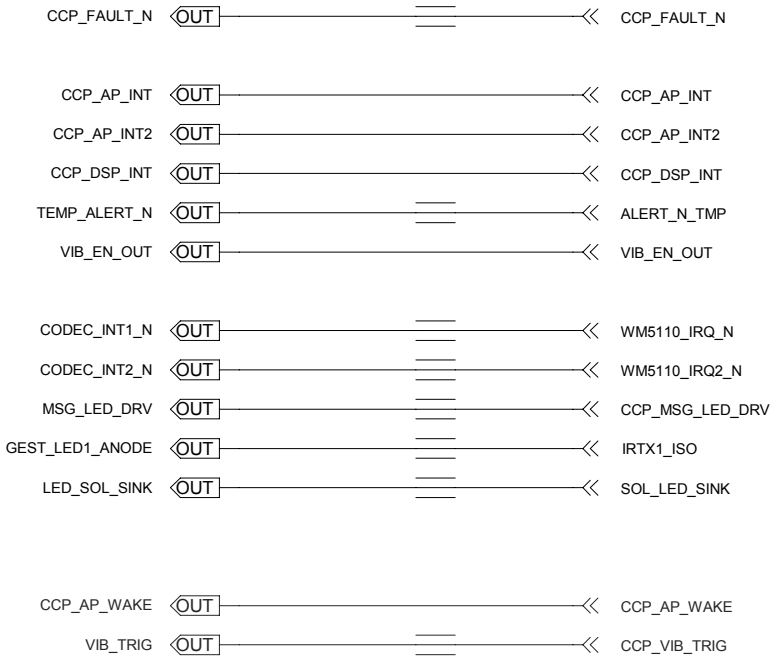
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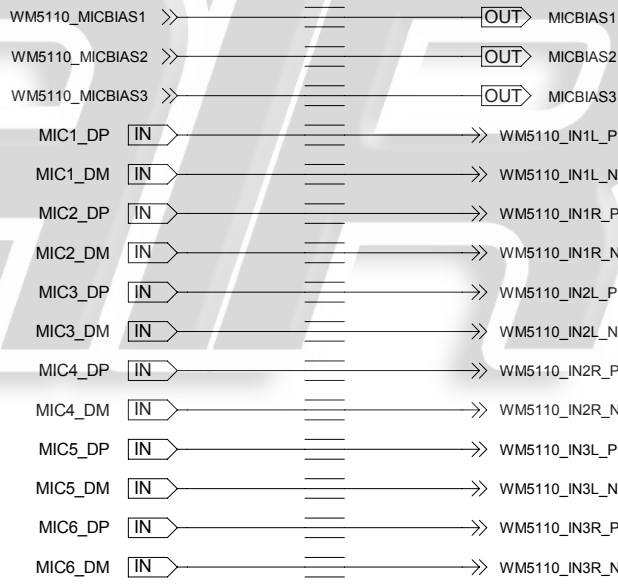
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OUTPUTS



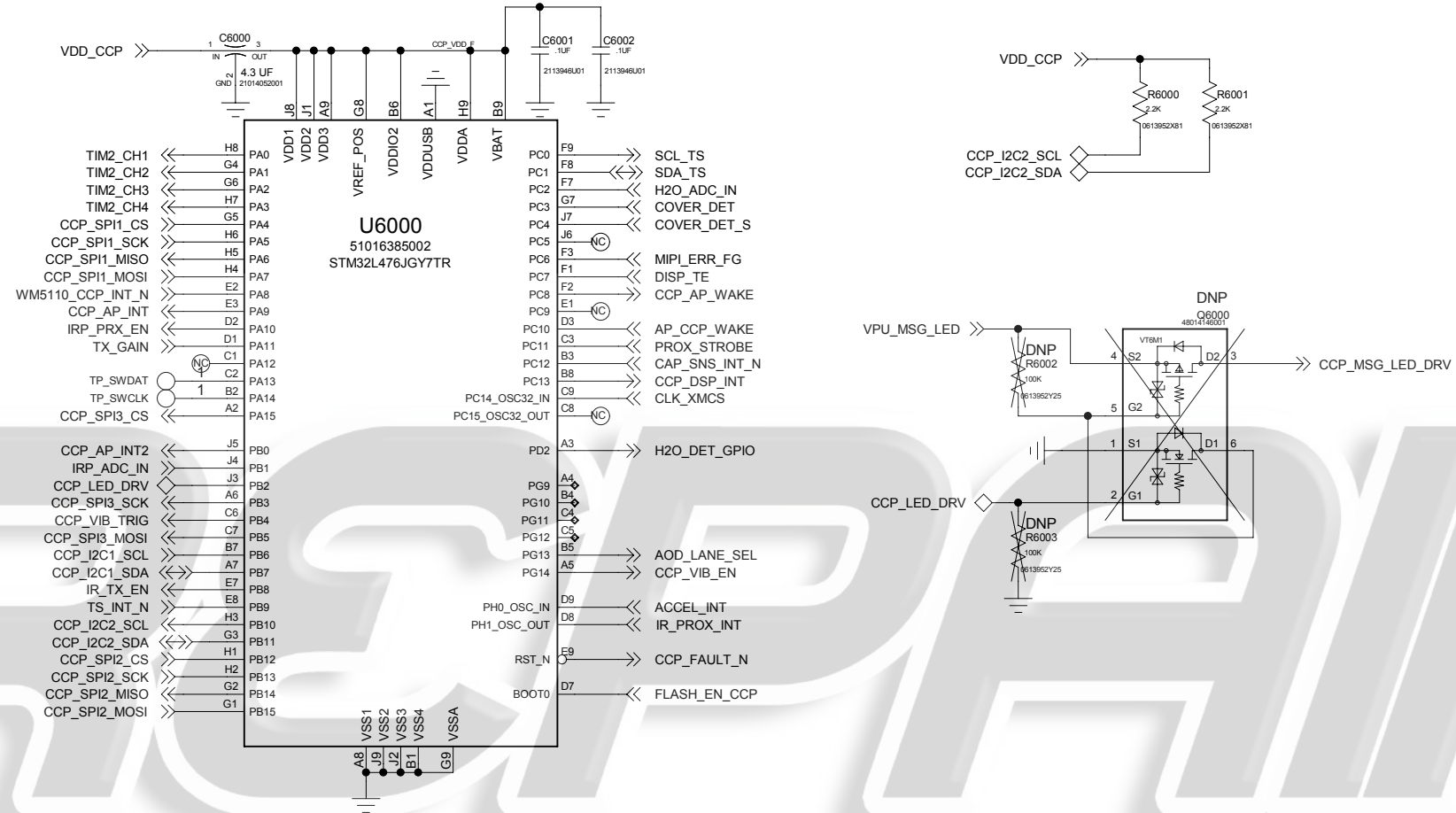
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APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
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		CAD FILE NAME:	
		PAGE 46 OF 76	

XMCS: CCP

REF: 6000-6049



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: JOHN GORSICA	BLOCK TITLE: xmcs		
LAST MODIFIED BY: NATE CONNELL Thu Jul 30 10:00:49 2015			
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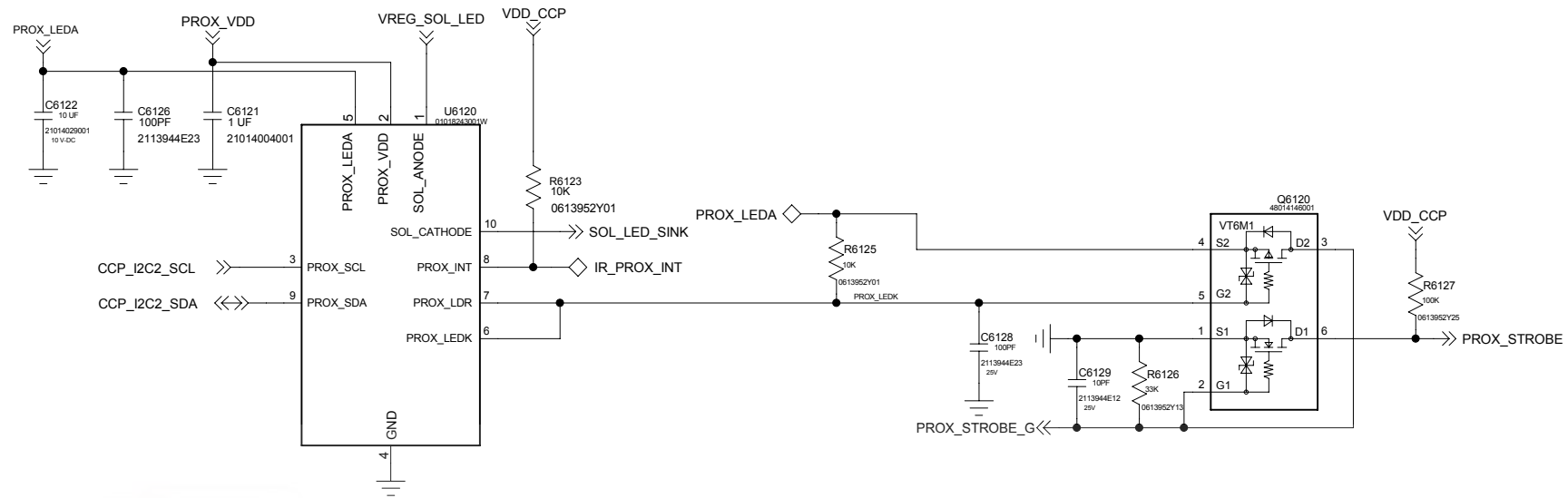
XMCS: SENSORS

REF: 6100-6199

PROX SENSOR/ALS

REF:6120-6129

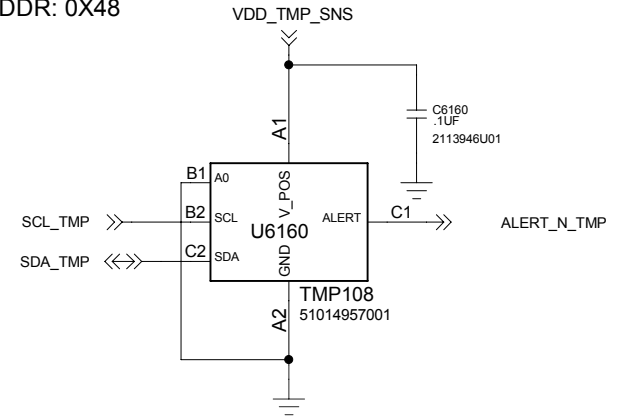
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BOARD TEMP SENSORS

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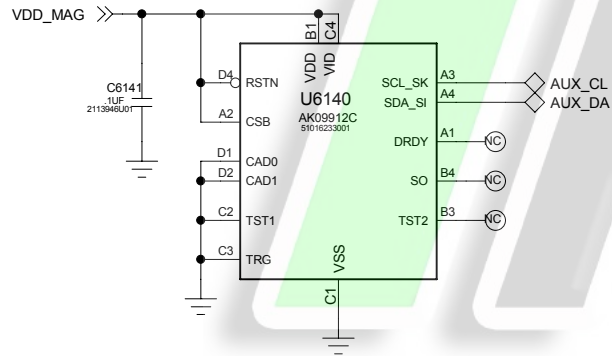
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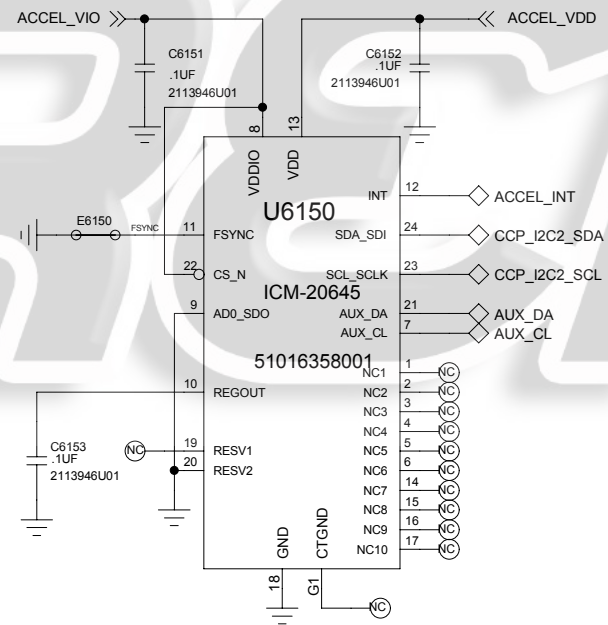
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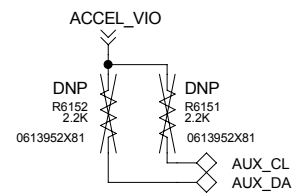
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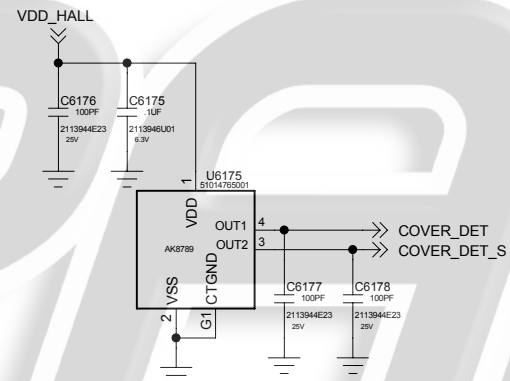


PRODUCTS MUST FOLLOW BUNKER HILL PCB ORIENTATION



HALL EFFECT

REF:6170-6179



2ND HALL EFFECT FOR SMART COVER SUPPORT

JOHN GORSICA/NATE CONNELL
APPROVALS

ENGINEER: C HERNANDEZ

LAST MODIFIED BY: C HERNANDEZ
Thu Jul 30 10:00:50 2015

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BLOCK TITLE:

xmcs

B	84016327001A <u>B</u> B <u>A</u> 680C100
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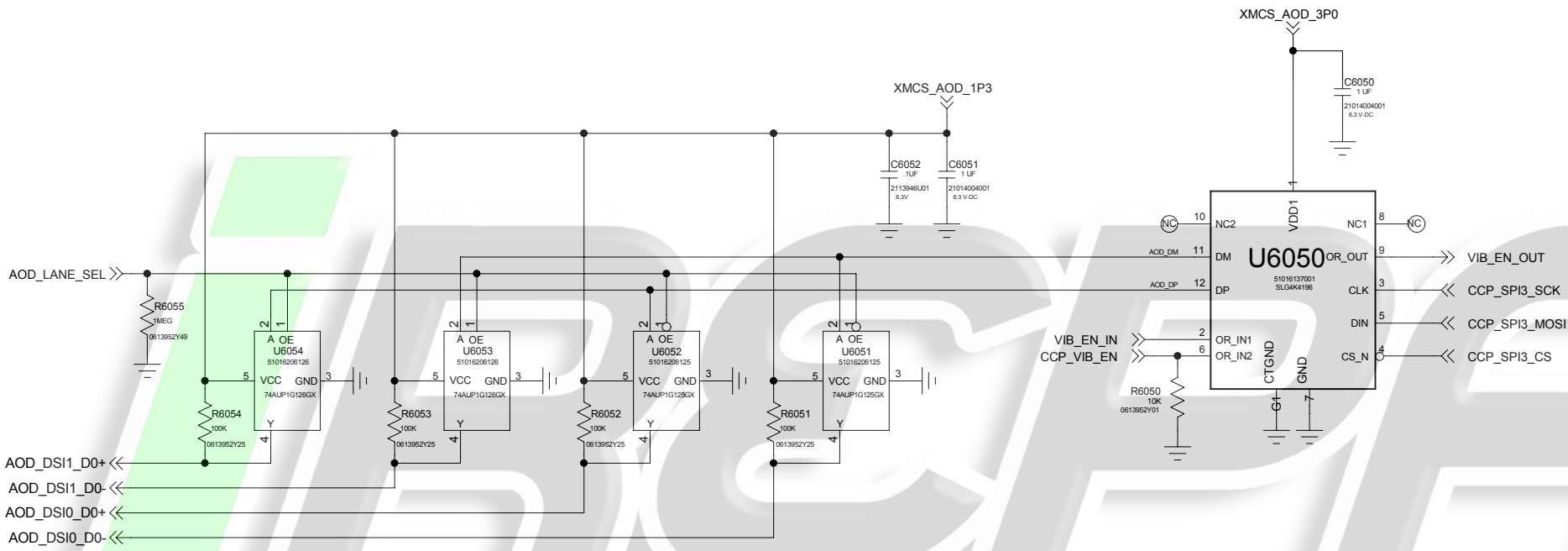
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PAGE 48 OF 76

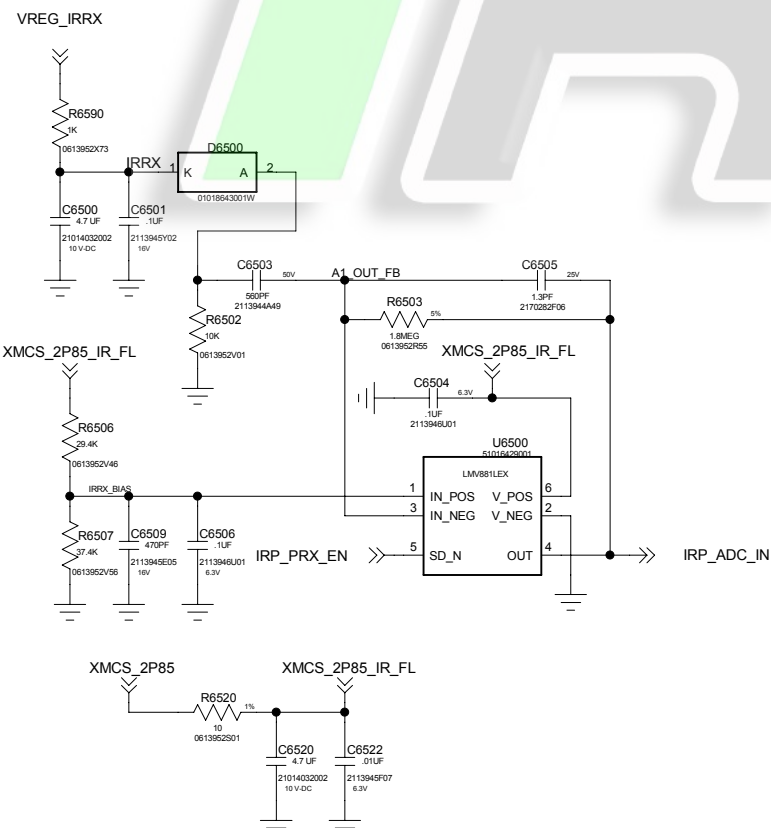
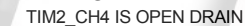
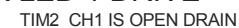
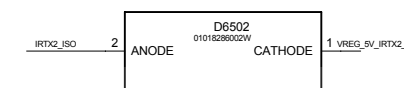
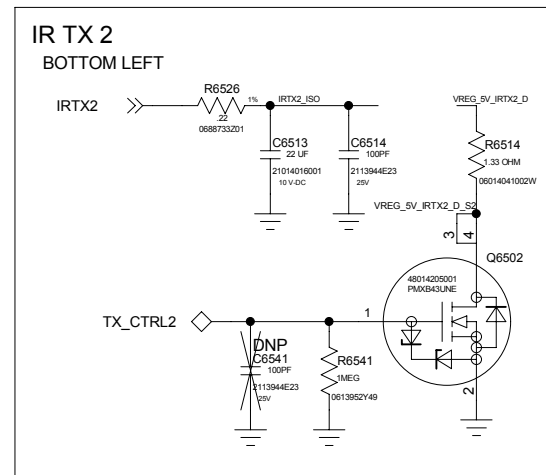
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REF: 6050-6059



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		REV 01	
		PAGE 49 OF 76	

REF 6500-6549



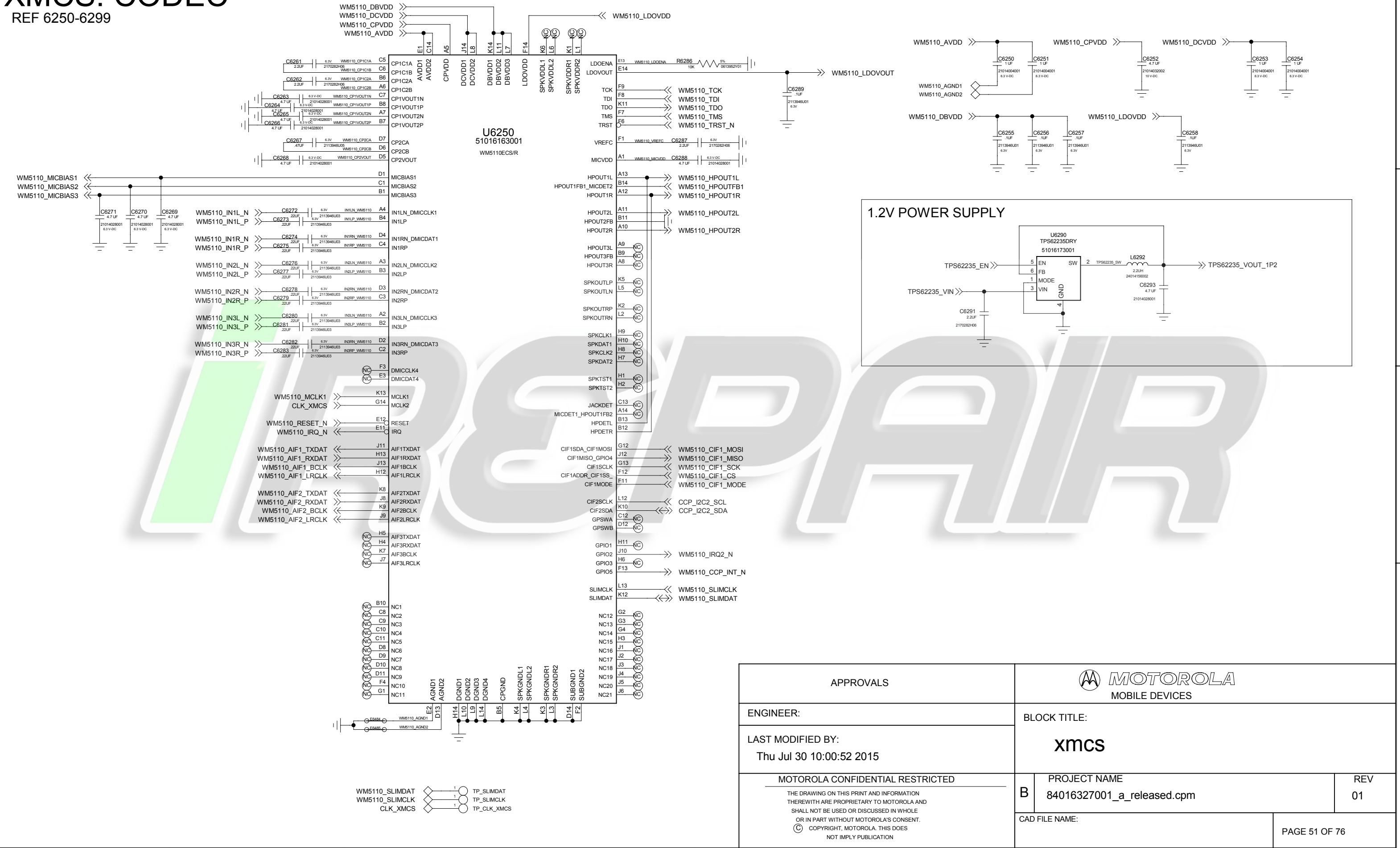
xmcs

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01

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XMCS: CODEC

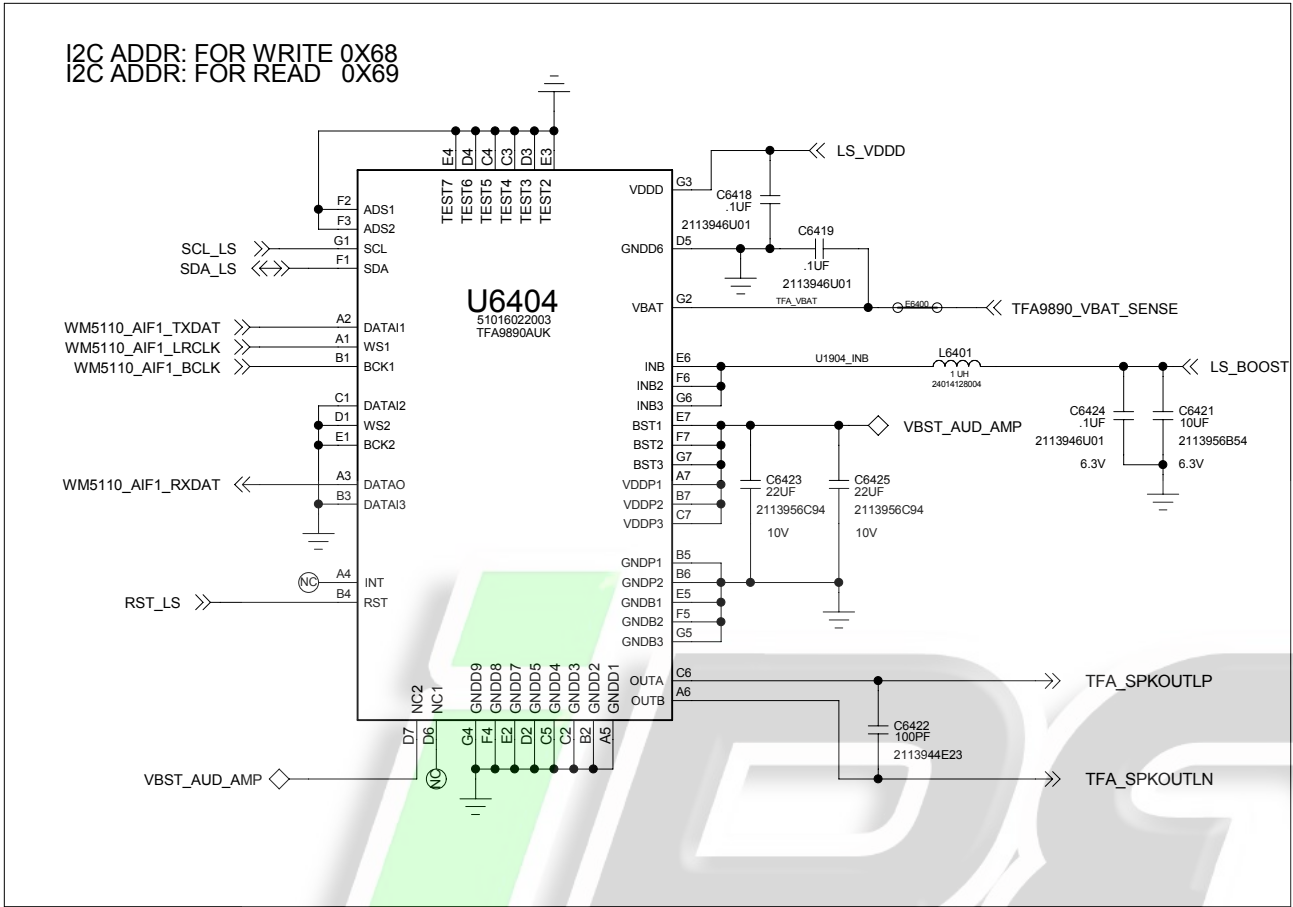
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		PAGE 51 OF 76	

XMCS: LOUD SPEAKER

REF 6400-6499



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CAD FILE NAME:			PAGE 52 OF 76

USR: PORTS

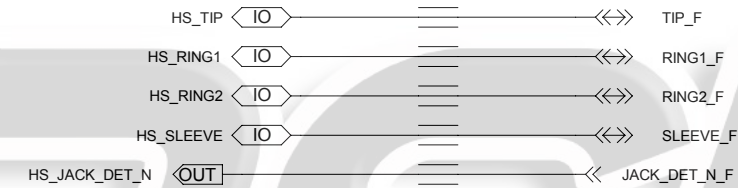
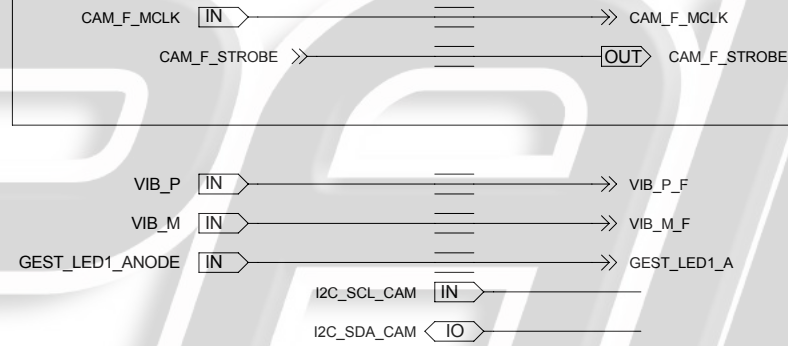
REF: 700-899

SUPPLIES

INTERFACES

GPIO - INPUT

GPIO - OUTPUT



APPROVALS

ENGINEER: NATE CONNELL

LAST MODIFIED BY: NATE CONNELL
Fri May 15 09:09:55 2015

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BLOCK TITLE:

user_input

PROJECT NAME
84016327001_a_released.cpm

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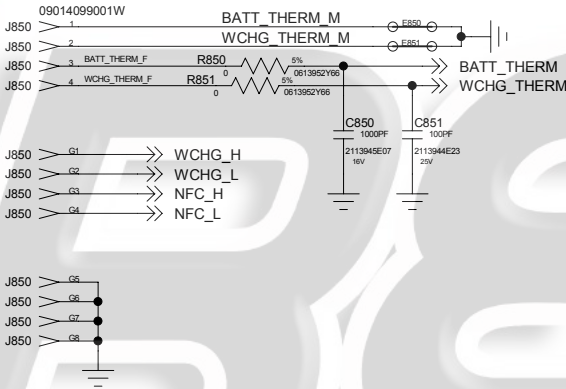
CAD FILE NAME:

PAGE 53 OF 76

USR: WLC/NFC FLEX

REF: 800-899

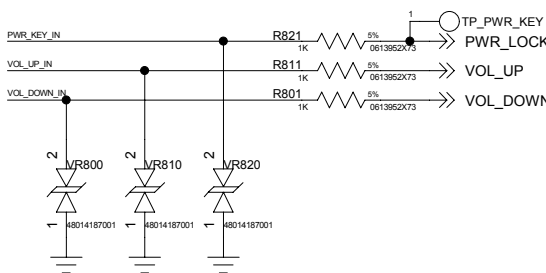
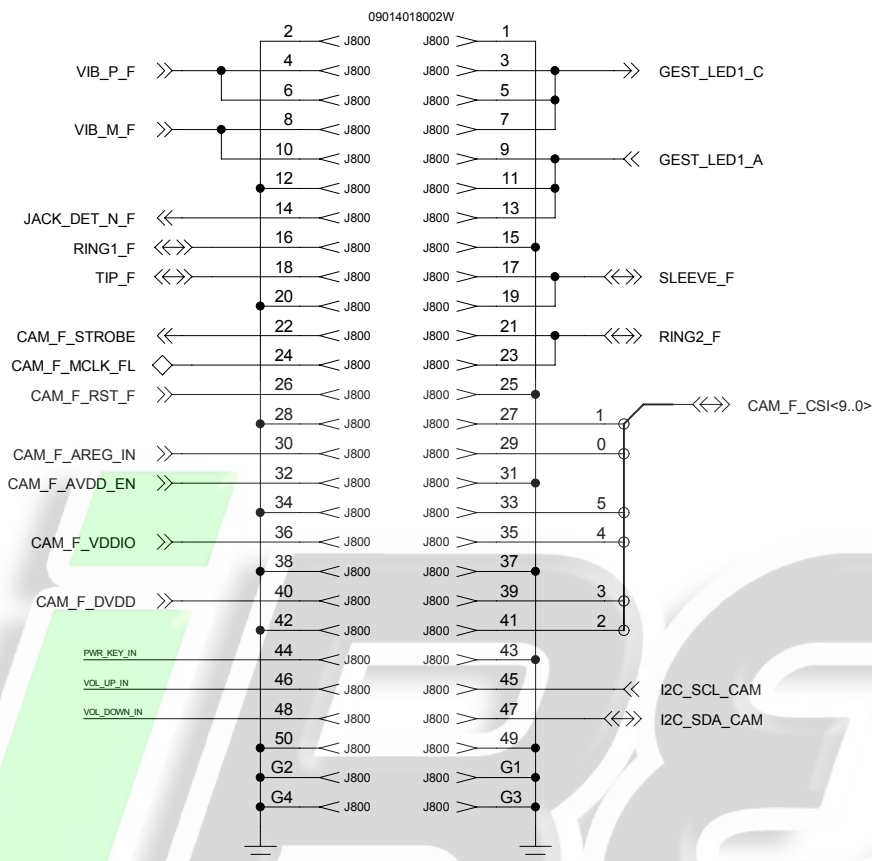
ROUTE BATT THERM_M AND WCHG_THERM_M AS DIFF PAIR WITH THERM
GROUND SHORT DOTS AT THERM INPUTS TO IC



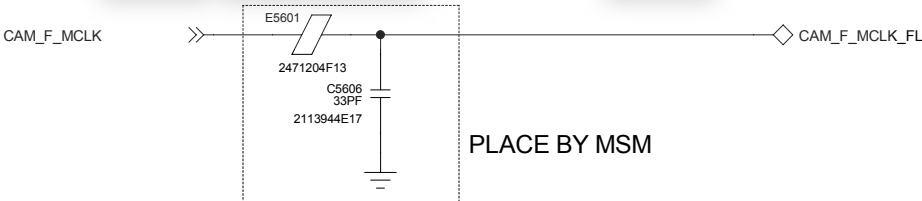
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ENGINEER: NATE CONNELL		BLOCK TITLE: user_input		
LAST MODIFIED BY: NATE CONNELL Thu Jul 30 10:00:54 2015				
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		B 84016327001_a_released.cpm		01
		CAD FILE NAME:		PAGE 54 OF 76

USR: TOP FLEX

REF: 800-899



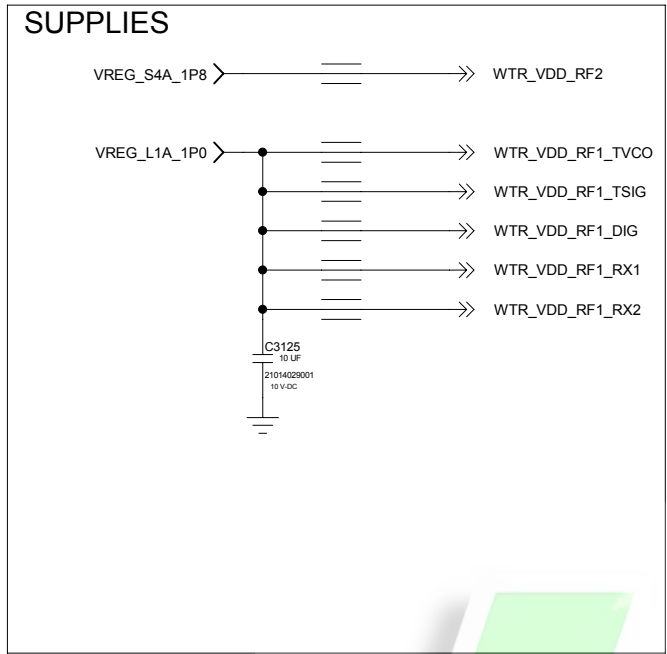
EMI/ESD FILTER



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LAST MODIFIED BY: NATE CONNELL Thu Jul 30 10:00:54 2015		user_input	
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		CAD FILE NAME:	PAGE 55 OF 76

WTR39X5: POWER

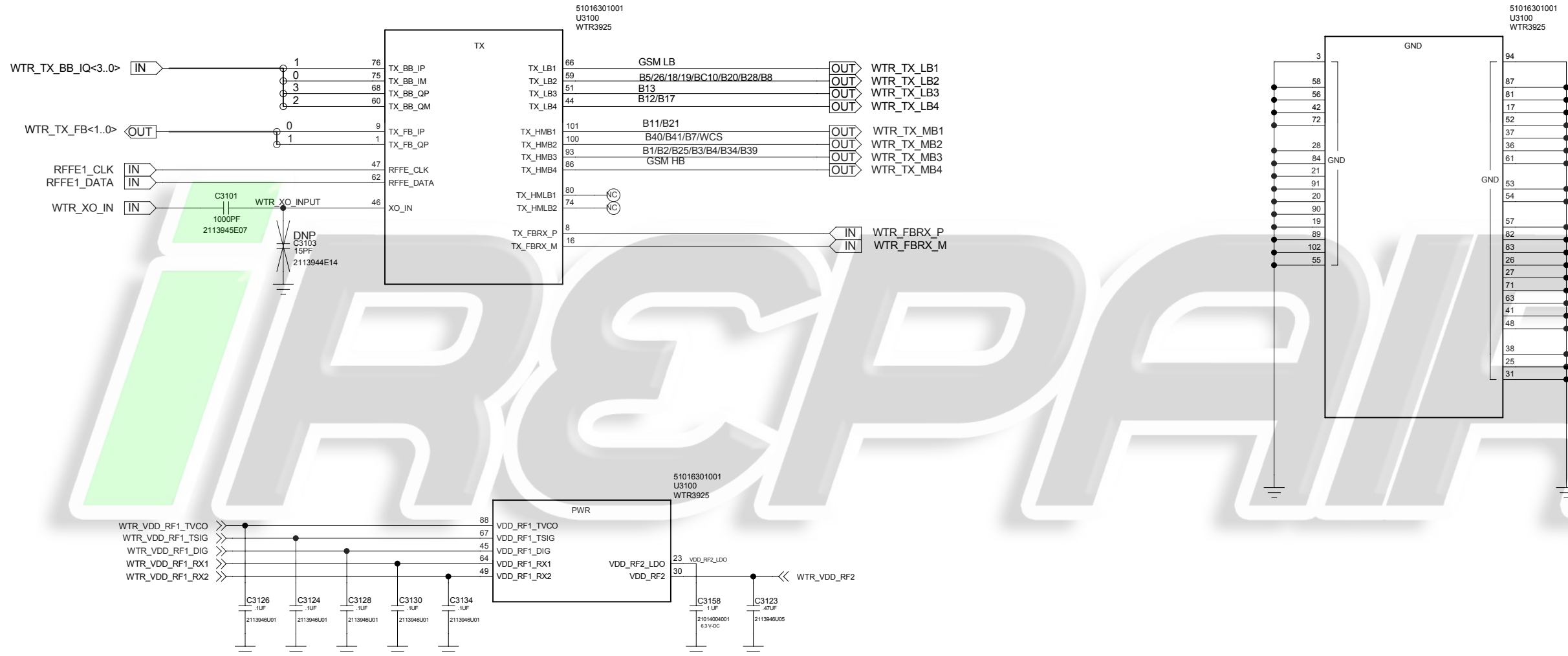
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APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: Thu Jul 30 10:00:56 2015			
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		B 80046327001_a_released.cpm	01
		CAD FILE NAME:	PAGE 56 OF 76

WTR39X5: TX/PWR/GROUND

REF 3100-3109



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY:		block_wtr39x5	
Thu Jul 30 10:00:55 2015			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		REV 01	
		CAD FILE NAME:	
		PAGE 57 OF 76	

WTR39X5: PRX/DRX/GPS

REF 3110-3119

D

D

C

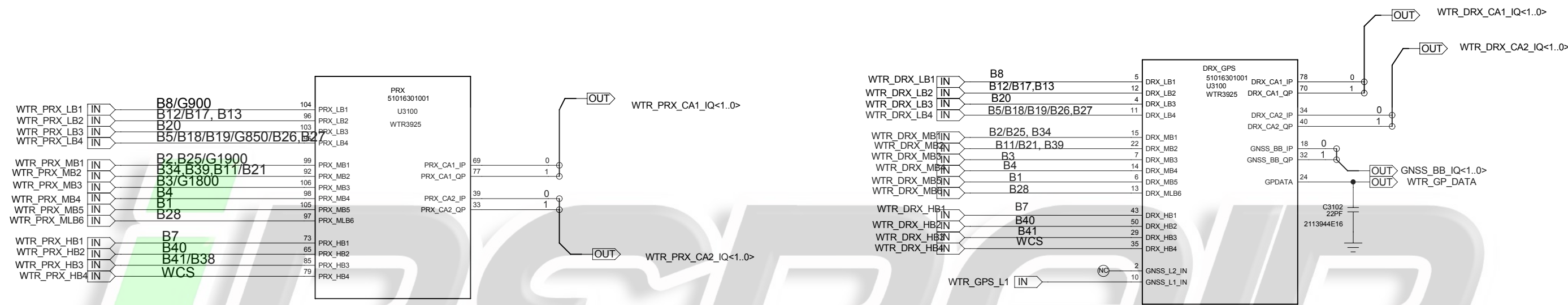
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B

B

A

A



QCOM'S NOTE:
WTR/WFR CA_IN/OUT INTERCONNECTION MAXIMUM TRACE LENGTH:
LOW BAND 12 MM; MID/ HI BAND 10 MM

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: block_wtr39x5	
LAST MODIFIED BY: Thu Jul 30 10:00:56 2015			
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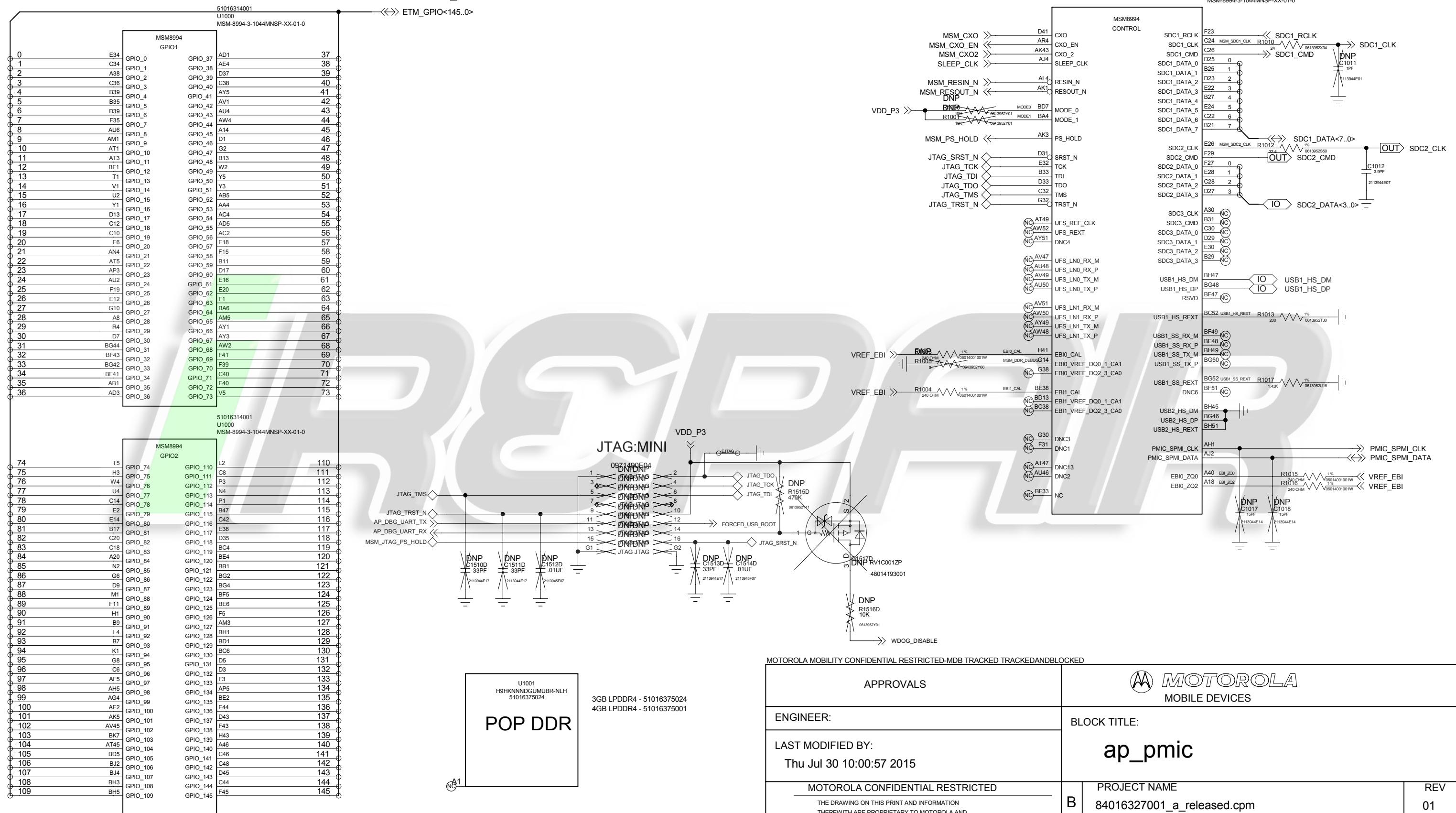
REFDES: 1200-1299

Figure 10: Power supply connections for the Q100. The diagram shows two columns of power supply connections. The left column includes VREG_S1A_S6A_1P0 to VDD_CORE, VREG_S2A_S12A_1P1 to VDD_MEM, VDD_MEM_2, VDD_MEM_3, VDD_USB_HS_CORE, and a network of resistors (R1204, R1205) and capacitors (C1206) connecting to VDD_MEM_BBXX_ADC and VDD_MEM_DAC. Below these are VREG_S4A_1P8 to VDD_DDR_CORE_1P8, VDD_P3, VDD_P7, VCCQ_EMMC, VREG_S7A_1P0 to VDD_MODEM, VREG_APCC0_1P0 to VDD_APC0, VREG_APCC1_0P988 to VDD_APC1, VREG_S2B_S3B_0P988 to VDD_GFX, VREG_LPDDR_1P1 to VDD_DDR_CORE_1P1, VDD_P1, and VREF_EBI. The right column includes VREG_L2A_1P25 to VDD_MIPI, VREG_L4A_1P225 to VDD_A1_1 and VDD_A1_2, VREG_L6A_1P8 to VDD_USB_1P8, VREG_L9A_1P8 to VDD_P5, VREG_L10A_1P8 to VDD_P6, VREG_L12A_1P8 to VDD_PLL2, VDD_PLL2_2, VDD_PCIE_1P8, VDD_P9, VDD_P11, VDD_UFS_1P8, VREG_L13A_2P95 to VDD_P2, VREG_L15A_1P8 to VDD_A2, VDD_QFPROM, VDD_A2_3, VREG_L20A_2P95 to VCC_EMMC, VREG_L24A_3P075 to VDD_USB_3P1, VREG_L26A_1P0 to VDD_EBI, VREG_L28A_1P0 to VDD_PLL1, VDD_PCIE_CORE, VDD_USB_SS_CORE, VDD_UFS_CORE, VREG_L31A_1P2 to VDD_P10, and SDC_UIM_VBIAS to VREF_SDC_UIM.

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NICK HACKETT	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: NICK HACKETT Thu Jul 30 10:01:07 2015			
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		CAD FILE NAME:	PAGE 59 OF 76

REFDES: 1000-1099

FOR PORTABLE DESIGNS RENAME ETM_GPIO BUS TO GPIO BUS



BASED ON 80-NJ051-41 REV. U

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MOTOROLA
MOBILE DEVICES

APPROVALS

ENGINEER:

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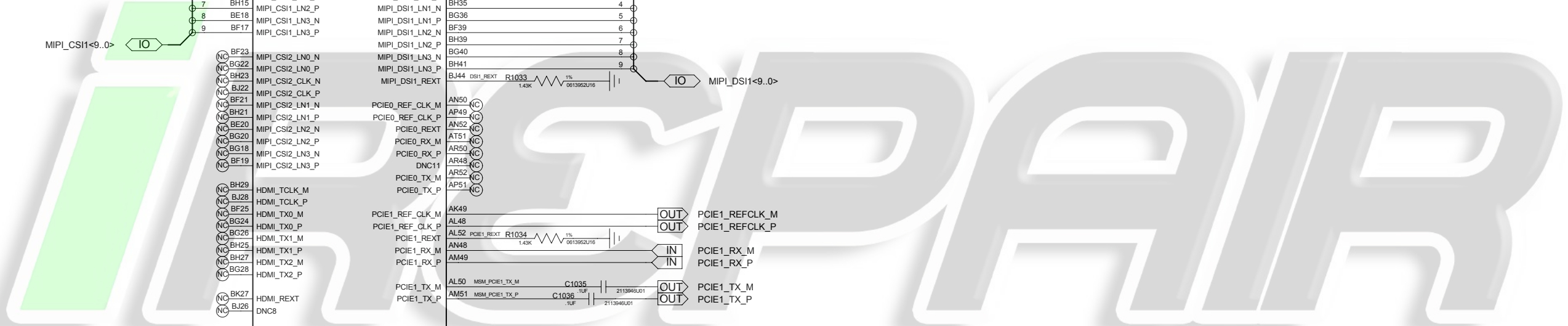
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REFDES: 1030-1039



8

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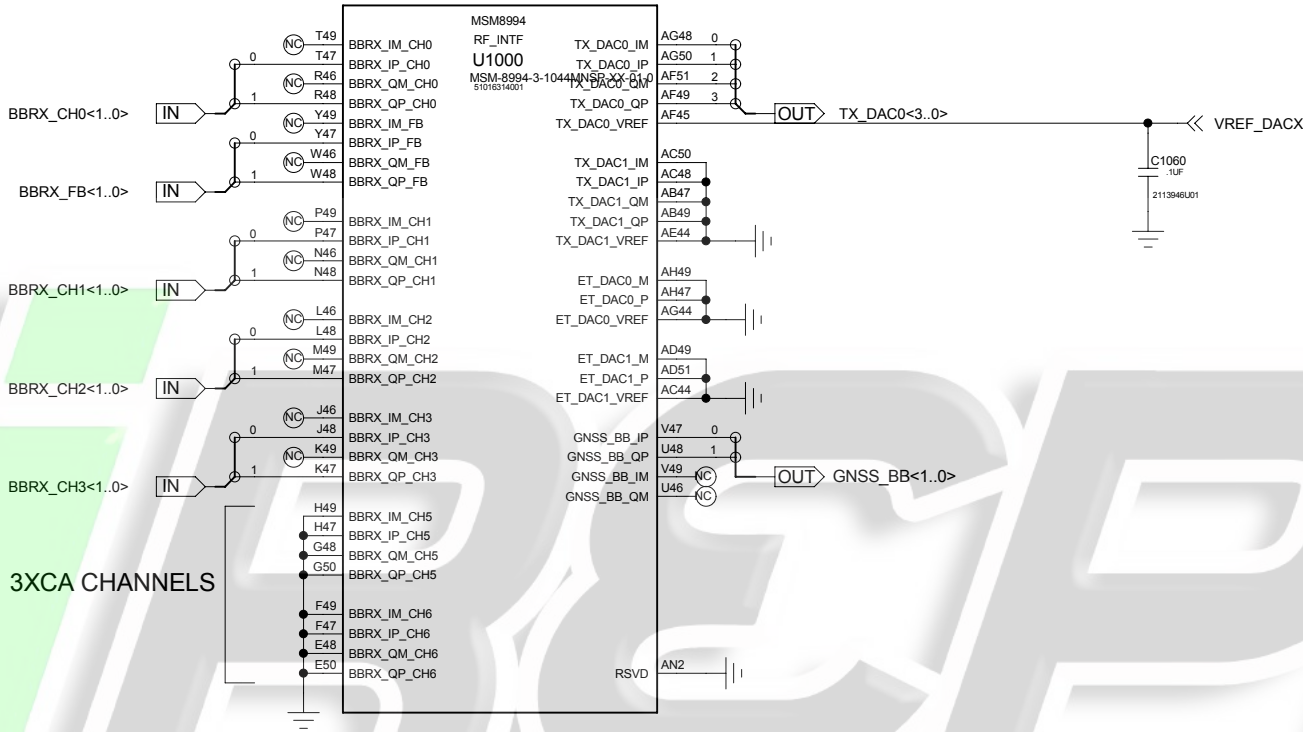
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MSM8994: RF INTERFACE

REFDES: 1040-1049

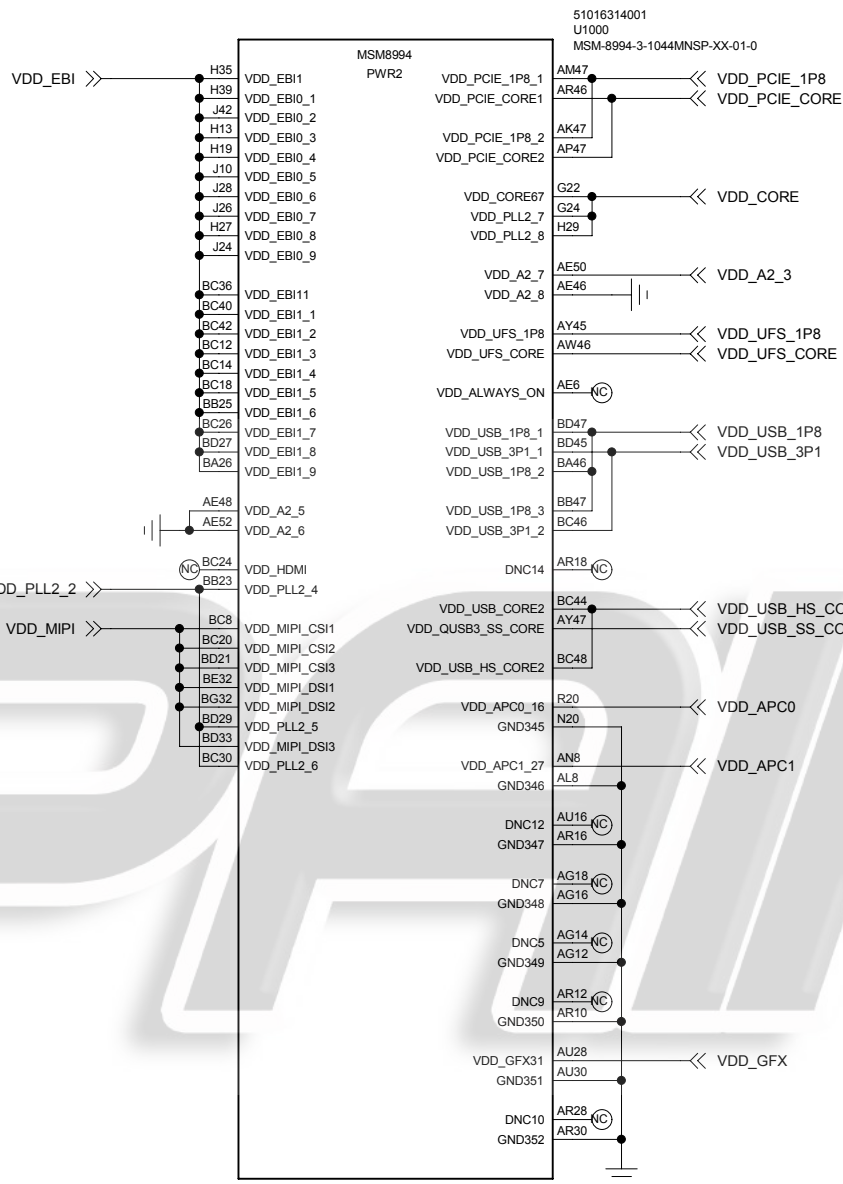
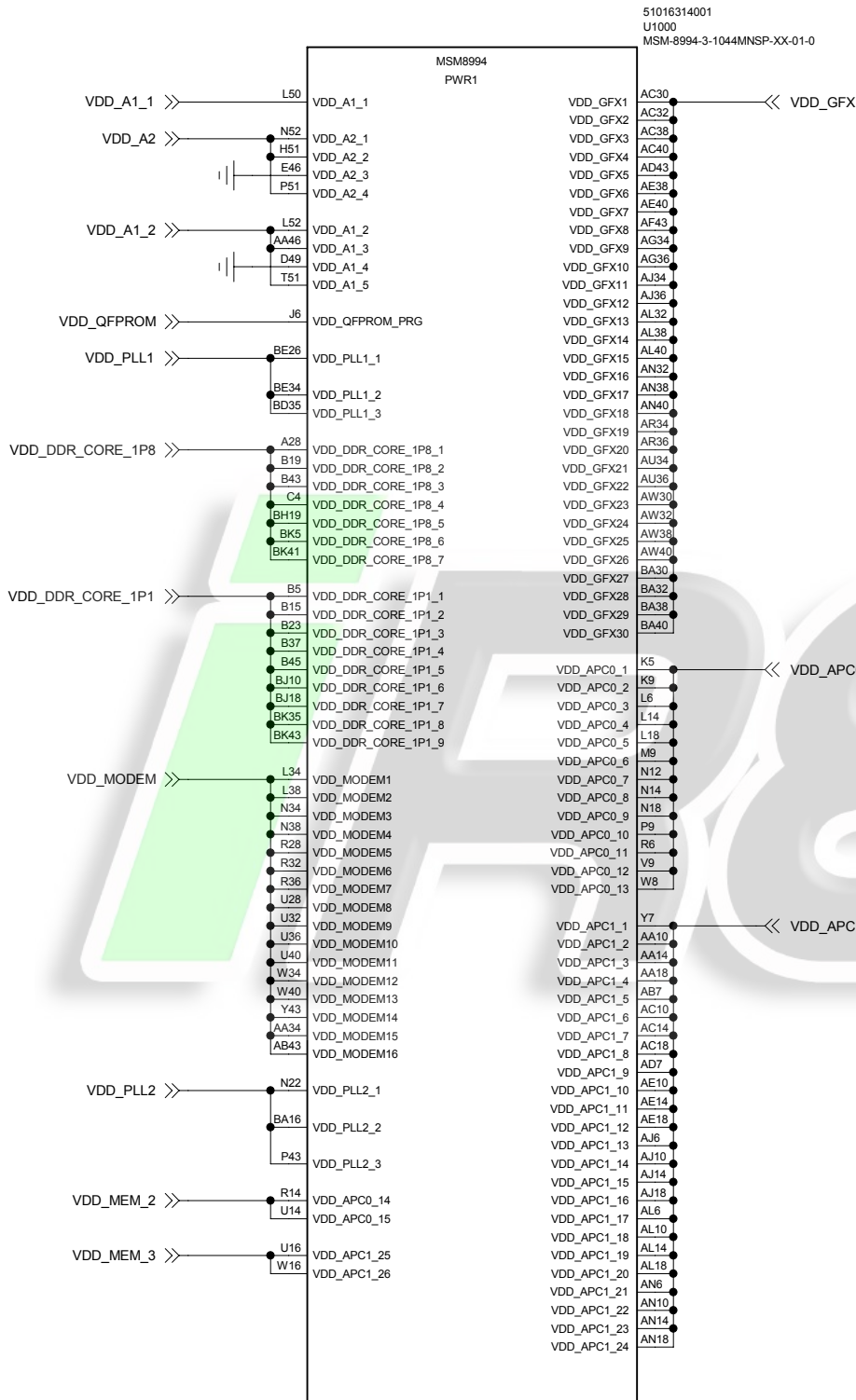


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BASED ON 80-NJ051-41 REV. U

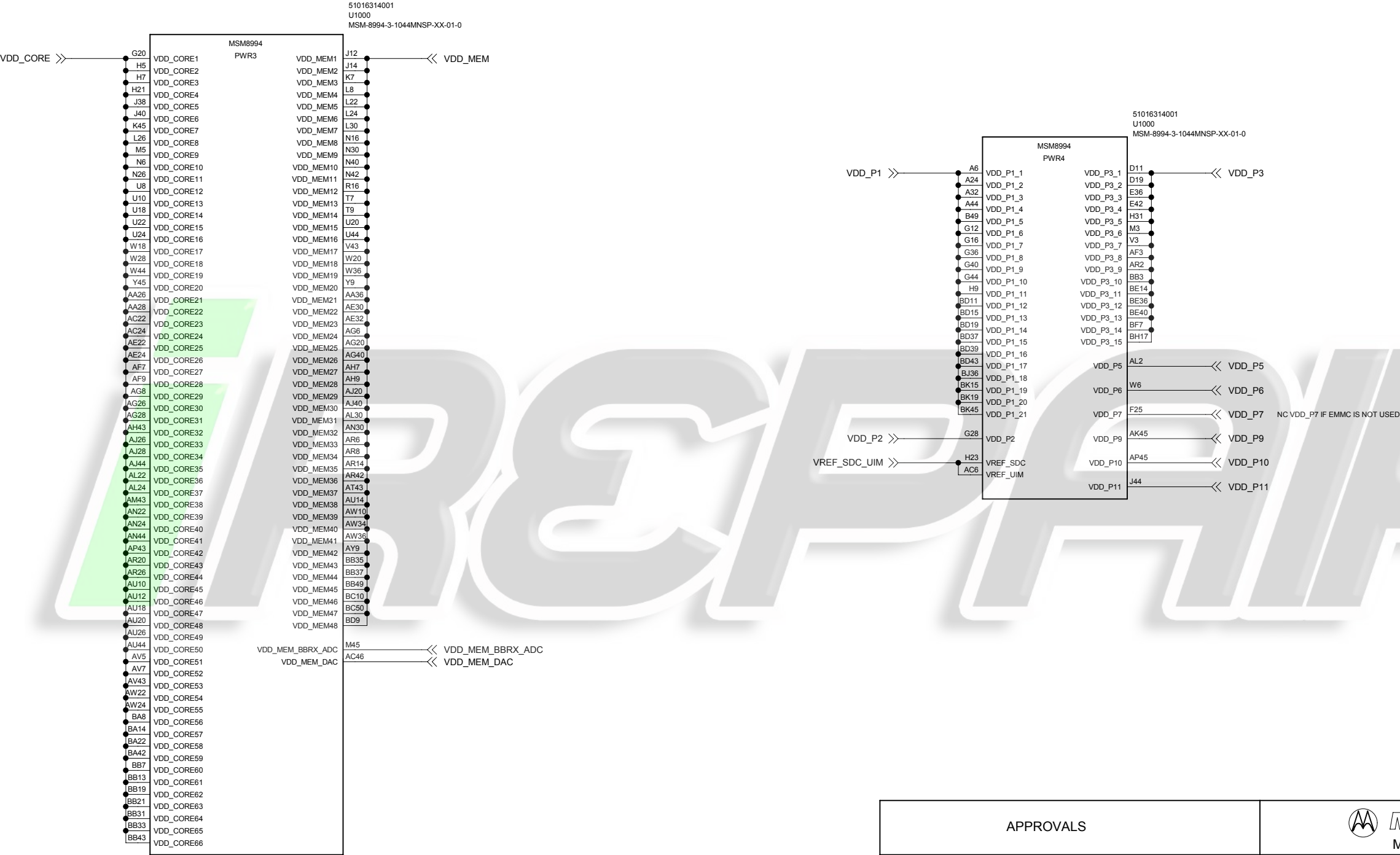
MSM8994: POWER 1



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		PAGE 63 OF 76	

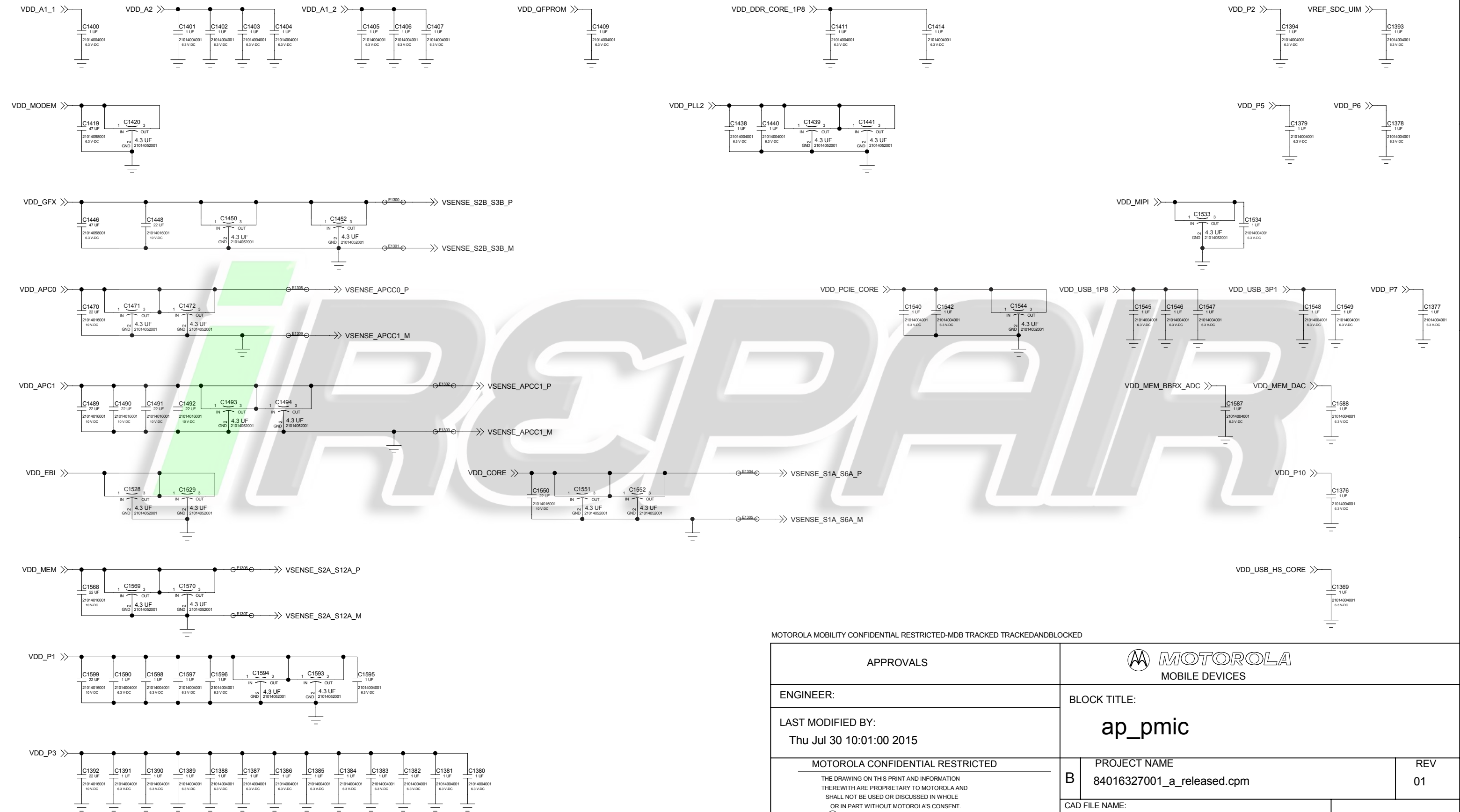
MSM8994: POWER 2



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MOTOROLA		BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: MOTOROLA Thu Jul 30 10:00:59 2015				
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		B 84016327001_a_released.cpm		01
UNBLOCKED		CAD FILE NAME:		PAGE 64 OF 76

MSM8994: BYPASS CAPS

REFDES: 1300-1599

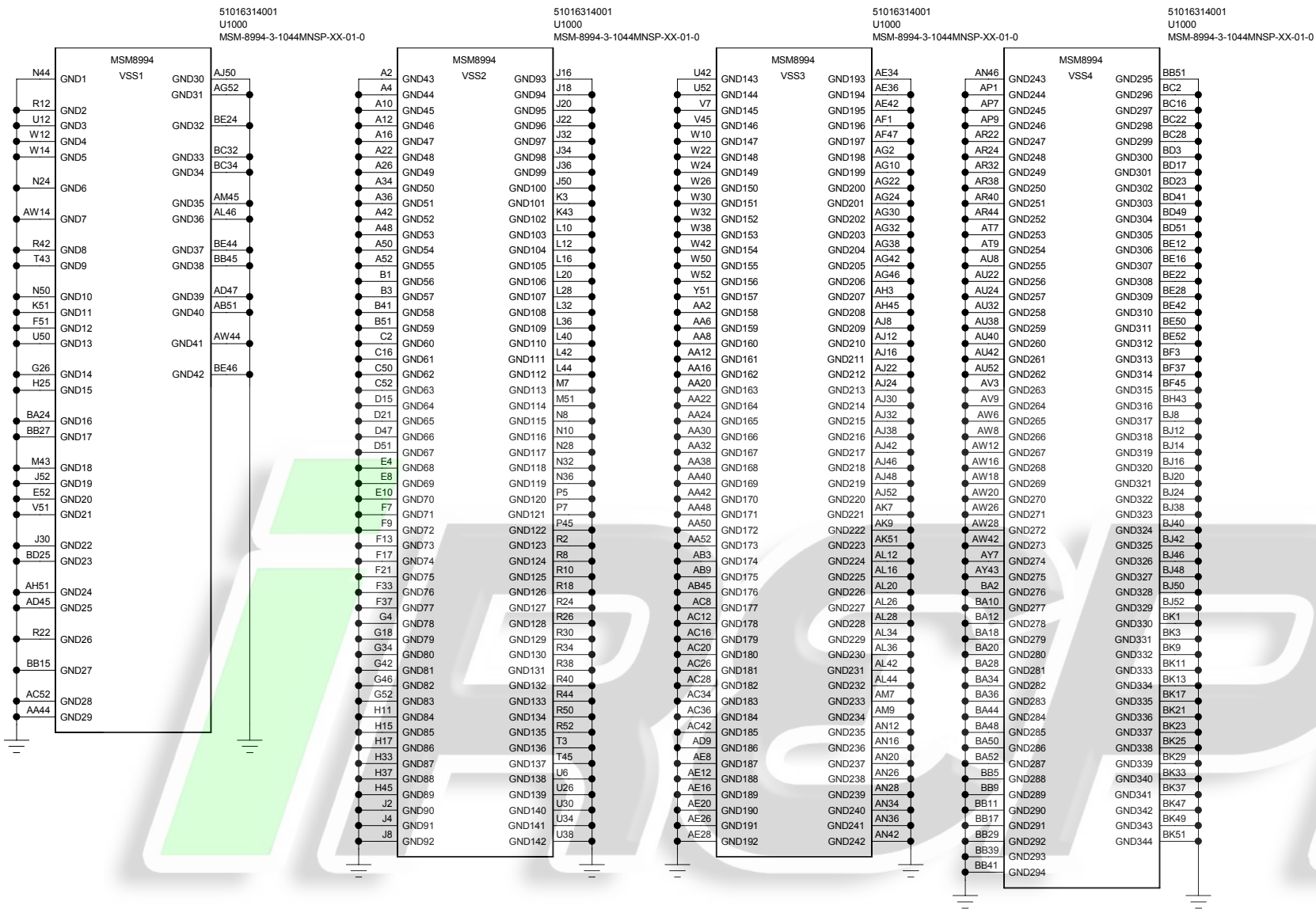


BASED ON 80-NJ051-41 REV. U

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Thu Jul 30 10:01:00 2015		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME B 84016327001_a_released.cpm	REV 01
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MSM8994: GND/NC



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Thu Jul 30 10:01:03 2015		ap_pmic	
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B		REV 01	
CAD FILE NAME:		PAGE 66 OF 76	

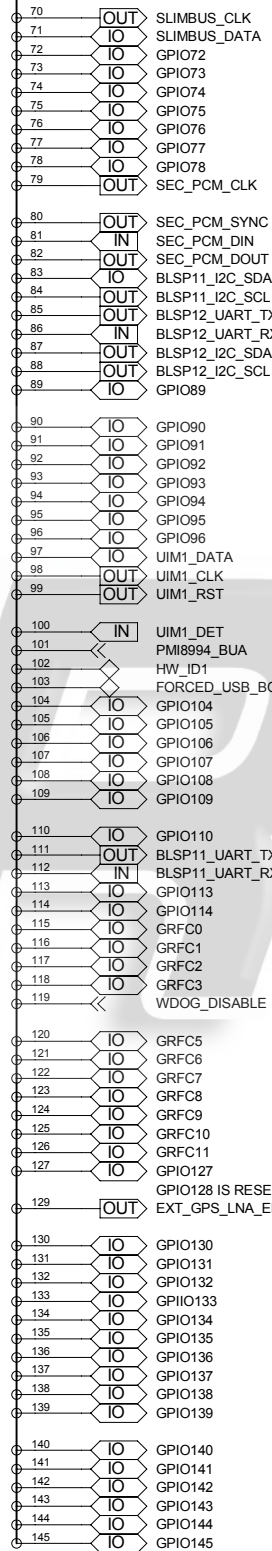
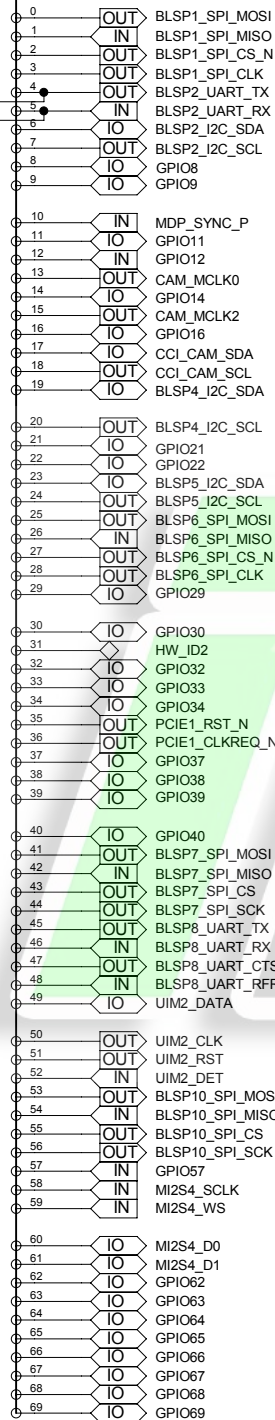
MSM8994: GPIO MAP

REFDES: 1100-1199

PURPLE=MOT CUSTOM
CYAN=QCOM REFERENCE

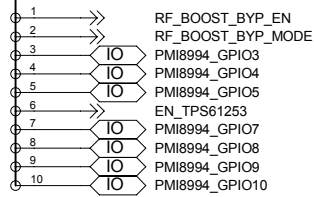
ETM_GPIO<145..0> <=>

AP_DBG_UART_TX
AP_DBG_UART_RX



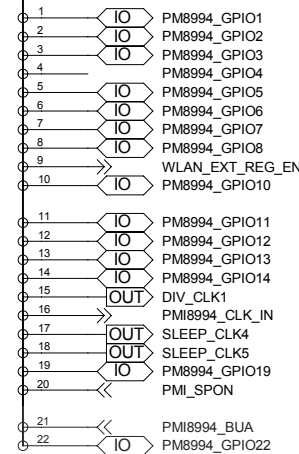
PMI8994_GPIO<10..1> <=>

PMI8994 GPIO



PM8994_GPIO<22..1> <=>

PM8994 GPIO



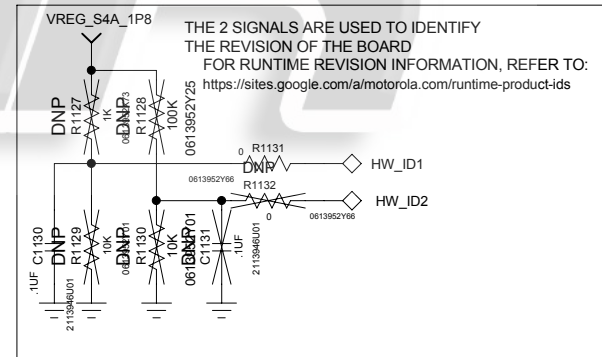
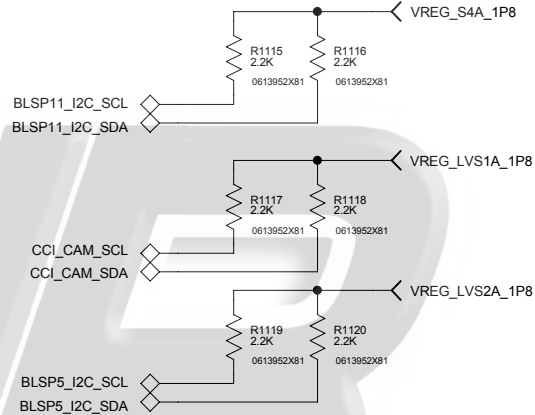
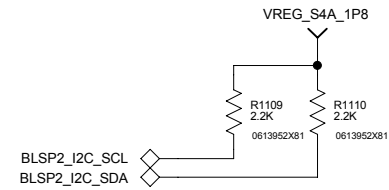
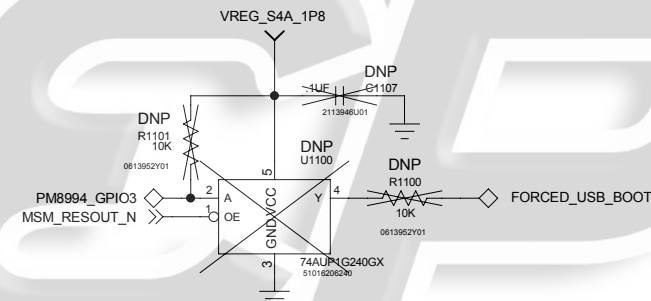
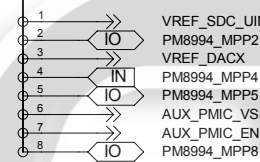
PMI8994_MPP<4..1> <=>

PMI8994_MPP



PM8994_MPP<8..1> <=>

PM8994_MPP



THE 2 SIGNALS ARE USED TO IDENTIFY
THE REVISION OF THE BOARD
FOR RUNTIME REVISION INFORMATION, REFER TO:
<https://sites.google.com/a/motorola.com/runtime-product-ids>

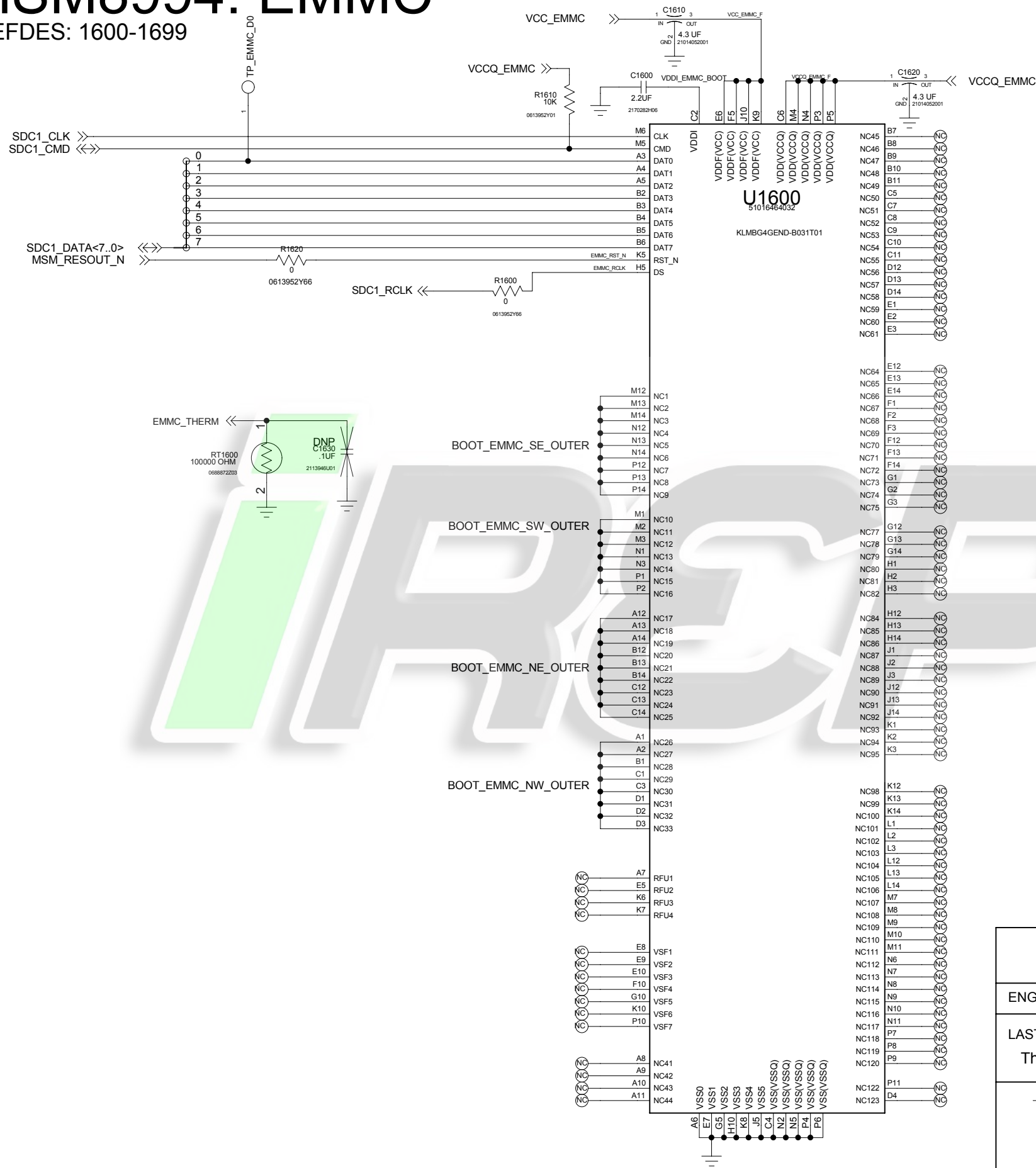
BASED ON 80-NJ051-41 REV. U

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: TIM MILLER Thu Jul 30 10:00:58 2015		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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		PAGE 67 OF 76	

MSM8994: EMMC

REFDES: 1600-1699



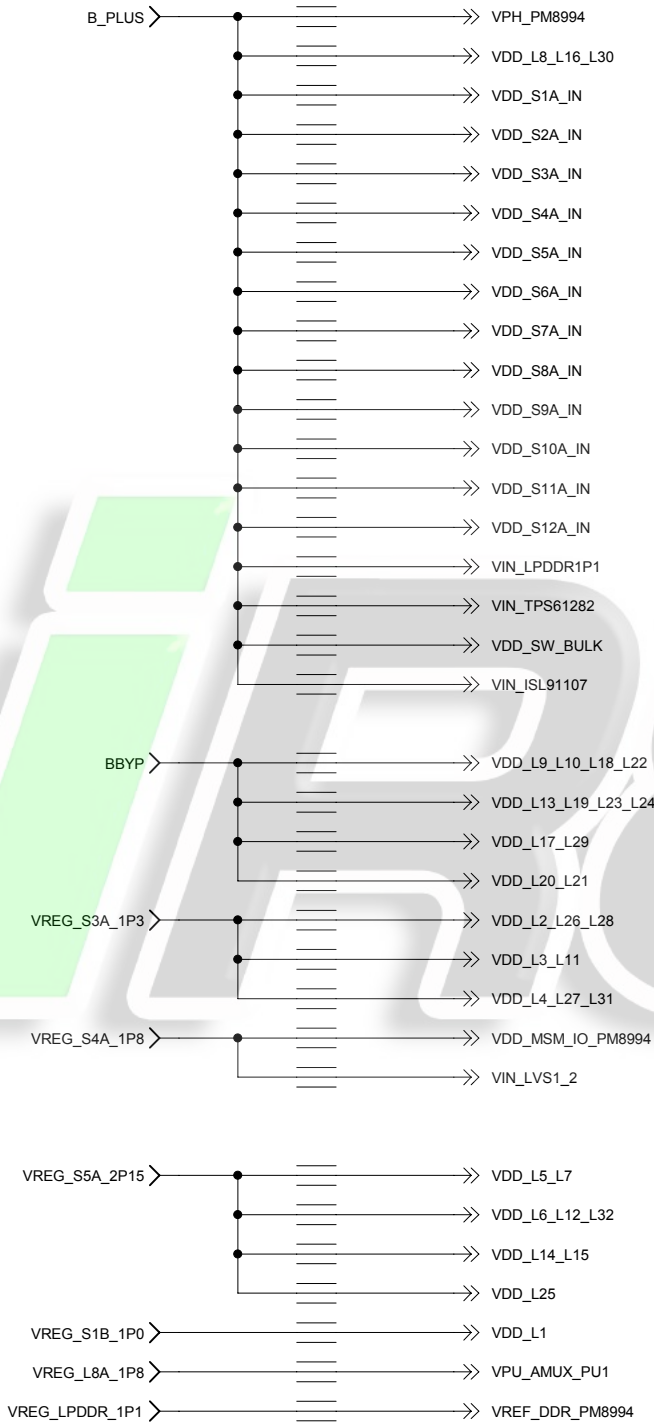
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APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MOTOROLA Thu Jul 30 10:01:02 2015		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		REV 01	
		PAGE 68 OF 76	

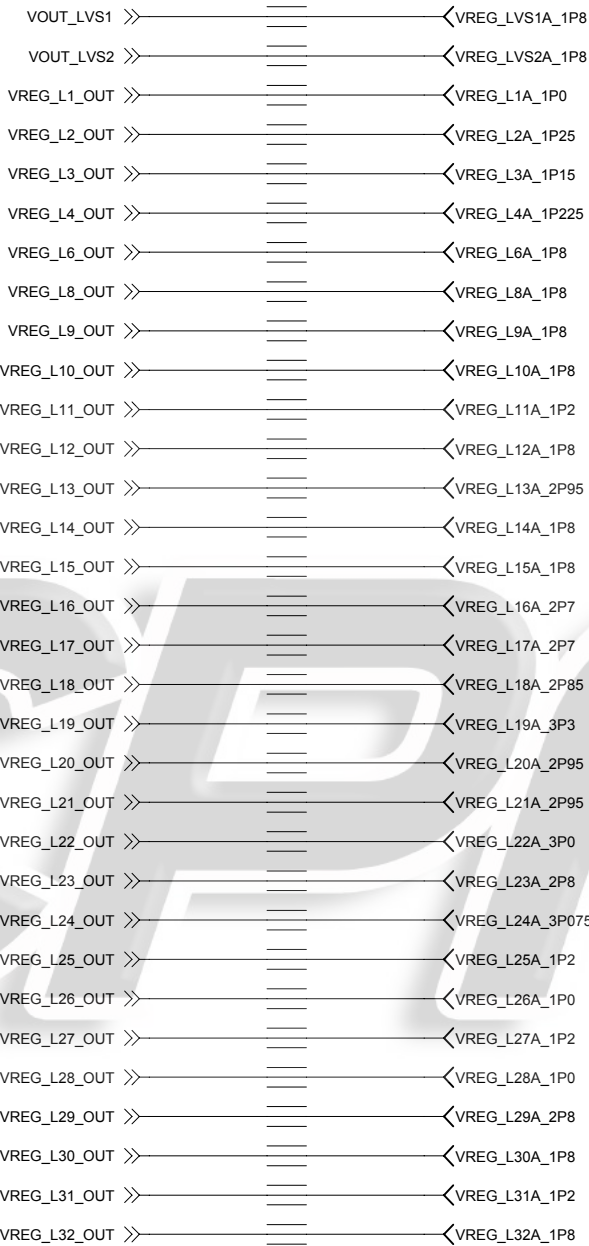
PM8994: POWER

REFDES: 2300-2399

INPUT POWER



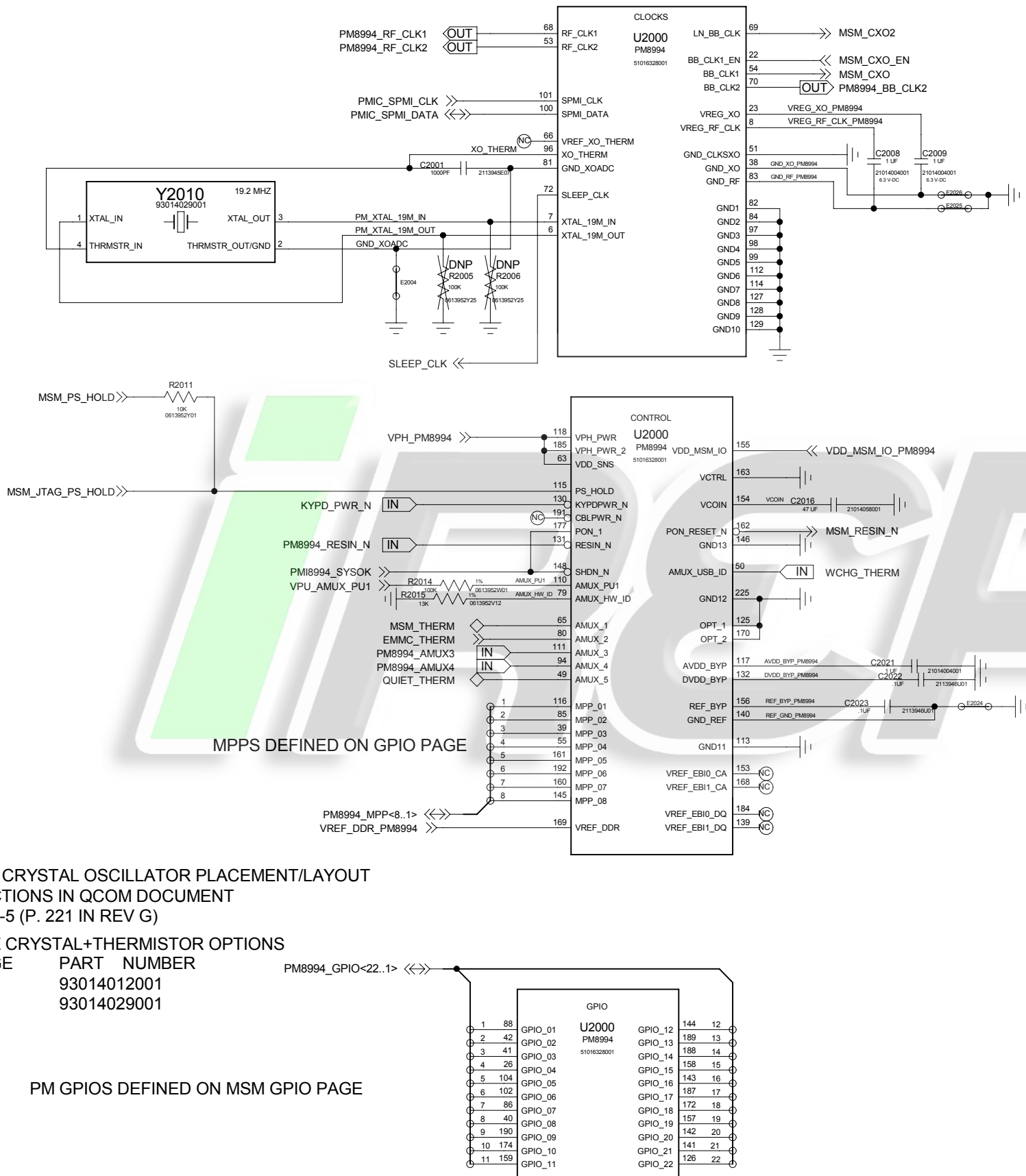
OUTPUT POWER



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NICK HACKETT		BLOCK TITLE:	
LAST MODIFIED BY: NICK HACKETT Fri May 15 09:09:42 2015		ap_pmic	
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		CAD FILE NAME:	
		PAGE 69 OF 76	

PM8994: CLK/CTRL/GPIO

REFDES: 2000-2099



FOLLOW CRYSTAL OSCILLATOR PLACEMENT/LAYOUT INSTRUCTIONS IN QCOM DOCUMENT 80-N5423-5 (P. 221 IN REV G)

*19.2MHZ CRYSTAL+THERMISTOR OPTIONS

PACKAGE	PART	NUMBER
2520	93014012001	
2016	93014029001	

PM GPIOs DEFINED ON MSM GPIO PAGE

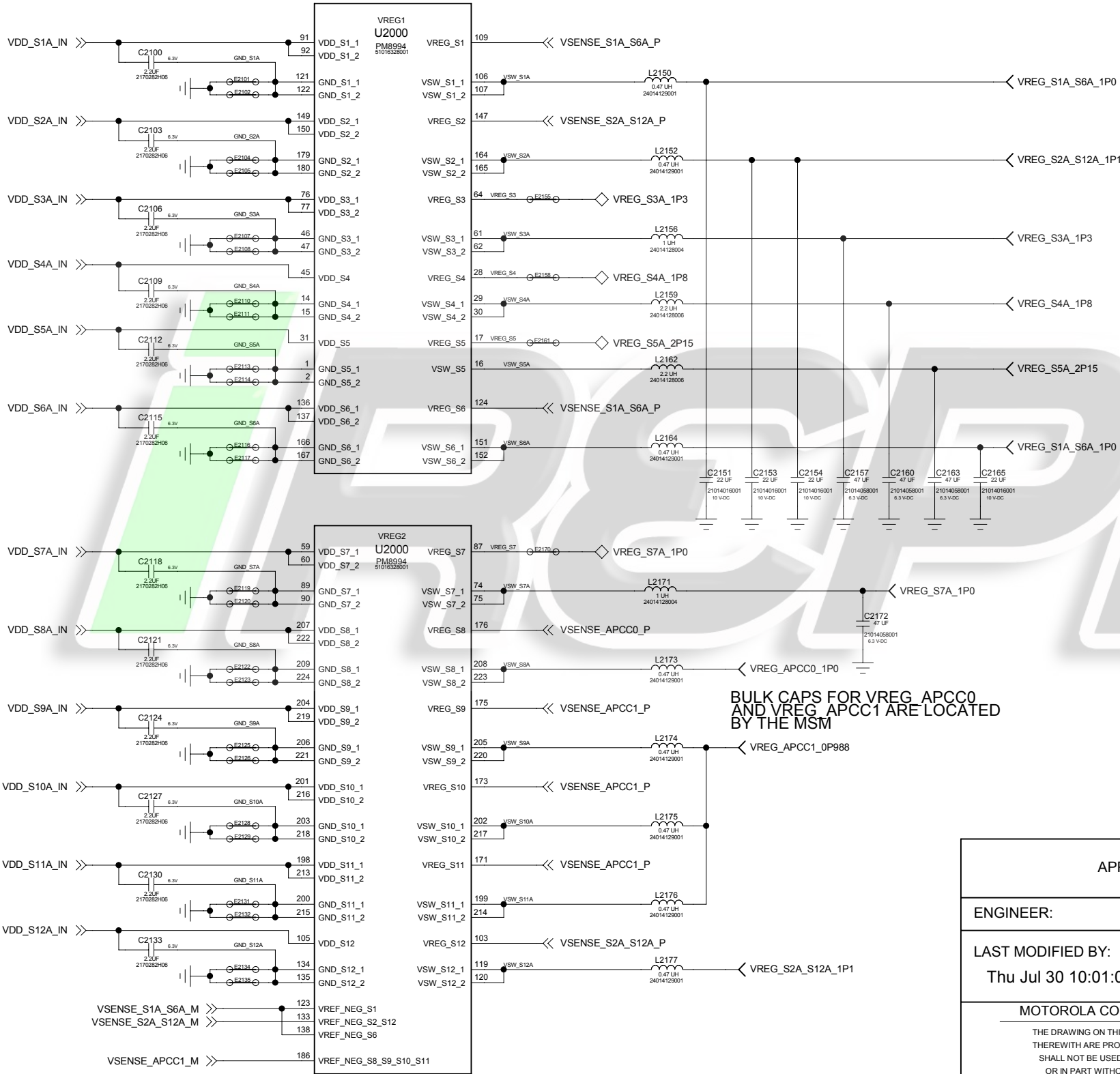
BASED ON 80-NJ051-41 REV. U

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

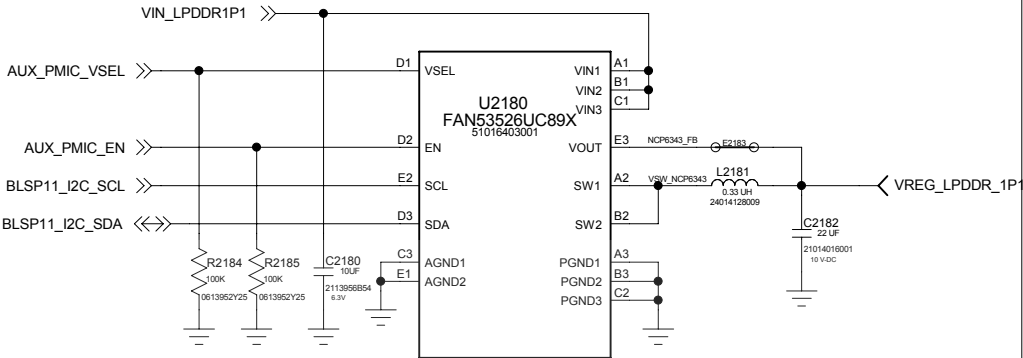
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Thu Jul 30 10:01:01 2015		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME 84016327001_a_released.cpm	REV 01
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PM8994: SMPS

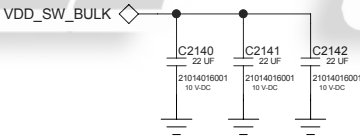
REFDES: 2100-2199



VREG_LPDDR1 EXTERNAL REGULATOR



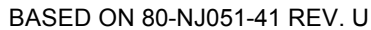
SWITCHER INPUT BULK CAPS



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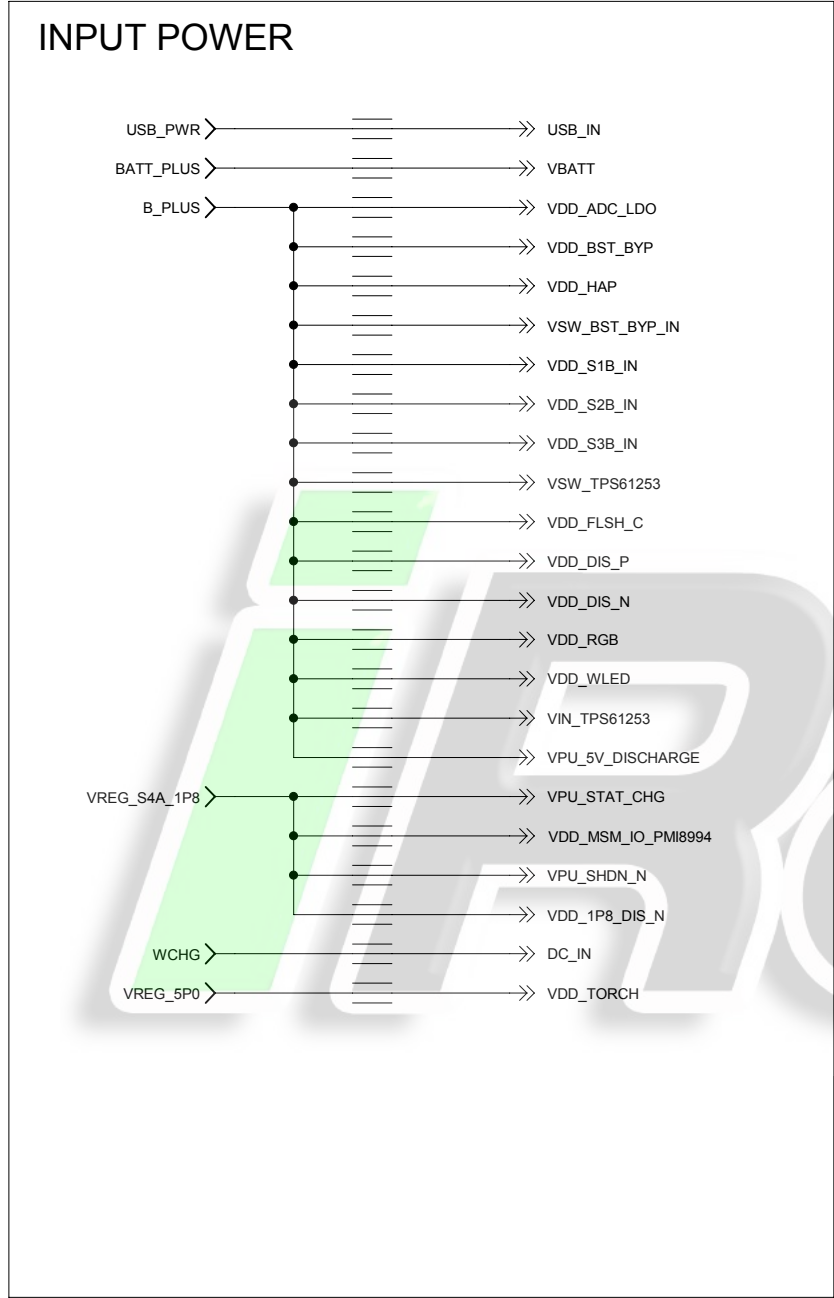
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ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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		CAD FILE NAME:	PAGE 71 OF 76

REFDES: 2200-2299

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PMI8994: POWER

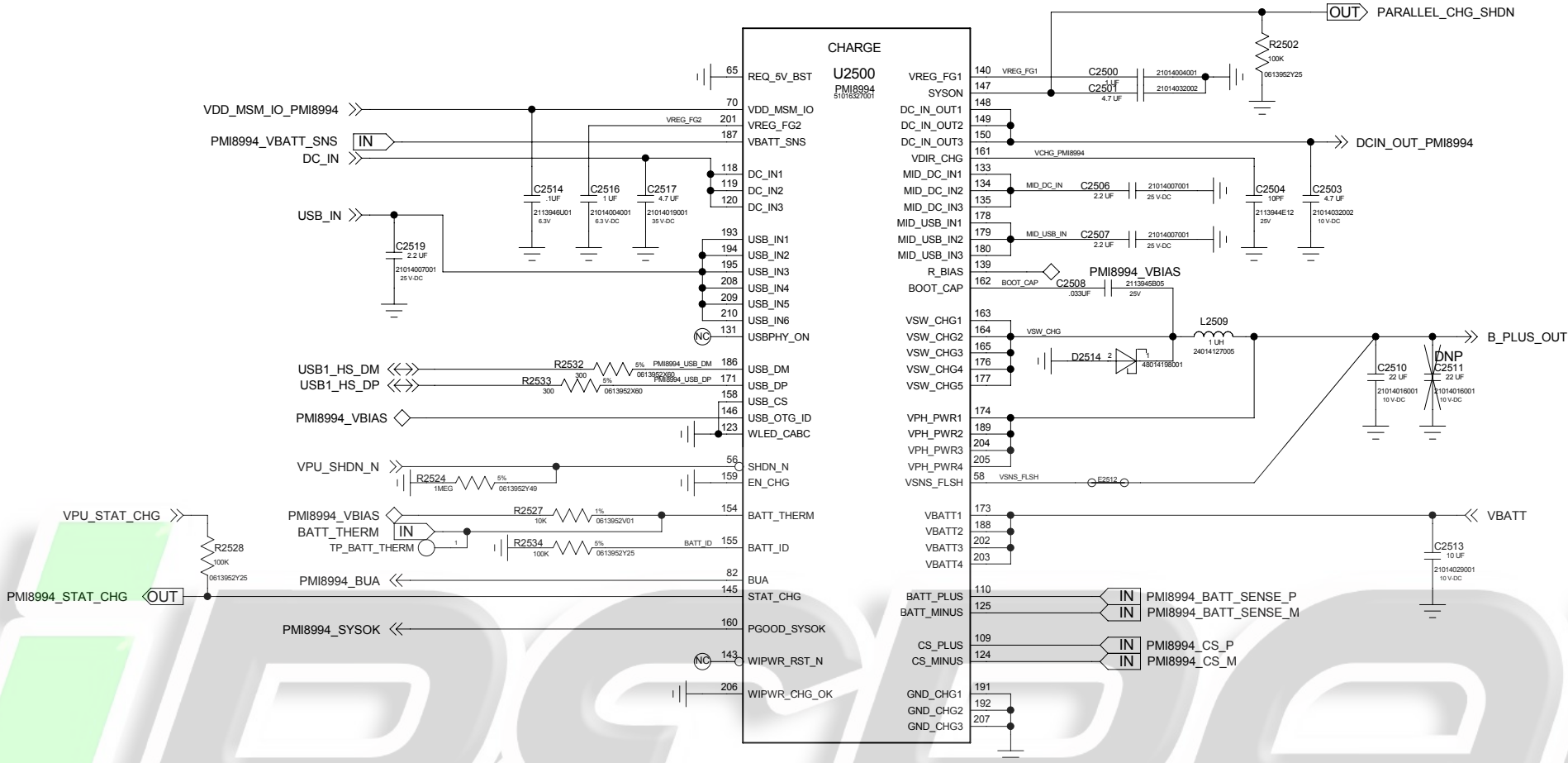
REFDES: 2400-2499



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: NICK HACKETT		BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: NICK HACKETT Fri May 15 09:09:46 2015				
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		B 84016327001_a_released.cpm		01
		CAD FILE NAME:		PAGE 73 OF 76

PMI8994: CHG/USB

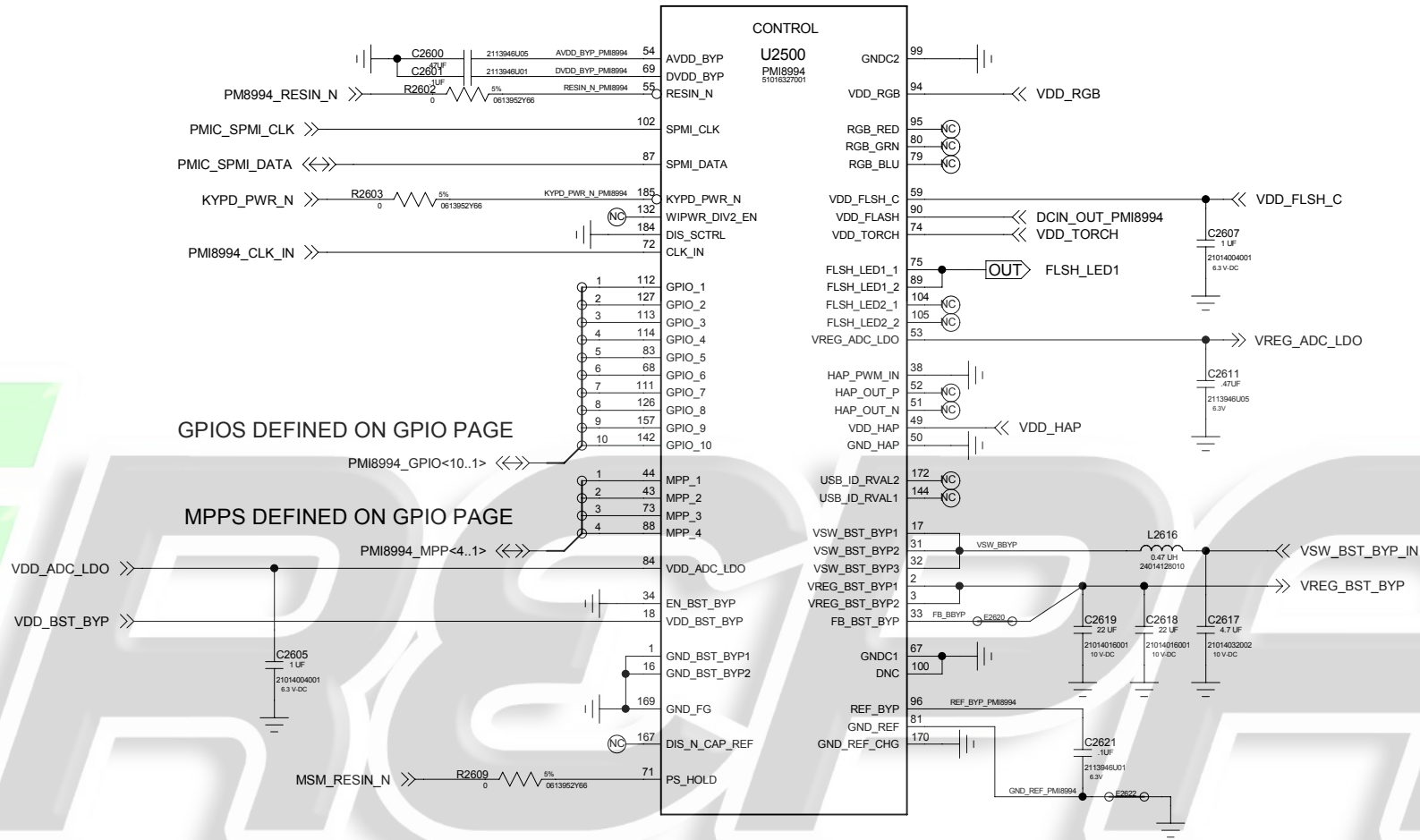
REFDES: 2500-2599



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Jul 30 10:01:05 2015		ap_pmic	
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		CAD FILE NAME:	
		PAGE 74 OF 76	

PMI8994: GPIO/LED/RGB

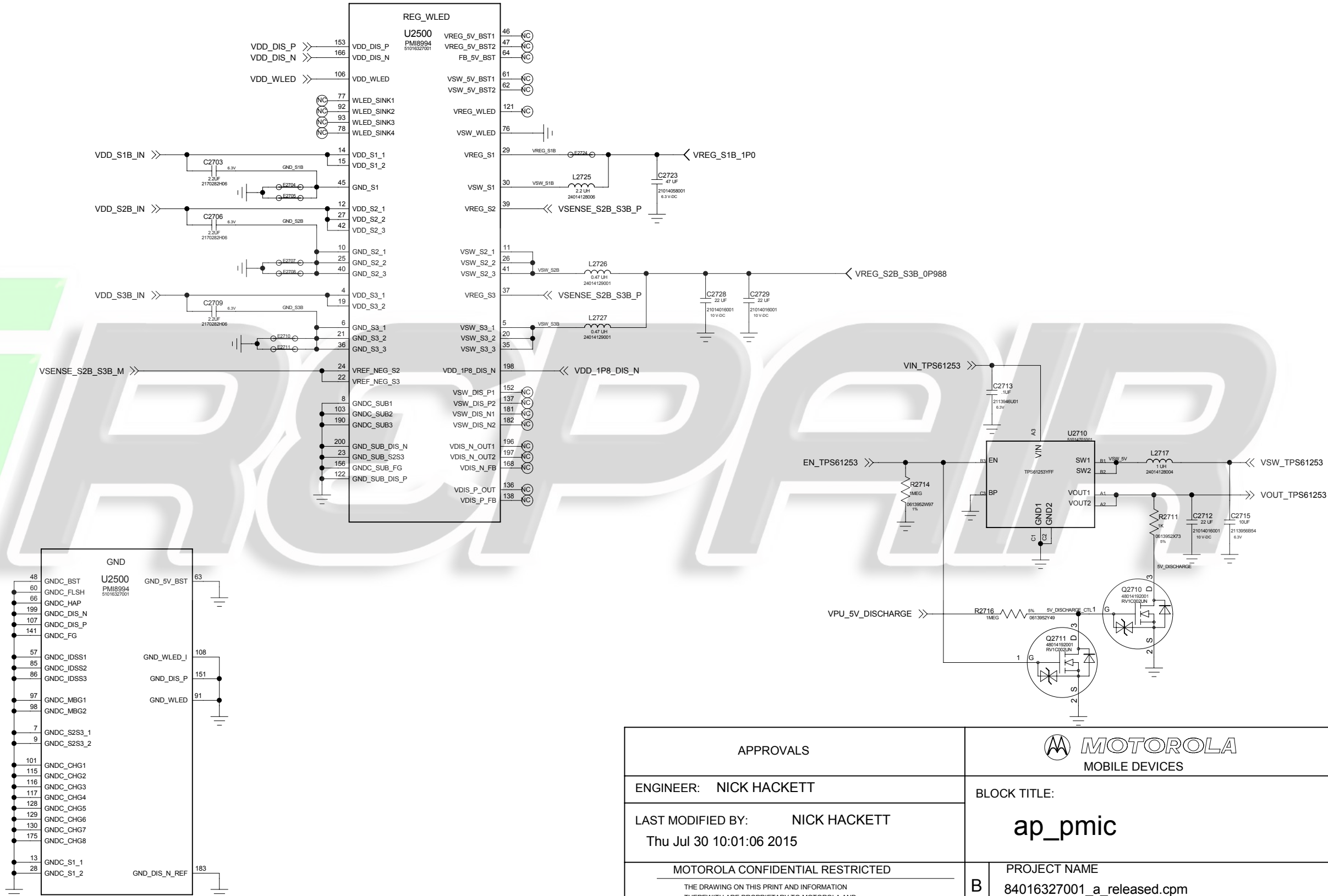
REFDES: 2600-2699



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Jul 30 10:00:59 2015		ap_pmic	
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CAD FILE NAME:			PAGE 75 OF 76

PMI8994: REF/WLED/GND

REFDES: 2700-2799



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NICK HACKETT		BLOCK TITLE:	
LAST MODIFIED BY: NICK HACKETT Thu Jul 30 10:01:06 2015		ap_pmic	
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CAD FILE NAME:		PAGE 76 OF 76	